PerforMAX®

Performance Management System

User Guide

Release 3.0.1

TRENDIUM™

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Contents

PerforMAX Overview

1

The following documentation provides information for getting started with PerforMAX™.

The topics in this documentation include the following:

- Conventions
- Before Using the Dashboard
- Starting the Dashboard
- Using the Dashboard
- Enhancing Search Ability
- Closing the Dashboard
- Displaying Version Number

This documentation provides instructions for using the GUI interface which is also known as the Dashboard.

In addition, you can find information on:

- How to configure PerforMAX so you can monitor and define reports on your domains
- How PerforMAX is used to generate scheduled and designer reports from collected data

The following information assumes that PerforMAX has been installed and configured on the host machine and is accessible via an Internet browser from the client workstations.

For information about installing PerforMAX, refer to the *PerforMAX Installation Guide*. For information about managing and administering PerforMAX, refer to the *PerforMAX System Administrator Guide*.

Introducing PerforMAX

PerforMAX is a performance management and reporting system that lets you gain real-time insight into your applications and networks. PerforMAX enables you to perform two major types of activities: planning, troubleshooting, and long-term problem analysis.

PerforMAX lets you establish performance profiles for applications running on your network and for the performance of the network itself. Once you create these performance profiles, PerforMAX enhances your ability to perform capacity planning by allowing you to run historical reports to analyze your network's performance. PerforMAX lets you monitor alerts for policy violations and proactively detect and then remove network bottlenecks before they impact the end-users of your network services.

In particular, PerforMAX provides four major features:

- Data collection from network devices, systems and applications
- Real-time alerts when performance policy violations occur
- Real-time charts for detailed troubleshooting
- Scheduled reports for validation of historical network performance

By using these four major features, you can understand the experience of your end users. PerforMAX lets you...

- See and react to problems before they reach a critical stage
- Detect changes in the behavior and performance of network traffic
- Establish a foundation for future monitoring and control
- Detect the presence of 'new' applications and unplanned usage automatically, including their sources and destinations

Managing Your Network Performance from One GUI

PerforMAX has been constructed to reflect the workflow involved in managing your network performance from end-to-end, including interaction with other OSSs. This is reflected in the structure of the Graphical User Interface (GUI) that you use in daily operations and in the core of PerforMAX. The architecture of the system is optimized against the tasks to be done.

The GUI, which is also called the Dashboard, allows users to select the functions from navigation buttons and drop-down menus that parallel typical workflows needed for domain management. See "Using the Dashboard" for more information.

The core of PerforMAX software resides within a server or group of servers in a n-tiered architecture that takes advantage of Java and CORBA to distribute functions throughout the system in very flexible ways.

PerforMAX's collection of performance and other time-based information is gathered from the following items:

- Network elements
- Network Management Systems (NMS)
- Element Management Systems (EMS)
- Data marts
- Servers
- Desktop PCs

Conventions

This section explains and gives examples of the terms and writing conventions used in this documentation.

The term *PerforMAX* refers to the software installed on the host machine. The host machine, then, is the machine on which PerforMAX is installed and running. The term

Dashboard refers to the client applet that accesses and displays the information stored in the PerforMAX database.

Instructions to select items from menus, enter text, click a button, or press a key look like this:

- Select the **Yes** option.
- Type the word **swap**.
- Click **OK**.
- Press **Enter**.

Dashboard

This section explains the look and feel of the Dashboard. It explains how to start and stop the Dashboard. It shows the main menu items that are found and a general explanation on how to navigate through the various items on the work menu.

Note: The **Back** button on the Internet Browser window is not supported and may result in the user's session being reset. Use the **Back** button on the PerforMAX screens to navigate backwards through the applications screens.

Before Using the Dashboard

Before accessing the Dashboard for the first time from a client machine, you must:

1 Download, install, and configure the *Java 2 Runtime Environment (JRE) 1.5 (recommended) or 1.4.2* from:

http://www.java.sun.com

For more information, see the *PerforMAX Installation Guide*. Contact your local IT Help Desk for assistance.

2 Download and install *Adobe Acrobat Reader.* Adobe Acrobat Reader 6.1 and above is recommended (7.0 is not supported). Acrobat Reader is available free of charge from Adobe at:

http://www.adobe.com/products/acrobat/readstep.html

Acrobat Reader is required to read documentation included with the Dashboard.

Connectivity to the host machine is required to run the Dashboard.

System Requirements

You can use the Dashboard with the Internet Explorer browser running on MS Windows on PCs.

Minimum Software/Hardware Requirements for PerforMAX Client Machine

Table 3. Client Machine Requirements

CLIENT MACHINE REQUIREMENTS	
Operating System	Microsoft Windows 95/98/NT/2000, Solaris 9.0 or higher
Display Setting — Color Palette	Microsoft Windows True Color
Java Runtime Environment (JRE)	JRE 1.5 or 1.4.2
Browser	Internet Explorer 6.0 or 6.0SP1
System Memory	Minimum: 512MB, Recommended: 1GB
Processor	Minimum: Pentium 3 1GHz

Notes:

- For proper operation of the Dashboard, you must set your PC’s display properties “Color Palette” setting to “True Color.” You may need to decrease your “Refresh Frequency” (resolution) to achieve this setting. Contact your local IT Help Desk for assistance.
- To avoid re-using an older installed version of the Java Runtime Environment (JRE) when connecting to, the PC verify that the “Enable Caching” check box is cleared.
- Launcher can be opened using Netscape 6.2 provided that there is no other instance of a Netscape browser with JRE running on the client system.

Clear Enable Caching

1. Click **Start** in MS Windows.
2. Click **Control Panel**.
3. Double-click **Java Plug-in**.
4. Select **Cache** tab.
5. Clear the **Enable Caching (Enabled)** check box.
6. Click **Apply** button.

Starting the Dashboard

To start the dashboard

1. Use your Web browser to navigate to the *Dashboard Login* page.

The Dashboard login page address is:

http://*host*:8080/

Where ***host*** =

name of the host where PerforMAX, is installed including its domain name)

For example, tampa.trendium.com

or

Where ***host*** =

host's IP address

For example, 192.168.201.60

If you do not know the host name or the IP address for your system, ask your system administrator. Port 8080 must be reserved for PerforMAX communications.

If Secure Sockets Layer (SSL) is configured and enabled for the PerforMAX GUI, use HTTPS instead of HTTP and the port configured when invoking the client (default port - 8443).

Example: **https://*server_host*:8443/**

After you access the dashboard. You can log in to the system.

Logging in to the System

To login to the system

1. From the PerforMAX login screen you will see the login dialog box.

   Note: If you do not see the login dialog box, close your browser and open a new browser window.

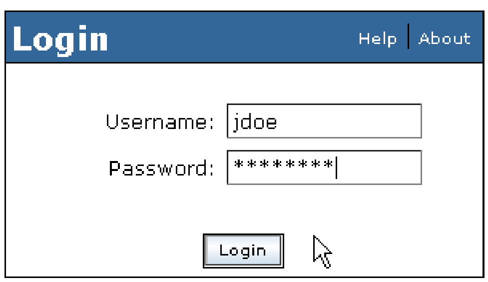


Figure 1-1: Dashboard Login

2. Enter your user name in the **Username** field.
3. Enter your password in the **Password** field.
4. Click **Login**.

   The Dashboard is displayed.

   On first-time login, the user may be required to change password from the pre-configured password. Upon initial login, the user may see the following screen.

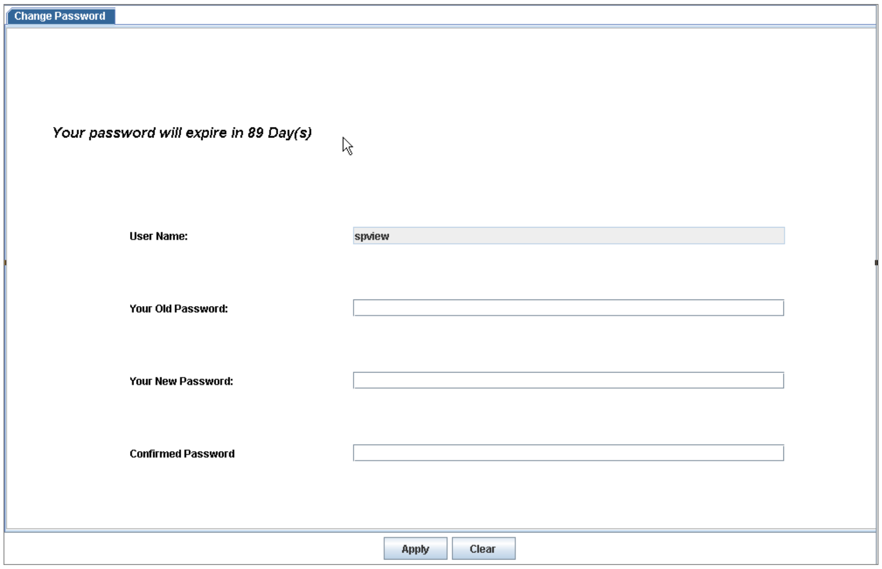


Figure 1-2: Change Password Screen

Change Password

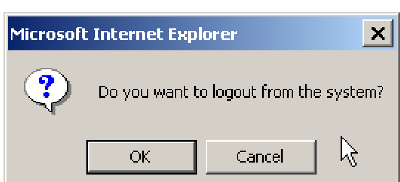
1. Enter your old password, new password, and confirmation of the new password.
2. Click **Apply**. Upon successful change, the user receives a password change confirmation.

Logging out of the System

To logout of the system

1. To logout of the system, close your Web browser or click **Log Out** on the active task bar.

   The Logout confirmation dialog is displayed:



2. Click **Yes** to logout or **Cancel** to remain logged in.

Using the Dashboard

This section provides a general overview of the features and user functions available from the main Dashboard as shown below. The main window Launch Pad contains the menu options that access applications functions.

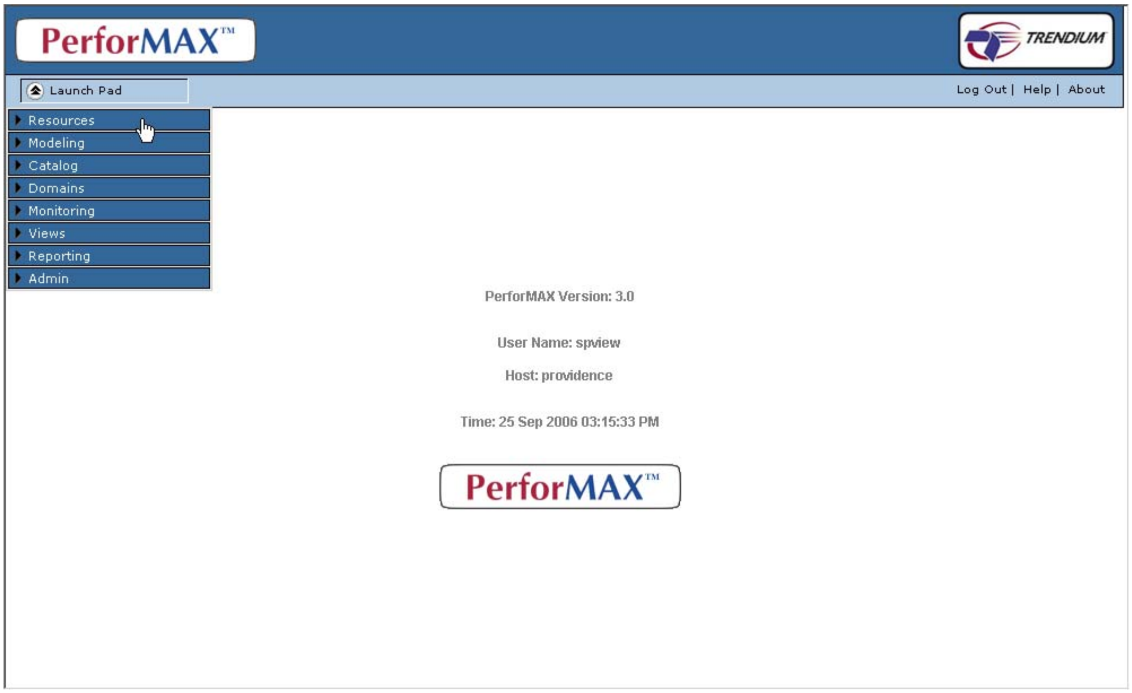


Figure 1-3: Dashboard

Main Menu

The *main menu options* categorize PerforMAX's management tools into logical groupings. They have been organized to allow users to work from top to bottom, as they move through the steps in a typical work flow. When selecting an item in the menu, its corresponding information is displayed in a separate display/access pane.

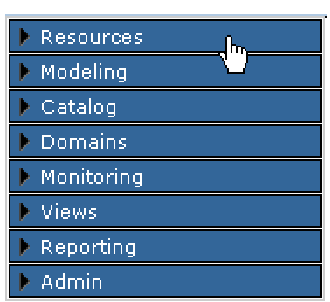


Figure 1-4: Main Menu

The main menu provides features for sales, engineering, marketing, system administration, and customer support.

Main Menu Option Descriptions
Resources, and **Domains** lists physical infrastructure and creates a context for connectivity between devices or processes.
Modeling and **Catalog** allow you to build policy templates and define offerings.
Monitoring allows you to configure various system administration features including system monitoring and system serviceability using system logs and maintenance scheduling. This feature allows you to manage licenses.
Views and **Reporting** provides real-time assessment and assurance of network performance. In addition, this component provides fault and incident management through its alerts and reporting features.
Admin allows users to perform basic management of their Dashboard accounts, such as the customization of their user preferences and passwords, as well as monitoring event logs.

Wizard Navigation

Many of the menu options display wizards to aid the user in quickly performing tasks. To help navigate using wizards, the following functions are available:

Display Filters

Display filters allow users to manage lists by scanning the database and displaying only those records matching the filter criteria.

Display/Access Pane

The *display/access* pane displays the information (for example, data records, reports, and so on) corresponding to the main menu item selected. This pane also provides access to entering, viewing, editing, and deleting database records.

Component-specific Buttons

Component-specific buttons are located at the bottom of the display/access screens and are only available when its corresponding menu item is selected from the main menu. Add, Delete, Edit, and View are examples of component-specific buttons. Typically, these buttons launch a wizard.

Enhancing Search Ability

When you are searching or filtering, you can use the percentage sign (**%**) as a wildcard. You can type it before and/or after a variable (such as a parameter or an IP address), or you can type the percentage sign (**%**) by itself to list all items available for a particular field.

While the actual name of the item you are trying to find can be entered in the primary filter field, it might be more convenient to enter a few of the characters of the name, and use the % sign after that to indicate that any character may occur after that point.

If the report list contains reports for the following domains:

ABC Corp

ABZ Communications

BBB Fiber

DEF Networks

Then the following shows which domains would be shown for a variety of entries in the Domain field.

ABC%	**AB%**	**%B%**
ABC Corp	ABC Corp	ABC Corp
	ABZ Communications	ABZ Communications
		BBB Fiber

Similar to the approach above using the percent (%) character, the “underscore” (_) character is a wildcard character typically used to match any *single* character.

The following example uses filtering reports by domains to explain this point.

If the report list contains reports for the following domains:

ABC Corp

ABZ Communications

BBB Fiber

DEF Networks

The top row in the below table shows the search string with both “%” and “_” characters, while the second and third row shows what is returned.

A_C%	**%_t%**	**DEF N_tworks**
A**B**C Corp	ABZ Communica**t**ions	DEF N**e**tworks
	DEF Ne**t**works	

Closing the Dashboard

1 To close the Dashboard, close your Web browser or click **Log Out** on the active task bar.

Displaying Version Number

In the event you need to contact Trendium's Technical Support Department, please note the version number that you are currently running.

To obtain the version number

1 Click the **About** button on the active task bar.

Cancel Long Operations

The "Cancel Long Operations" feature provides both a "Progress Indicator" and "Cancel Option" dialog.

Additionally, PerforMAX can be configured for Automatic Cancellation operations. If Automatic Cancel is enabled, retrieval operations that extend beyond the specified time are automatically canceled.

Progress Indicator

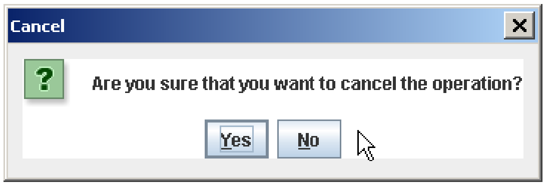
Database retrieval operations vary in time.

To inform the user that the operations are in progress, a Progress Indicator is displayed after two seconds of retrieval operations (for example, retrieving Monitoring information from the database). The following figure shows the Progress Indicator.

Cancel Option

A Progress Indicator is displayed after two seconds of retrieval operations.

Click **Cancel** and the following Cancellation Confirmation dialog is displayed:



1 Click **Yes** and the current operation is cancelled and the user is returned to the invoking screen. Click **No** and the current operation will continue to process.

Resources

2

The following documentation provides information and procedures for resources.

Some of the topics in this documentation include the following:

- Understanding Resources and Resource Managers
- Resource Procedures
- Resource Manager Procedures
- Understanding Resource Interfaces
- Resource Interface Procedures
- Understanding Connections
- Connections Procedures
- Discovering Device Interfaces

Understanding Resources and Resource Managers

A resource is any device, server, application, or database, deployed in a service offering. It applies to physical, logical, and virtual devices, software, and services with which PerforMAX communicates. Each resource collects and stores a specific set of resource metrics (such as the current bandwidth, unavailable seconds, and cell-delay variation) depending of the type of resource.

A resource manager is a software application or file that contains data from multiple resources. Frequently, resource managers are third-party applications such as element or network managers or specialized data collectors.

See "Resources" and "Resource Managers" for more information.

Resources

A resource is any device (for example, Cisco 7200), server, application (for example, Exchange or Remedy) that is service capable. That is, a resource is any source of raw data. Resource types, then, are categorizations of resources based on their characteristics. Resource types are listed in PerforMAX by manufacturer, product name, model, and version (for applications).

PerforMAX has a collection agent for each resource type that is responsible for collecting resource metrics from all the resources of that type. Resource metrics are collected for not only the resource type but also the process code (technology) supported by the device type. The following table gives an example of different resource metrics that can be collected from Cisco 72XX series for ATM and IP traffic.

Technology Code	Cisco Router 72XX series Resource Metric
ATM	AAL5 InOctets.
ATM	AAL5 OutOctets
IP	An estimate of the interface current bandwidth in bits per second.
IP	The length of the output packet queue (in packets).
IP	The number of inbound packets that contained errors preventing them from being deliverable to a higher-layer protocol.
IP	The number of inbound packets which were chosen to be discarded even though no errors had been detected.
IP	The number of non-unicast packets delivered to a higher-layer protocol.
IP	The number of outbound packets that could not be transmitted because of errors.
IP	The number of outbound packets which were chosen to be discarded even though no errors had been detected.
IP	The number of packets received via the interface which were discarded because of an unknown or unsupported protocol.
IP	The number of subnetwork-unicast packets delivered to a higher-layer protocol.
IP	The size of the largest datagram which can be sent/received on the interface specified in octets.
IP	The total number of octets received on the interface - including framing characters.
IP	The total number of octets transmitted out of the interface; including framing characters.
IP	The total number of packets that higher-level protocols requested be transmitted to a non-unicast address
IP	The total number of packets that higher-level protocols requested be transmitted to a subnetwork-unicast address.

These resource metrics are then processed by resource adapters and converted from their resource-specific format into a generic format known as quality metrics. See "Understanding Resource and Quality Metrics" for more information.

Because PerforMAX communicates with any resource type based on any technology/process, you can customize the collection of Quality of Service data to meet each individual assessment and assurance needs. To collect data from resource types, you must purchase and license a customized Technology Code and Resource Adapter for each type from Trendium. See "Understanding Technology Codes" for more information.

You can access a detailed resource type description containing product, name, manufacturer, category, model, version, protocol, supported collection intervals, and description information by viewing the resource type record. The resource type list provides all of the information that the Technology Codes list provides. However, it also provides the version, protocol, and supported collection intervals.

After licensing a resource type, you can add specific instances of the resource type for resource monitoring, assessment, and assurance purposes.

Resource types can only be viewed and are not subject to modification. However, you can add resource types by purchasing additional resource adapter licenses.

See "Resource Procedures" for more information.

Resource Procedures

A managed resource is any network, process or technology that supports a service, such as a content server, router, application, optical, or switch. Before PerforMAX can collect data, you must license the appropriate technology code and as mentioned in the section "Understanding Resources and Resource Managers". Then, the service provider can register the resource with PerforMAX using the Resource.

The registration process includes entering the resource's name, alias, type, address, status, location, and description. Once registered with PerforMAX, the resource is displayed in the Stand-Alone Resources list. In addition, the resource's associated interfaces are automatically added to their respective lists.

After creating a resource, you can modify, view, and delete resource records. In addition, you can regather, or refresh, the resources associated with a device to ensure PerforMAX contains accurate resource information. This section contains the following procedures:

- Displaying a Resource List
- Adding a Stand-Alone Resource Record
- Editing a Resource Record
- Viewing a Resource Record
- Deleting a Resource Record
- Importing a Resource Record
- Regathering a Stand-Alone Resource

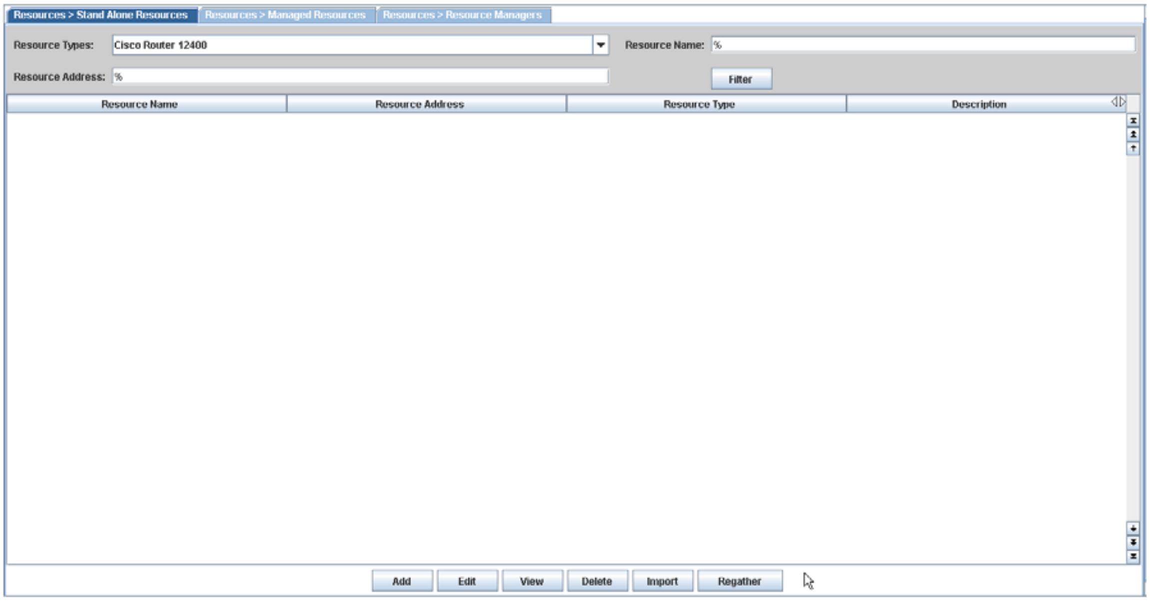
Displaying a Resource List

You can display a list of resources based on selected criteria by using the filtering feature.

To display a resource list

1 From the **Resources** menu, select **Resources**.

The following screen is displayed:



2 Select the tab corresponding to the category of resource for which a list is to be displayed: **Stand-Alone Resources** or **Managed Resources**.

If you...	**then...**
select **Stand-Alone Resources**,	go to "To display a stand-alone resources list"
select **Managed Resources**,	go to "To display a managed resources list"

To display a stand-alone resources list

a. On the **Stand-Alone Resources** tab, select the **Resource Types** from the drop-down list.
b. In the **Resource Name** field, type the resource name or, to list all resources meeting the filtering criteria, accept the default (**%**).
c. In the **Resource Address** field, type an address or, to list all resource managers meeting the filtering criteria, accept the default (**%**).
d. Click **Filter.**

The Stand-Alone Resources list is displayed.

To display a managed resources list

a. On the **Managed Resources** tab, select the **Resource Manager Type** from the drop-down list.
b. Select the **Resource Manager Name** from the drop-down list.
c. In the **Resource Name** field, type the resource name or, to list all resources meeting the filtering criteria, accept the default (**%**).
d. In the **Description** field, type the description of the resource manager or, to list all descriptions meeting the filtering criteria, accept the default (**%**).

e. Click **Filter**.

The Managed Resources list is displayed.

Note: You can use the percentage sign (**%**) as a wildcard. You can type it before and/or after a variable (such as a parameter or an IP address), or you can type the percentage sign (**%**) by itself to list all items available for a particular field.

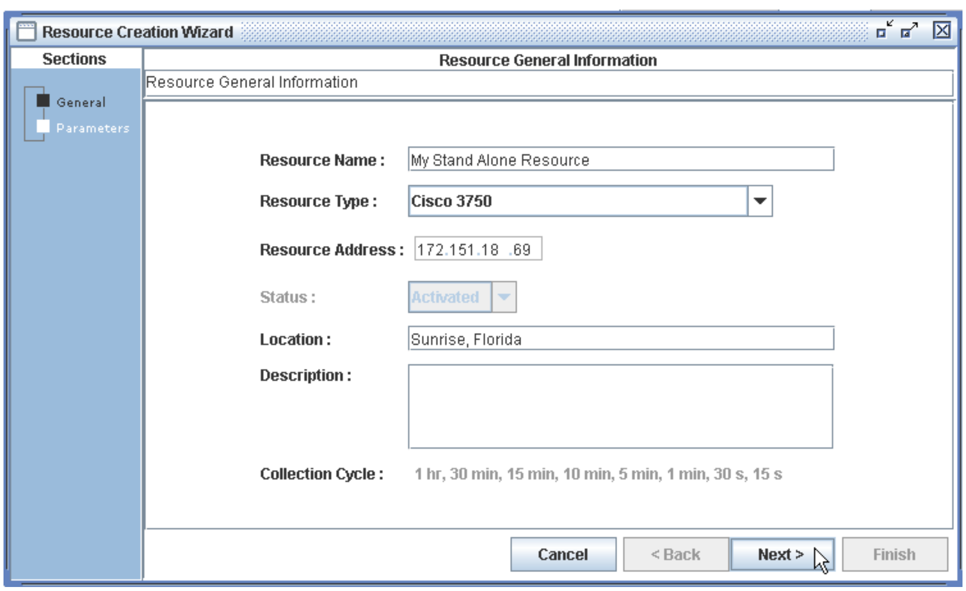
Adding a Stand-Alone Resource Record

You can add a stand-alone resource record to the database. This procedure does not apply to managed resources. Adding stand-alone resources provides PerforMAX with the information necessary to collect and evaluate Quality of Service data associated with a specific resource.

To add a stand-alone resource record

1 On the **Stand-Alone Resources** tab, click **Add**.

The following screen displays:



2 Type a resource name in the **Resource Name** field.

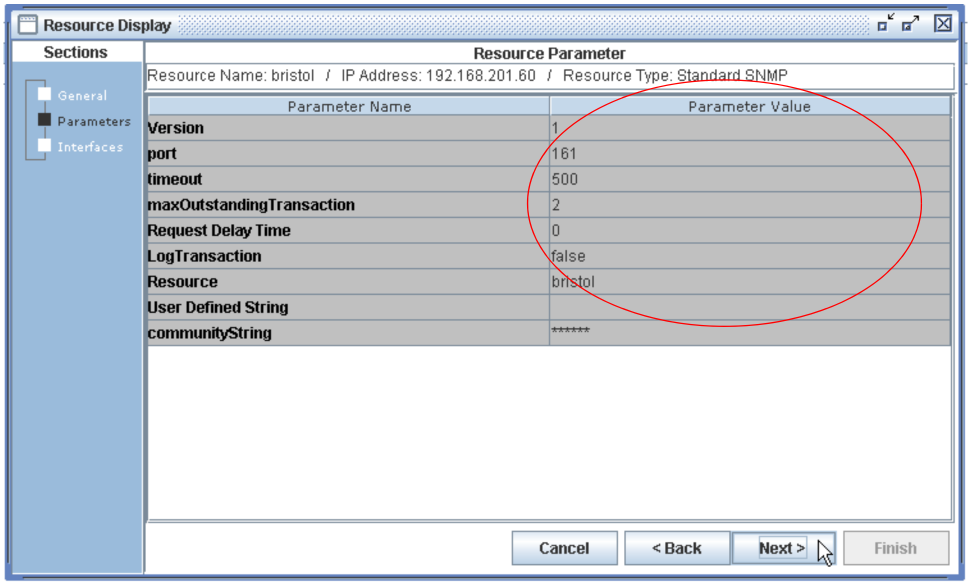
3 In the **Resource Address** field, type the resource's address.

4 In the **Location** field, type the resource's location.

5 In the **Description** field, type additional comments that are appropriate for the resource you are adding (optional).

6 Click **Next**.

A screen similar to the following is displayed:



Note: Resources will not accept any parameter values that are outside the "acceptable" range of values found in the adapter configuration file.

To modify a resource parameter

1. Double-click the **Parameter Value** to activate the option.
2. Enter the desired **Parameter Value**.
3. Click on the **Parameter Name** pane to confirm the modification.
4. Click **Finish**.

Note: The above procedure may also be performed to modify Resource Managers as well as Editing a Resource or Creating/Editing a Resource Manager.

Editing a Resource Record

You can edit existing resource records after generating a resource list. The edit feature allows you to make modifications to certain resource fields.

To edit a stand-alone or managed resource record

1. Select the tab corresponding to the category of resource to be edited: **Stand-Alone Resources** or **Managed Resources**.
2. Select, or highlight, the row corresponding to the resource to be edited.
3. Click **Edit**.
4. Make the appropriate edits.

   Note: You can only edit the **Resource Name, Location**, and **Description** fields.

5 Click **Next** to review the resource parameters.

6 Click **Finish** to save your edits.

The edited Resource record is saved and displayed on the appropriate tab.

Viewing a Resource Record

You can view the details of a resource record after generating a resource list. The view feature provides read-only access to the data fields used to create a resource record. This procedure applies to both stand-alone and managed resources.

To view a resource record

1 On the **Stand-Alone Resources** or **Managed Resources** tab, select the row corresponding to the resource to be viewed.

2 Click **View**.

The Resource record displays in read-only format.

3 Click **Next** and **Back** to navigate through the screens.

4 Click **Cancel** to exit the screen.

The Resource record list redisplays.

Deleting a Resource Record

You can delete a resource record from the database. This procedure applies to both stand-alone and managed resources.

In general, you cannot delete any resources if there are any kind of dependent resources or connections.

To delete a resource record

1 Select, or highlight, the row corresponding to the resource to be deleted.

2 Click **Delete**.

A confirmation message asks "Are you sure you want to delete the selected item?"

3 Click **Yes** to confirm the deletion.

Note: You can only delete a resource record that is not associated with any other dependencies.

Importing a Resource Record

You can import comma delimited (**.csv**) files into PerforMAX. Before you import a **.csv** file you should make sure that the file's fields match the fields in PerforMAX.

To import resource records

1. Click **Import**.
2. Navigate to the location where the import file is located.
3. Select the import file and click **Open**.
4. Click **Submit**.

Note: Import files are in CSV (Comma Separated Value) format. '#' in the first column introduces a comment line which is ignored by the GUI. The file is divided into multiple blocks, each block starts with a header which defines a parameter name list. A special character "!" at the beginning of a line indicates a header.

Following the header, each row is a CSV list corresponding to the parameters defined in the header. A value can be empty which is indicated by two continuous commas or two commas with only spaces in between. Following is a sample import file layout.

```
# Import file layout
# First header
! parameter1, parameter2, ..., parameterN
value11, value12,..., value1n
value21, value22,..., value2n
...
# Second header
! parameter1, parameter2, ..., parameterM
value11, value12,..., value1m
value21, value22,..., value2m
```

Regathering a Stand-Alone Resource

You can regather or update a stand-alone resource. Use this feature when you have deleted resources in your network so the resources listed in PerforMAX match your actual network configuration.

To regather a stand-alone resource

1. On the **Stand-Alone Resources** tab select the row corresponding to the stand-alone resource to be regathered.
2. Click **Regather**.

New resources appear on the Stand-Alone Resources tab.

Resource Managers

The term *resource manager* refers to any application that collects data from a service resource, such as a network management system (NMS), an element management system (EMS), or a data mart.

A resource manager is a software application that manages multiple resources virtually. Frequently, resource managers are third-party applications, such as the Cisco WAN Manager and PolicyPoint.

More specifically, a resource manager is any application that collects raw data from service resources, such as a network management system (NMS), element management system (EMS), or data mart. PerforMAX interfaces with several resource managers, and thereby virtualizes the resources (for example, devices, routers, and so on) so that data can be collected for assessment and assurance purposes. In addition, PerforMAX enables resource monitoring of managed resources.

You cannot directly access the resources that the resource manager controls: you must access them through the resource manager. See Figure 2-1.

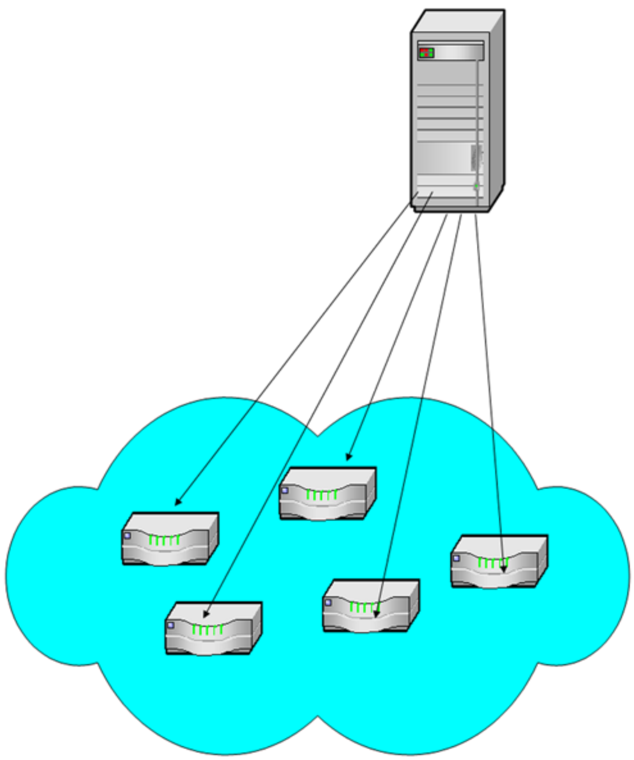


Figure 2-1: Resource Manager

Note: Before PerforMAX can collect data from a managed resource, you must first license the appropriate technology code and resource adapter. Then, you must register an instance of a resource manager so that the corresponding managed resources are discovered and added to the system.

For more information about resource types, see “Understanding Resources and Resource Managers”. For more information on technology codes, see “Understanding Technology Codes”.

Once registered with PerforMAX, the resource manager appears in the Resource Managers list. In addition, the resource manager's associated resources such as devices and interfaces automatically populate their respective lists.

PerforMAX lets you manage any registered resource manager provided it can communicate with PerforMAX. However, you must have the appropriate drivers for all resources and the resource manager installed in PerforMAX.

See "Resource Manager Procedures" for more information.

Resource Manager Procedures

You must configure a resource manager before you can add it to PerforMAX. A Resource Manager consolidates the management of potentially hundreds of devices. Every device in PerforMAX has its own resource record and IP address.

Information about resource managers is stored as resource manager records in the applicationdatabase.

The first time you add a resource manager to PerforMAX, it provides the applicationwith a list of all of the resources it manages. However, the resource manager does not dynamically update this list as its number of resources increases or decreases. To update this list, you must re-discover the resources or *regather* the resources.

When you regather resources, PerforMAX goes into the specified resource manager and checks if devices have been added or deleted. Then, the application updates with a list of all devices, regardless of whether the devices are active or inactive.

The following topics describe the procedures for managing resources:

- Displaying a Resource Manager List
- Adding a Resource Manager Record
- Editing a Resource Manager Record
- Viewing a Resource Manager Record
- Deleting a Resource Manager Record
- Regathering Resources Associated with a Resource Manager

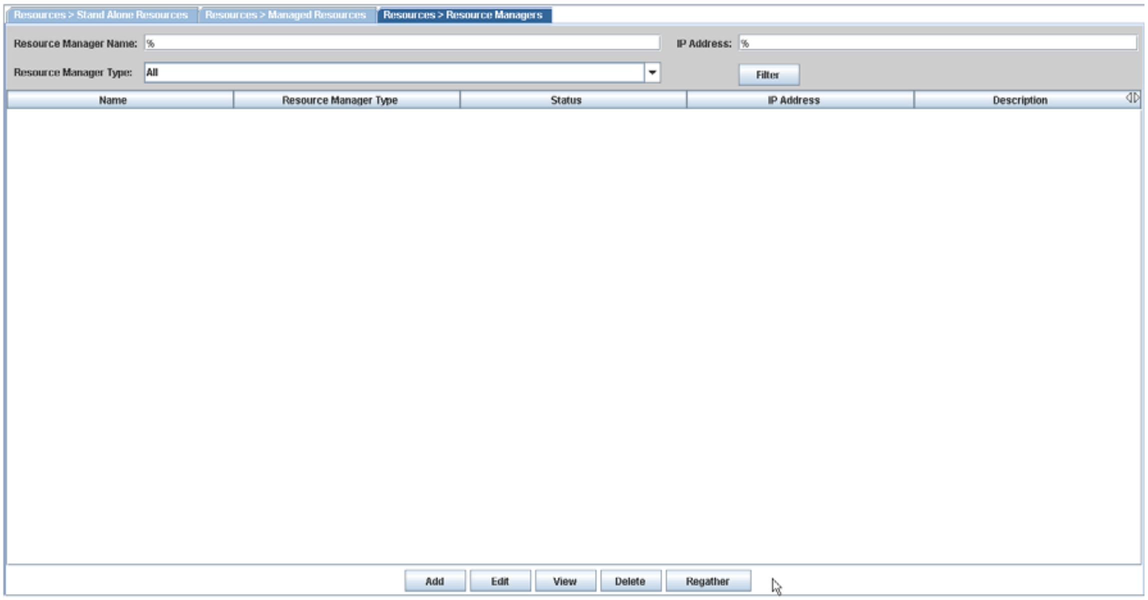
Displaying a Resource Manager List

You can display resource manager list based on selected criteria by using thefiltering feature.

To display a resource manager list

1. From the **Resources** menu button, select **Resources.**
2. From the default screen, select the **Resource Managers** (tab).

The following screen is displayed:



3 In the **Resource Manager Name** field, type the resource manager name or, to list all resource managers meeting the filtering criteria, accept the default (**%**).

4 In the **IP Address** field, type an IP address or, to list all resource managers meeting the filtering criteria, accept the default (**%**).

5 Select the **Resource Manager Type** from the drop-down list.

6 Click **Filter**.

The Resource Manager list is displayed.

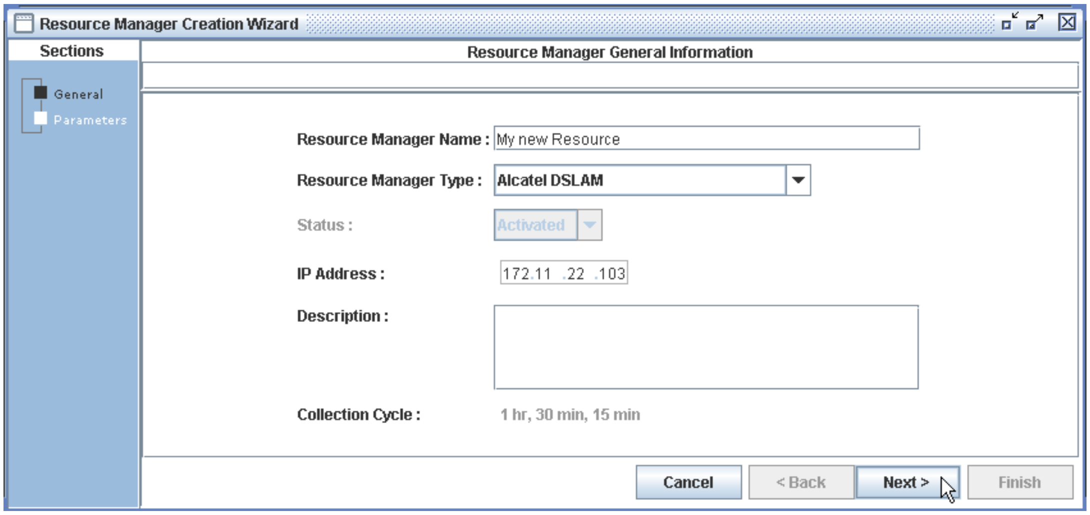
Adding a Resource Manager Record

You can add a resource manager record to the database. This provides PerforMAX with the information necessary to collect data from the resource manager's associated resources (for example, devices, switches, and so on).

To add a resource manager record

1 From the resource manager list, click **Add**.

The following screen is displayed:



2. Type a resource manager name in the **Resource Manager Name** field.
3. Select the **Resource Manager Type** from the drop-down list.
4. In the **IP Address** field, type the resource manager's IP address.
5. In the **Description** field, type additional comments that are meaningful for your implementation.
6. Click **Next** to review the resource parameters.
7. Click **Finish**.

   The new Resource Manager appears on the Resource Manager list.

Editing a Resource Manager Record

You can edit an existing resource manager record after generating a resource manager list. The edit feature allows you to make modifications to certain resource manager fields.

To edit a resource manager record

1. From the resource manager list, select, or highlight, the row corresponding to the resource manager record to be edited.
2. Click **Edit**.
3. In the **General Information** screen, you can make changes to the **Resource Manager Name** and **Description** fields only.
4. Click **Next** to review the resource parameters.
5. Click **Finish** to save your changes.

   The changes are saved in the Resource Manager record.

Viewing a Resource Manager Record

You can view the details of a resource manager record in read-only format after generating a resource manager list.

To view a resource manager record

1. From the resource manager list, select, or highlight, the row corresponding to the resource manager to be displayed.
2. Click **View** to review General Information.

   The Resource manager record displays in read-only format.
3. Click **Next** and **Back** to navigate through the screens.
4. Click **Cancel** to exit the screen.

   The Resource Manager list is redisplayed.

Deleting a Resource Manager Record

You can delete a resource manager record from the database.

Note: You can only delete a resource manager record if it does not have other elements depending on it in PerforMAX**.**

To delete a resource manager record

1. From the resource manager list, select, or highlight, the row corresponding to the resource manager to be deleted.
2. Click **Delete**.

   A confirmation message asks "Are you sure you want to delete the selected item?"
3. Click **Yes** to confirm the deletion.

   The Resource Manager record is removed from the Resource Manager list.

Regathering Resources Associated with a Resource Manager

You can use the regather feature to update any lists of resources associated with a resource manager. Use this feature when you have deleted resources or resource interfaces that have been auto discovered so the resources listed in PerforMAX match your actual network configuration.

To regather resources associated with a resource manager

1. From the resource manager list, select, or highlight, the row corresponding to the resource manager to be regathered.
2. Click **Regather**.

   New resources appear within the Resource Manager record.

Discovering Device Interfaces

Device interfaces are also discovered automatically. Typically, the Resource Manager feature identifies interfaces by the bay, shelf, slot, card, port, and channel.

The Resource Manager consolidates the collection of data from resources. It can manage hundreds of interfaces and services, whether they are physical services or logical ones.

It is important to note that registering a resource manager does not imply PerforMAX is collecting data for the associated resources. Rather, registering a resource manager provides PerforMAX with the information to virtualize the managed resources. You can then use these managed resources (and discovered interfaces) to create resource/interface collection groups for resource monitoring purposes.

You can also modify, view, delete, and regather managed resources associated with a resource manager.

PerforMAX contacts resource managers to get information about resources. This information comes in various formats, including comma-delimited format. PerforMAX uses resource manager information to obtain information about resource and statistics.

PerforMAX can interface with several resource managers, including Cisco WAN Manager and PolicyPoint. In doing so, PerforMAX graphically represents the managed resources (for example, devices, routers, and so on).

Before PerforMAX can collect data from a managed resource, you need to first license the appropriate technology code and resource adapter as mentioned in the section "Understanding Resources and Resource Managers". Then, you must register an instance of a resource manager so that PerforMAX can discover the corresponding managed resources and add them to the system.

You add a resource manager by accessing the Resource Manager (by clicking **Add** on the Resource Managers pane) and entering the resource manager's name, type, status, address, and description.

Registering a resource manager provides PerforMAX with the information to virtualize the managed resources. A service provider can then use these managed resources (and discovered interfaces) to create resource/interface collection groups for resource monitoring purposes.

You can also view, modify, delete, and regather managed resources associated with a resource manager.

Understanding Resource Interfaces

A resource interface cannot be added directly to or deleted from PerforMAX. Rather, a resource interface is discovered when its corresponding is registered with PerforMAX. Once discovered, the resource interface is displayed in the Resource Interfaces list.

You can modify a resource interface's description using the Interface Modification pane. In addition, the Interface Modification pane displays information that cannot be edited including the interface reference name, interface name, interface type, resource name, point of attachment, and status.

It is important to note that discovering a resource interface does not imply that PerforMAX is collecting data for that resource interface. Rather, discovering a resource interface provides PerforMAX with the information necessary to virtualize the resource interface. You can then use these resource interfaces to create interface collection groups for resource monitoring purposes.

See "Resource Interface Procedures" for more information.

Resource Interface Procedures

The resource interfaces can be used to build connections which can then be assessed. You can perform the following resource interface related tasks:

- Displaying a Resource Interface List
- Editing a Resource Interface Record
- Viewing a Resource Interface Record
- Deleting a Resource Interface Record

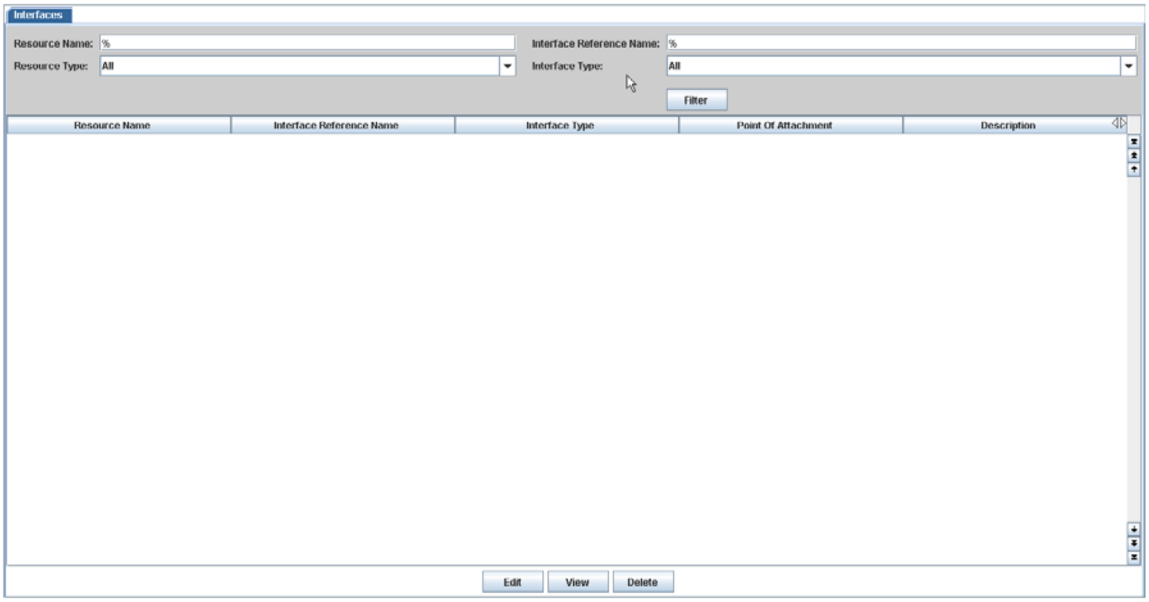
Displaying a Resource Interface List

You can display a list of resource interfaces based on resource name, resource interface type, and resource interface reference name by using the filtering feature.

To display a resource interface list

1 From the **Resource** menu, select **Interfaces**.

The Interfaces screen is displayed:



2. In the **Resource Name** field, type the resource name or (to list all) accept the default (**%**).
3. In the **Resource Type** field, select the resource type from the drop-down list or (to list all) accept the default (**All**).
4. In the **Interface Reference Name** field, type the interface reference name or (to list all) accept the default (**%**).
5. In the **Interface Type** field, type the interface type or (to list all) accept the default (**%**).
6. Click **Filter**.

   The Resource Interface list is displayed.

Note: You can use the percentage sign (%) as a wildcard. You can type it before and/or after a variable (such as a parameter or an IP address), or you can type the percentage sign (%) by itself to list all items available for a particular field.

Editing a Resource Interface Record

You can edit resource interface record descriptions after generating a resource interface list.

To edit a resource interface record

1. On the resource interface list, select, or highlight, the row corresponding to the resource interface to be edited.
2. Click **Edit**.

   Note: You can only edit the Interface Description.
3. Make the appropriate edits.

4 Click **Finish** to save your edits.

The changes to the Resource Interface record are saved.

Viewing a Resource Interface Record

After generating a resource interface list, you can view the details for a resource interface record in read-only format.

To view a resource interface record

1. On the resource interface list, select, or highlight, the row corresponding to the resource interface to be viewed.
2. Click **View**.
3. Click **Next** to view Interface Endpoint information.
4. Click **Cancel** to exit the wizard.

You are returned to the Resource Interface list.

Deleting a Resource Interface Record

You can remove, or delete, a resource interface record from the database as long as there are no dependencies elsewhere in the system.

To delete a resource interface record

1. Select, or highlight, the row corresponding to the resource interface to be deleted.
2. Click **Delete.** A confirmation message is displayed:

   "Are you sure you want to delete the selected item? All End Points associated with this interface will be deleted."

3. Click **Yes** to confirm the deletion.

Understanding Connections

Connections are the binding of performance class and connection type within a network. Once connections are defined, you can offer performances whose properties are governed by the connection contained in the connection template upon which they are based.

Connection Types defines how SAPs are connected in a connection. The types of connections are:

- Cloud
- Tunnel
- Funnel and

- Single-point

Connections Procedures

The Connections wizard allows the user to **Display**, **Add, Edit, View**, and/or **Delete** a connection.

- Displaying a Connections List
- Adding a Connection
- Editing a Connection
- Viewing a Connection
- Deleting a Connection

Displaying a Connections List

You can display a list of connections based on connection name and/or performance class by using the filtering feature.

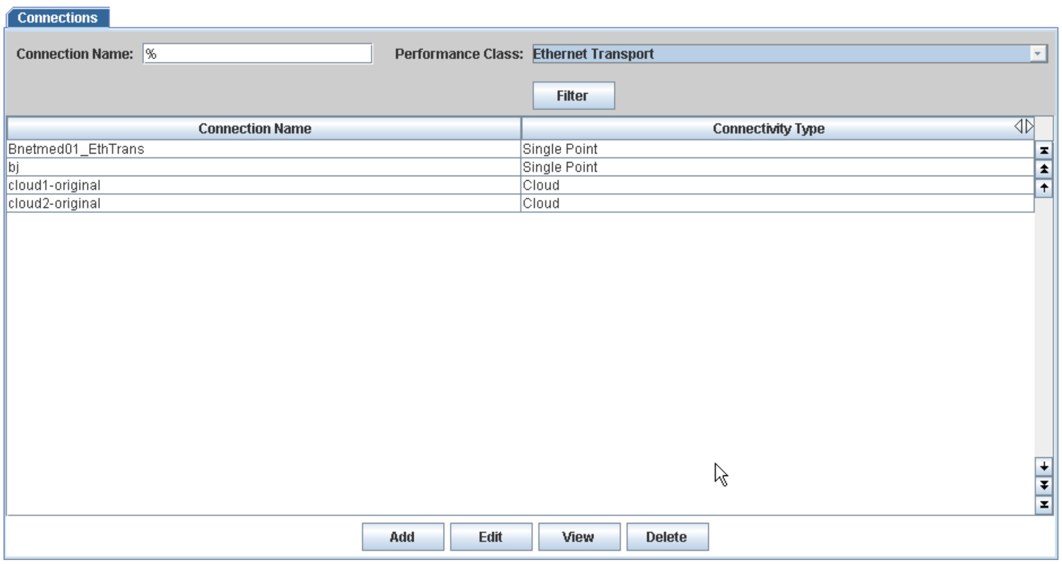
To display a connections list

1. From the **Resource** menu, select **Connections**.

   The Connections screen is displayed.

2. In the **Connection Name** field, type the name or (to list all) accept the default (**%**).
3. From the **Performance Class** drop-down list select a performance class.
4. Click **Filter**.

   A screen similar to the following is displayed:



Adding a Connection

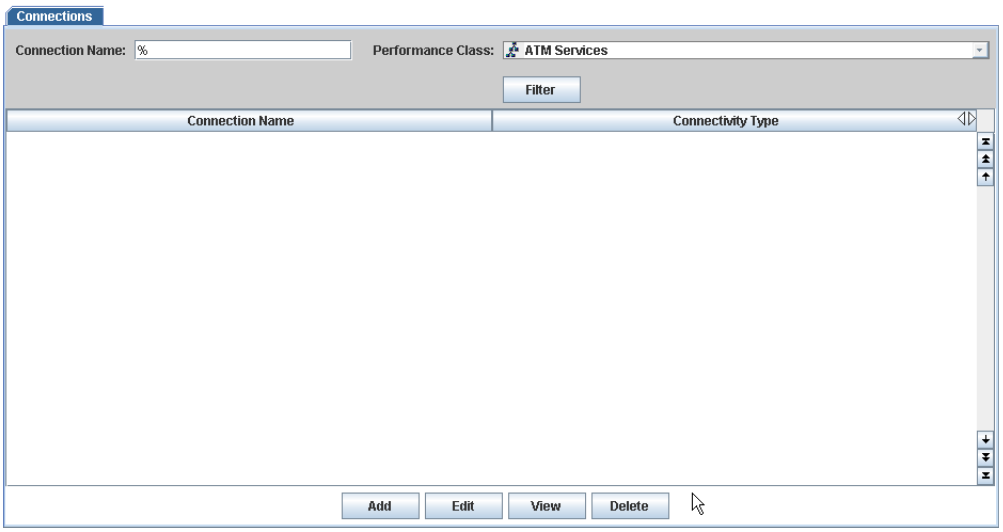
You can add a connection by using the add feature. Users can add a connection based on:

- Connectivity Type
- Performance Class
- End Points

To add a connection

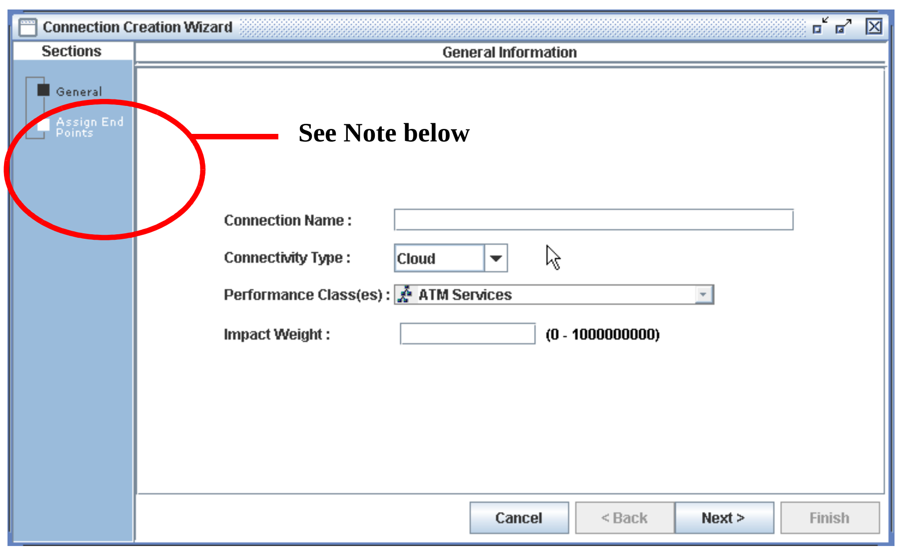
1 From the dashboard, select **Resources** > **Connections**.

A screen similar to the following displays:



2 Select **Add**.

The following screen is displayed:



3 In the **Connection Name** field, type a name for the connection being created.

4 From the **Connectivity Type** drop-down list, select one of the following connectivity types:

- Cloud
- Tunnel
- Funnel
- Single Point

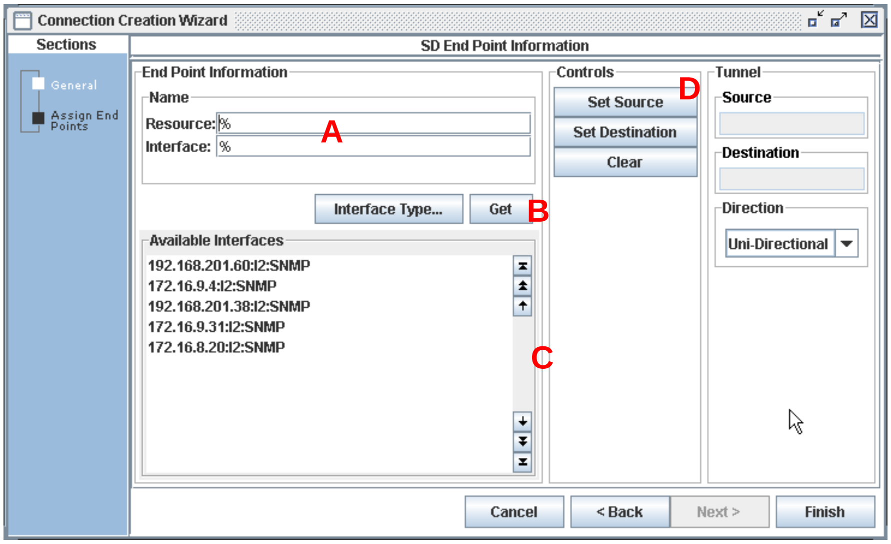
See "Connectivity Types" for detailed information.

5 From the **Performance Class(es)** drop-down list, select a performance class.

6 In the **Impact Weight** field, type an impact weight for the domain.

7 Click **Next**.

A screen similar to the following is displayed:



8 Type a **Resource Name, Interface Name** or **End Point** to list the available resources, interfaces or end points (names are available under the Resources navigation button) in the *End Points Information* section. (A above.), click **Get**. Alternately, using the % wildcard, click **Get** to list All available resources, interfaces, or end points.

9 Select the desired **End Point** in the Available End Points window (C above) to specify an end point. Continue selecting end points to complete the cloud. If selecting different **Connectity Types** on the *General Information screen,* the Control area (D above) of the wizard displays the following Connectivity Types.

Connectivity Types

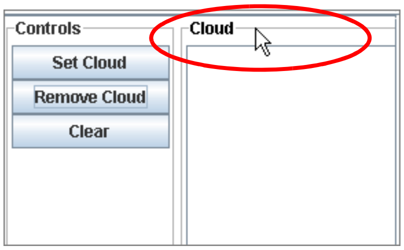


Figure 2-2: Connectivity Type - Cloud

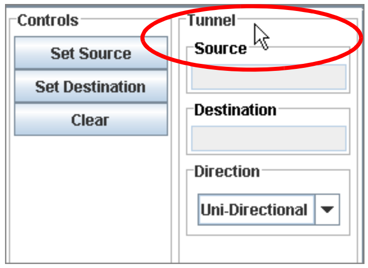


Figure 2-3: Connectivity Type - Tunnel

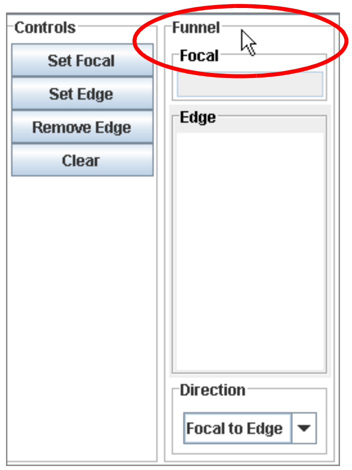


Figure 2-4: Connectivity Type - Funnel

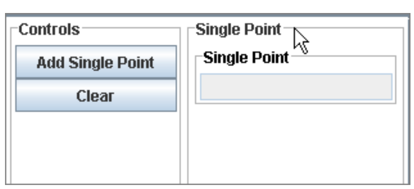


Figure 2-5: Connectivity Type - Single Point

- **Cloud** connections - sets a **Group** of end points.
- **Tunnel** connections - sets the **Source** and **Destination** end points.

Note: If you select **Tunnel** and **Uni** or **Bi-Directional** for the **Direction**, then the calculation is the same.

If you select **Bi-Directional**, then the data value can be sent in both (two) directions.

- **Funnel** connections - sets the **Focal** and **Edge** end points.
- **Single Point** - sets the single point using the **Add Single Point** option.

10 Select the **Direction** (**Uni-Directional** or **Bi-Directional**) using the drop-down option menu.

11 Click **Finish**.

Editing a Connection

To edit a connection

Procedures for editing a connection are similar to those for “To add a connection”. However, not all parameters can be edited. See this topic for detailed information.

Viewing a Connection

You can view a connection in read-only format after generating a list.

To view a connection

1 From the generated list, select, or highlight, the connection that you want to view.

2 Click **View**.

3 Click **Next** and **Back** to navigate through the screens.

4 Click **Cancel** to exit the screen.

The Connections list redisplays.

Deleting a Connection

You can delete a connection after generating a list.

To delete a connection

1 From the generated list, select, or highlight, the connection that you want to delete.

2 Click **Delete**.

3 Click **OK** at the *Confirmation Dialog box* to complete the delete operation.

Understanding Technology Codes

Technology codes represent the bearer service (for example, SONET, ATM, DWDM) or application (for example, data center). Each technology code, and its associated bearer service, contains a unique set of Parameter(s) that is supported by PerforMAX.

PerforMAX displays only the Technology Codes that are available for a particular Resource Type.

In order to collect data for a specific technology/process, you must purchase the appropriate license (which adds the corresponding technology code to PerforMAX).

Each technology/process contains its own set of supported quality-oriented data parameters. In addition, for each technology/process, there is a unique PerforMAX process code. When you license a technology/process, the corresponding technology code is added to the Technology Code list.

The following are examples of some of the technologies/services that aresupported:

- Cable
- Optical ()
- Wireless
- /
- Application
- Server
- Ethernet
- DSn/En

Using the information derived by the technology codes, the Service Technology Manager measures the traffic performance from the resources within the technology/ service in real-time. This information appears in the Dashboard.

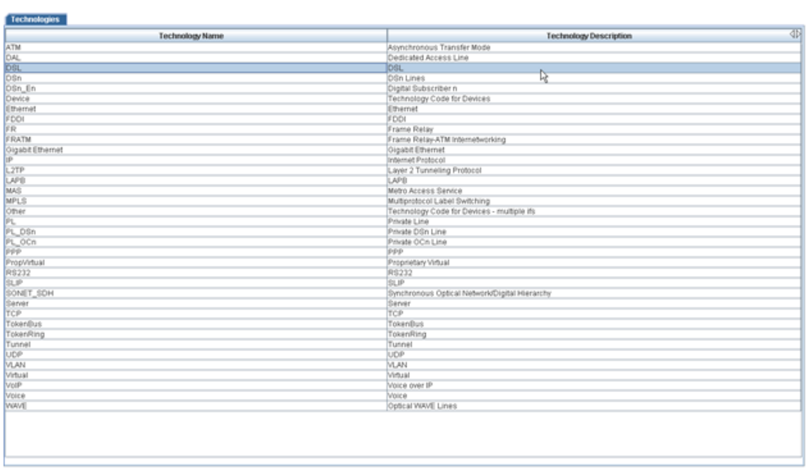
Displaying the Technology Code List

You can display the list of technology codes supported by PerforMAX.

To display the technology code list

1 From the **Resources** menu, select **Technologies**.

The Technology Code list is displayed:


Technologies

Technology Name	Technology Description
ATM	Asynchronous Transfer Mode
DAL	Dedicated Access Line
DSL	DSL
DSn	DSn Lines
DSn_En	Digital Subscriber n
Device	Technology Code for Devices
Ethernet	Ethernet
FDDI	FDDI
FR	Frame Relay
FRATM	Frame Relay-ATM Internetworking
Gigabit Ethernet	Gigabit Ethernet
IP	Internet Protocol
L2TP	Layer 2 Tunneling Protocol
LAPB	LAPB
MAS	Metro Access Service
MPLS	Multiprotocol Label Switching
Other	Technology Code for Devices - multiple ifs
PL	Private Line
PL_DSn	Private DSn Line
PL_OCn	Private OCn Line
PPP	PPP
PropVirtual	Proprietary Virtual
RS232	RS232
SLIP	SLIP
SONET_SDH	Synchronous Optical Network/Digital Hierarchy
Server	Server
TCP	TCP
TokenBus	TokenBus
TokenRing	TokenRing
Tunnel	Tunnel
UDP	UDP
VLAN	VLAN
Virtual	Virtual
VoIP	Voice over IP
Voice	Voice
WAVE	Optical WAVE Lines

Modeling 3

The following documentation provides information and procedures for modeling.

The topics in this documentation include the following:

- What is Modeling?
- Understanding Resource and Quality Metrics
- Building a Model
- Quality Metric Procedures
- Resource Metric Procedures
- Managing Key Performance Indicators (KPIs)
- KPI Procedures
- Managing Performance Classes

What is Modeling?

The term modeling refers to the process of correlating the infrastructure, network elements, and your network performance.

For PerforMAX to make comparisons between sets of resource data, it takes the data that your devices, servers, and other resources normally export, maps the naming conventions for the data and then correlates it to other related data.

The following key components make up the application’s modeling capabilities:

- **Resource Metrics** — Resource Metrics are the raw data produced by the device that indicate the data transmission rates and so on.
- **Quality Metrics** — Quality metrics are an abstraction related to a certain technology and bound to a group of resource metrics.
- **Key Performance Indicators** — KPIs are user defined expressions that generate a value, which can be used to measure the network performance.

Understanding Resource and Quality Metrics

The term Quality Metrics refers to the data generated by various parts or end-points of a device.

More specifically, resource metrics represent the Quality Metrics data supported by a network resource (for example, switch, device, content server, and so on).

Creating Standardized Metric Naming Conventions

Because different manufacturers use their own naming conventions for their metrics, PerforMAX needs to convert these metrics in a resource into a standard, common format. For example, Cisco might describe the metric that measures octets as an inOctets, whereas Cienna might call the same metric a MagicInOctets.

Before PerforMAX can compare the metrics from these two devices, it must give both of the metrics a common name. These common names are referred to as Quality Metrics.

In other words, for PerforMAX to compare two resource metrics it must have established a direct relationship or *mapping* between the underlying technology and quality metrics in PerforMAX.

Access the Quality Metrics List to display the list of supported quality metrics for each technology type.

When you display a list of quality metrics, you can filter the list so that only the metrics you want to see appear.

When you filter quality metrics, you can select a technology code and parameter name, scanning all quality metric records in PerforMAX's database, and displaying only those records meeting the specified criteria.

You can add, modify, and view quality metrics using the Quality Metric wizard. Finally, you can delete quality metrics so that the application no longer collects data for the mapped resource metrics.

When displaying a quality metric, you see the parameter name, the quality metric, the technology code, the unit of measurement, and a description of the metric.

Building a Model

By performing the tasks in this section in a defined sequence, you can create a model. While each task can be performed independently, building a model relies on the sequence because elements created in the first task provide required data for elements created later.

Quality Metric Procedures

For Quality Metrics, you can perform the following procedures:

- Displaying a Quality Metric List
- Adding Quality Metrics
- Editing Quality Metrics
- Viewing a Quality Metrics Record
- Deleting Quality Metrics

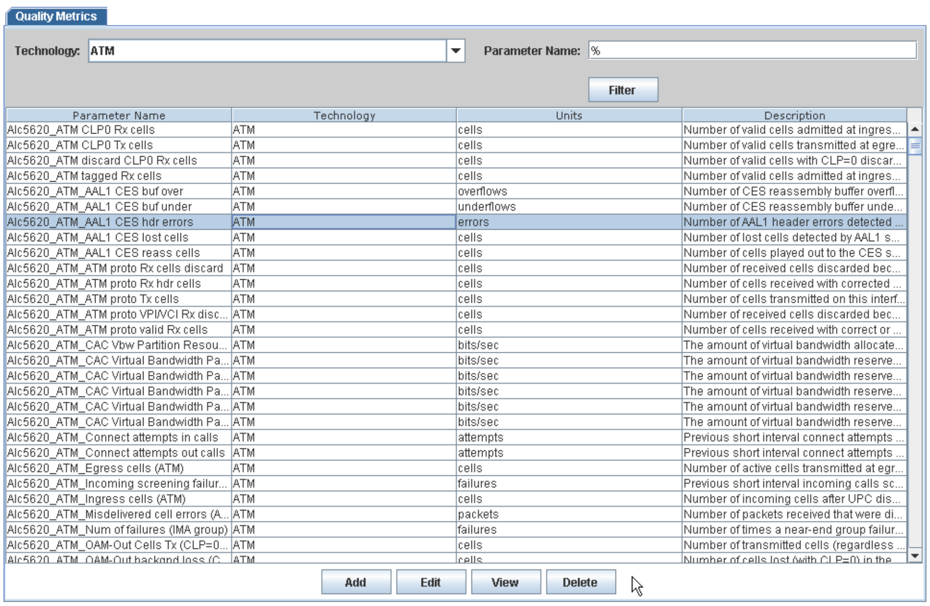
Displaying a Quality Metric List

You can display a list of Quality Metrics meeting a specific criterion, such as a Technology by performing the following steps.

To display a quality metric list

1. From the **Modeling** menu, select **Quality Metrics**.
2. Select the **Technology** from the drop-down list.
3. In the **Parameter Name** field, enter the parameter name or accept the default (**%**), to list all parameters for the selected technology code.
4. Click **Filter**.

   A screen similar to the following is displayed:



Adding Quality Metrics

You can add Quality Metrics to PerforMAX.

Note: The association of Quality Metrics to Resource Metrics is done in the Resource Metrics screen.

To add a quality metric

1. On the quality metric list, click **Add**.

   The Quality Metrics Creation Wizard is displayed.

2. Enter a quality metric name in the **Name** field.
3. In the **Description** field, type a description of the quality metric.
4. Select a **Technology** from the drop-down list.
5. Select a **Unit** from the drop-down list.

6 Click **Finish**.

The new quality metric is added to the Quality Metric list.

Editing Quality Metrics

You can change all data items of a Quality Metric after it has been created.

To edit a quality metric

1. Display a list of quality metrics and select the Quality Metric to be edited.
2. Click **Edit**.

   The Quality Metrics Modification screen is displayed.
3. Make changes in any of the available fields.
4. Click **Finish**.

   You are returned to the Quality Metrics list.

Viewing a Quality Metrics Record

You can view the details of a Quality Metrics record.

To view a quality metrics record

1. From the quality metrics list, select, or highlight, the row corresponding to the quality metrics record to be viewed.
2. Click **View**.

   The Quality Metrics parameters are displayed in view format only.
3. Click **Cancel** to close the quality metrics record display.

   You are returned to the Quality Metrics list.

Deleting Quality Metrics

You can delete a Quality Metric after it has been created, provided that it is not referenced in elsewhere in PerforMAX.

To delete a quality metric

1. Display a list of quality metrics and select the quality metric to be deleted.
2. Click **Delete**.
3. Click **Yes** to confirm the deletion.

Note: If the quality metric you want to delete is referenced elsewhere, an Exception Notification message will display indicating that you cannot delete this quality metric.

Resource Metric Procedures

For Resource Metrics, you can perform the following procedures:

- Displaying a Resource Metrics List
- Adding Resource Metrics
- Editing Resource Metrics
- Viewing a Quality Metrics Record
- Deleting Resource Metrics

Displaying a Resource Metrics List

You can display a list of Resource Metrics based on resource type, Technology and Name using the filtering feature.

To display a resource metrics list

1 From the **Modeling** menu, select **Resource Metrics**.

The Resource Metrics default screen is displayed.

2 Select the **Resource Type** from the drop-down list.

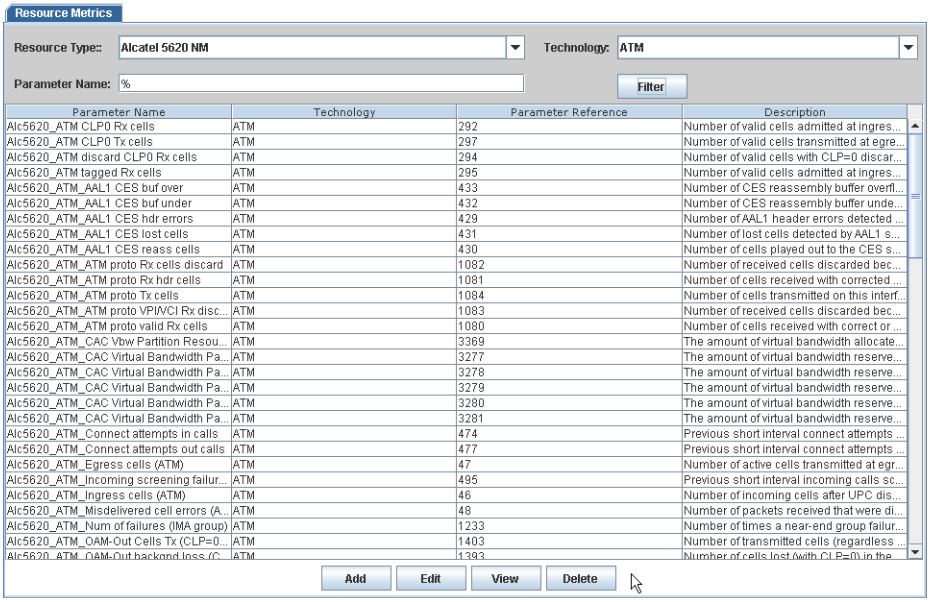
3 Select the **Technology** from the drop-down list.

4 In the **Parameter Name** box, enter the parameter name or, to list all parameters for the selected resource type/technology code, accept the default (**%**).

When you are typing a parameter name, type the name or use the percentage sign (**%**) when typing a partial name.

5 Click **Filter**.

A screen similar to the following displays:



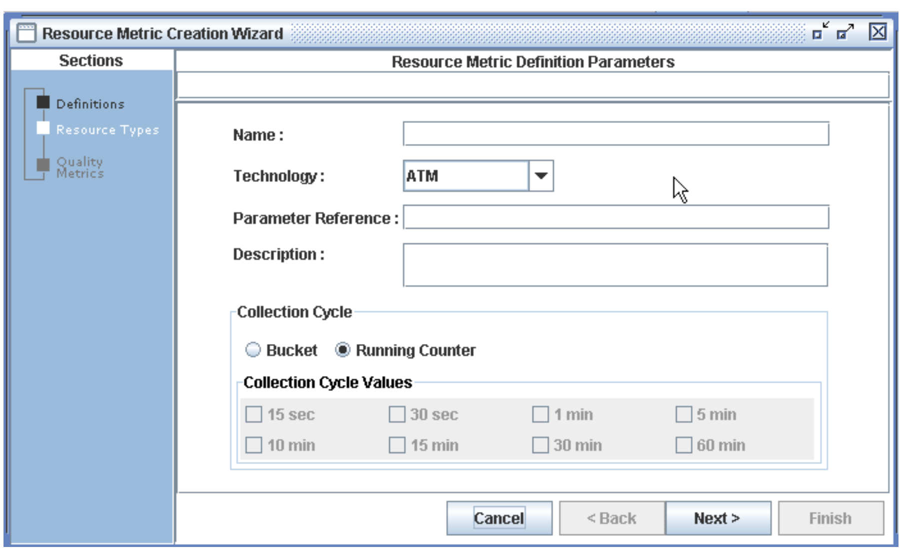
Adding Resource Metrics

You can add a Resource Metric by performing the following steps.

To add a resource metric

1 From the resource metrics list, click **Add**.

The following screen is displayed:



2 In the **Name** field, type a unique name for the Resource Metric to be added.

3 Select a **Technology** (technology code) from the drop-down list for the resource metric you are creating.

4 In the **Parameter Reference** field, type the parameter reference. The Parameter Reference provides additional Resource Metric identification parameters.

5 In the **Description** field, type any text that describes the resource metric (optional).

6 In the **Collection Cycle** section, select one of the option buttons and proceed to the next step.

7 Do one of the following:

If you choose the...	then go to...
Bucket option button,	step 8
Running Counter option button,	step 9

Note: Bucket - new value every time, the bucket is emptied on every collection.

- Running counter - cumulative data.

8 In the **Collection Cycle Values** section, select a time interval to indicate the frequency the resource is to be queried for events.

9 Click **Next**, and continue to "To add a resource metric: device types (part B)".

To add a resource metric: device types (part B)

1 From the **Available Resource Types** list, select a type, and click **Add**.

2 Repeat step 1 until all the required device types have been added.

3 Click **Next**, and continue on "To add a resource metric: quality metrics (part C)."

To add a resource metric: quality metrics (part C)

1 From the **Available Quality Metrics** list, select the quality metric to be associated with this resource metric. Multiple selections are performed by using **Shift** + **Select** or **CTRL** + **Select**.

2 Repeat step 1 until all you have included all of the quality metrics to be associated with this resource metric.

3 Click **Finish**.

Editing Resource Metrics

You can change all data items of a resource metric after it has been created.

To edit a resource metric

1 Display a list of resource metrics and select the resource metric to be edited.

2 Click **Edit**.

The Resource Metric Modification wizard appears.

3 Make changes in any of the available fields.

4 Click **Finish**.

You are returned to the Resource Metrics list.

Viewing a Resource Metrics Record

You can view the details of a Resource Metrics record.

To view a resource metrics record

1. From the Resource Metrics list, select, or highlight, the row corresponding to the resource metric to be viewed.
2. Click **View**.

   The Resource Metrics parameters are displayed in view format only.

3. Use the **Next** and **Back** buttons to navigate through the screens.
4. Click **Cancel** to exit the wizard.

   You are returned to the Resource Metric list.

Deleting Resource Metrics

You can delete a quality metric after it has been created, provided that it is not referenced elsewhere in PerforMAX.

To delete a resource metric

1. Display a list of resource metrics and select the resource metric to be deleted.
2. Click **Delete**.

   Note: If the resource metric you are trying to delete is referenced elsewhere, the **Exception Notification** dialog box appears. This dialog box indicates that you cannot delete this resource metric.

Using the MIB Browser

Management Information Bases (MIBs) are a collection of objects or definitions that define the properties of the managed objects. The MIB browser is used to view and operate on data available through a SNMP agent on a managed device.

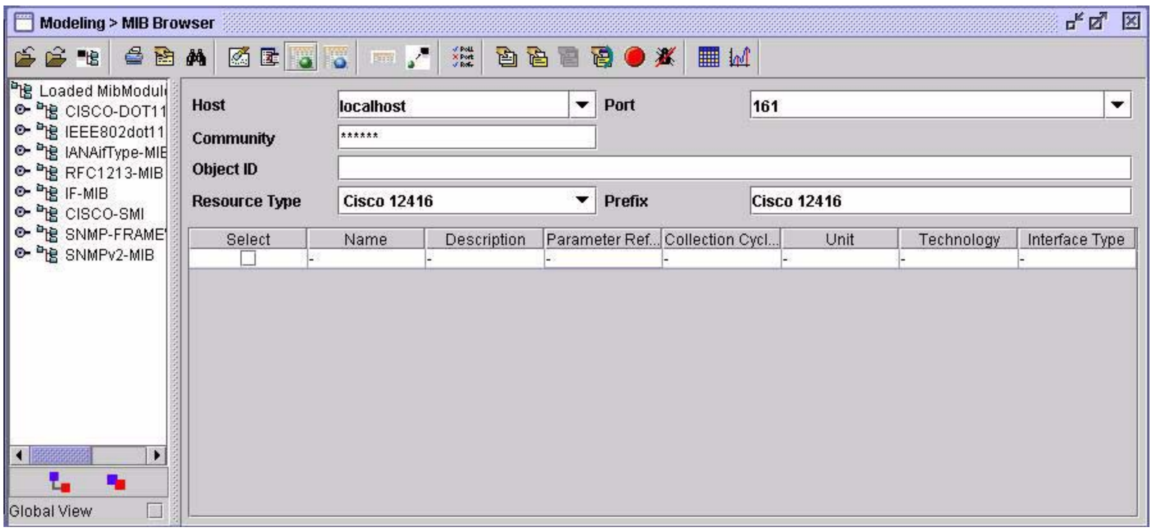
The MIB Browser specifically allows for the selection, viewing and control of performance data variables. MIB browser lets user specify the resource metrics and interface types to be collected. Users may view MIBs, upload MIBs, display attributes for a specific device type and use SNMP functions.

MIB Procedures

To display the MIB browser

1 From the **Modeling** menu, select **MIB Browser**.

A screen similar to the following displays:



Toolbar options include:

Toolbar Option	Description
Load MIBS	Allows the user to load MIB(s) from the client computer to the MIB Browser client. This is a local operation.
Upload MIBS	Allows the user to load MIB(s) to the server.
Resource Type - MIB Mapping	User can select a device type to load the MIBs. Device type to MIBs mapping is stored at server and this mapping can be updated
Print	Allows the user to print the MIB tree, SNMP Results, or the MIB results.
MIB Node Description	Used to view the details of a MIB attribute.
Find MIB Node	Used to search for a loaded MIB by name.
Clear View	Used to delete all selected resource metrics entries displayed in the MIB Browser information pane.

Toolbar Option	Description
SNMP Operation	Used to initiate SNMP Get, Get Next, and Get Bulk operations.
Resource Metrics Modeling	Used to View and Add resource metrics.
Interface Types Modeling	Used to View and Add interface metrics.
Compact Rows	Used to display only checked rows of resource/ interface metrics.
Submit Rows Data	Used to initiate Add resource/interface metrics
SNMP Settings	Used to configure SNMP parameters
Get SNMP Variable	Operation to “get” all objects under the selected MIB.
Get Next SNMP Variable	Similar to the SNMP GET operation, but retrieves the value of the next Object ID in the MIB tree
Get Bulk SNMP Variable	Used to retrieve voluminous data from a large table
SNMP Walk	Gets the values of all nodes that come after the selected node from the agent.
Stop SNMP Query	Stops current SNMP operation.
Debug	Used to debug SNMP operations.
View Table Data	Displays the results of a Table Get in tabular format.
View Graph	Displays the results of a Table Get in graphical format.

Uploading MIBs

MIBs are one of the basic inputs that are required for working with MIB browser. Users can upload MIBs to the server. Once uploaded MIBs will be available for all users.

MIBs may be uploaded in one two ways:

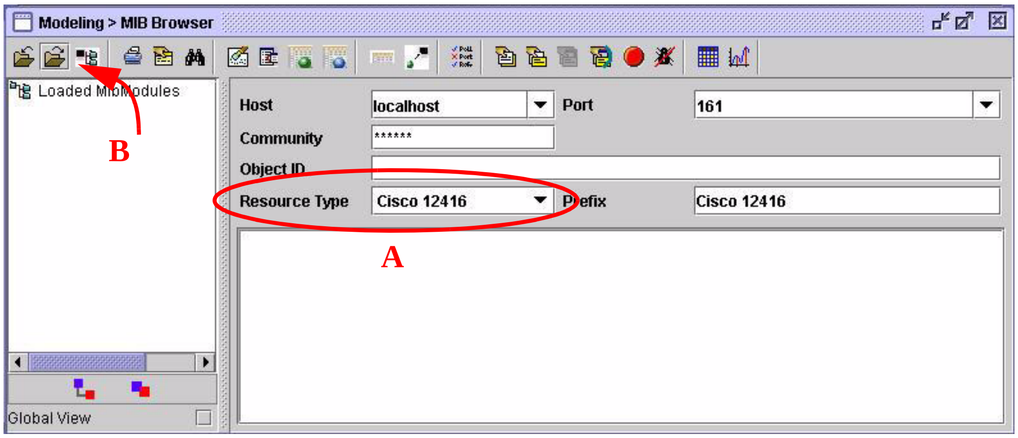
1. **Load MIBs -** Allows the user to load MIB(s) from the client computer to the MIB Browser client. This is a local operation.
2. **Upload MIBs -** Allows the user to load MIB(s) to the server.

To upload MIB files

1 Select **Modeling** from the dashboard.

2 Select **MIB Browser**.

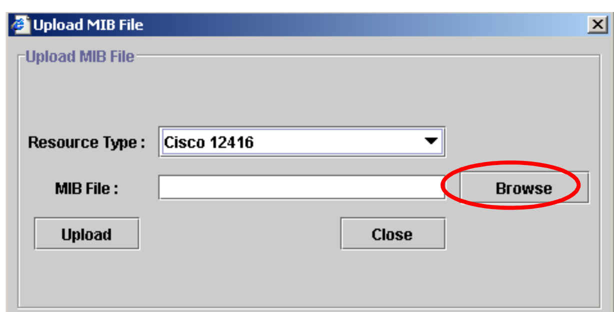
The MIB Browser screen is displayed:



3 Select **Resource** (A above).

4 Click **Upload MIBs** to browse for the MIBs (B above).

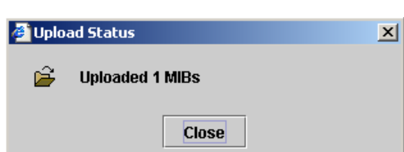
The Upload MIB File dialog box is displayed:



5 Click **Browse** to locate the MIB file.

6 Click **Upload.**

The MIB file is uploaded and upon successful completion, the MIB Status dialog is displayed:

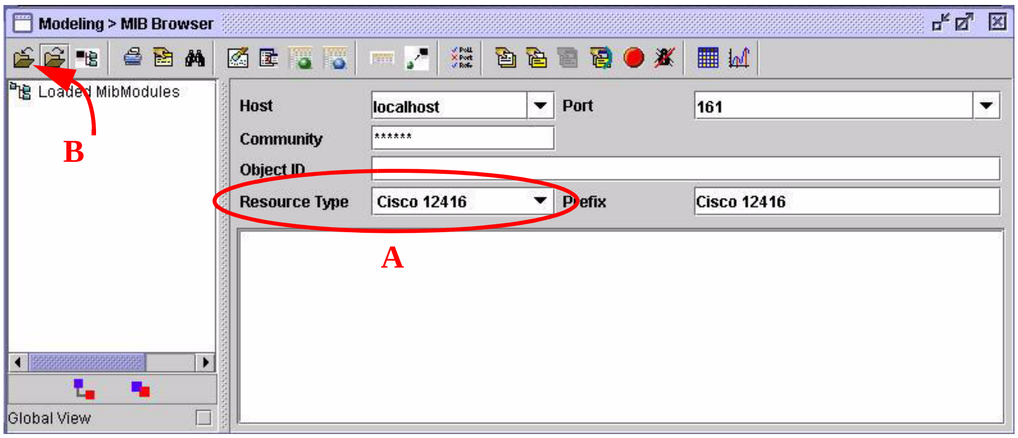


To load MIB files to the local machine

1 Select **Modeling** from the dashboard.

2 Select **MIB Browser**.

The MIB Browser screen is displayed:



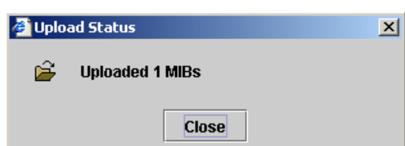
3 Select **Resource** (A above).

4 Click **Load MIBs** to browse for the MIBs. (B above).

5 Click **Browse** to locate the MIB file.

6 Click **Upload.**

The MIB file is uploaded and upon successful completion, the MIB Status dialog is displayed:



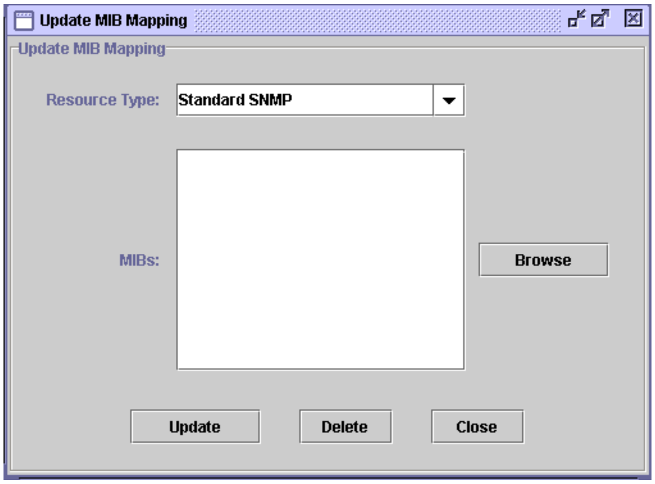
Adding a Resource Type

Users can select a resource type to load the MIBs. Resource type to MIBs mapping is stored on server and the mapping can be updated, if user wants to change MIBs associated with a resource type. Any update in the mapping will be available for all the users.

To update resource types - MIBs mapping

1 Click on **Resource Type - MIB Mapping** toolbar icon.

2 Select **Resource Type**.



3 Select **MIBs** to be associated.

4 Click **Update**.

Note: If there is a large number of MIBs associated with a device type, then it requires a large amount of memory to load those MIBs. Therefore, if you are going to be working with around 200 - 600 MIBs, you should make sure that your machine has a minimum of 256MB RAM. However, any user can work without any problem using less than 200 MIBs.

Resource Metrics Modeling Using the MIB Browser

Resource metrics that are collectable for a resource type will be depicted on the MIB tree with different icons.

To view resource metrics

1 Select a resource type from the **Resource Type** combo box.

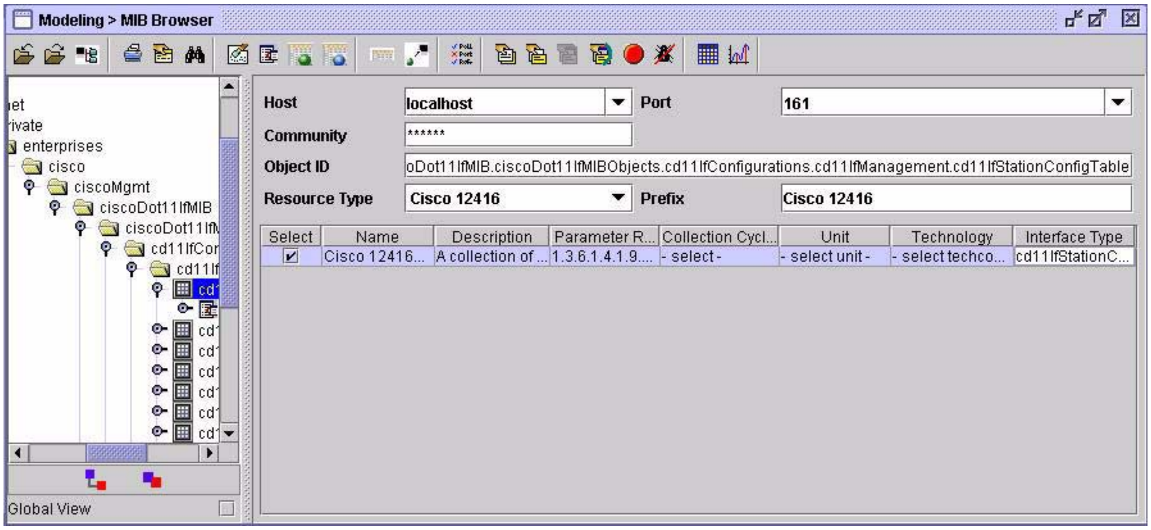
2 Browse through the MIB tree to see various collectable resource metrics.

Adding Resource Metrics

Resource metrics that need to be collected can be specified from the MIB browser screen.

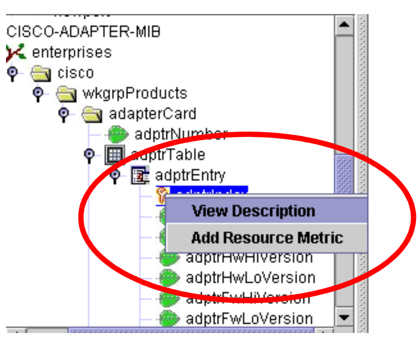
To specify resource metrics data

Snapshot of a screen that is used for adding resource metric is shown:

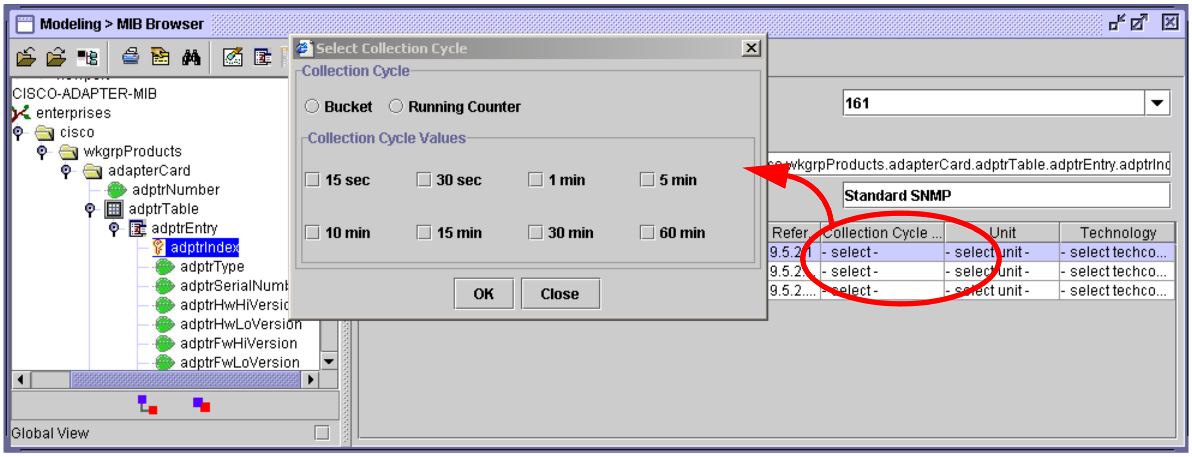


1 Select any **MIB node** or **MIB group node**.

2 Double-click or right-click option on the **MIB node** to add the resource metric.



3 Select a table cell to edit the field.



4 If you don't want to add any selected resource metric, deselect those rows from table and click on **Compact Rows** toolbar icon. To delete all selected entries, click on **Clear** toolbar icon.

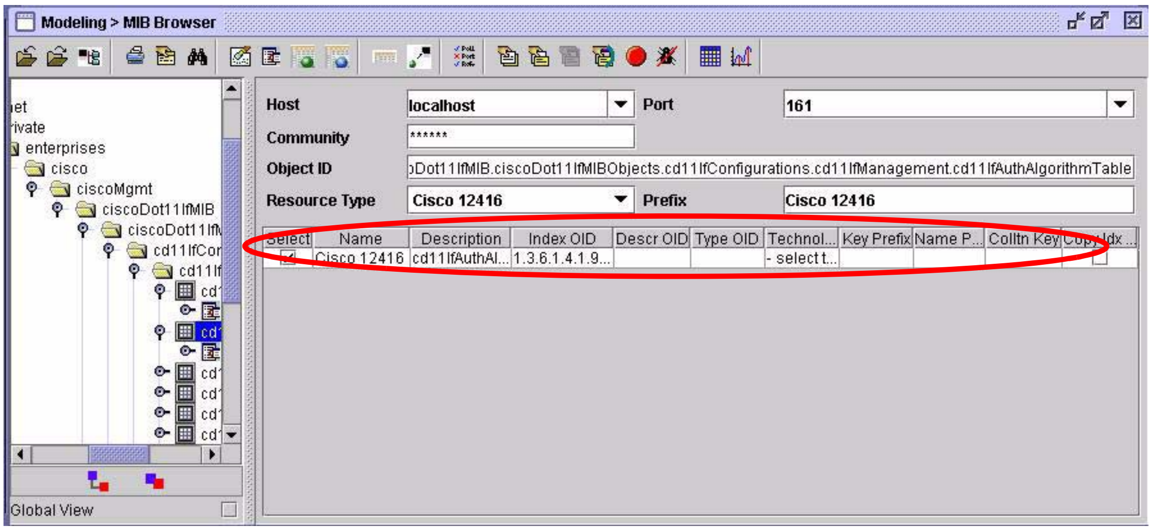
Interface Type Modeling

Adding Interface Types

Interface types can also be selected from the MIB browser screen.

To add interface types

1 Click on **Interface Types Modeling** toolbar icon.



2 Select a tabular node from MIB tree and double-click or right-click on it. If selected table has a type column (column ending with “type”), then a row for each possible value of type column will be shown.

3 If selected column has any description column (column ending with desc, descr), then OID of that column will come in “desc OID” table column. Shown OID description can be modified by selecting table rows and then by right-clicking on any table column of MIB tree.

4 MIB browser automatically derives index OID on the table. But this can be changed by first selecting table rows, and then right-clicking on index fields of MIB tree.

5 To modify other fields of interface types, edit table cell.

6 If you want to delete any selected interface types, deselect the rows from the table and click on **Delete**. To delete all selected entries, click on **Clear** toolbar icon.

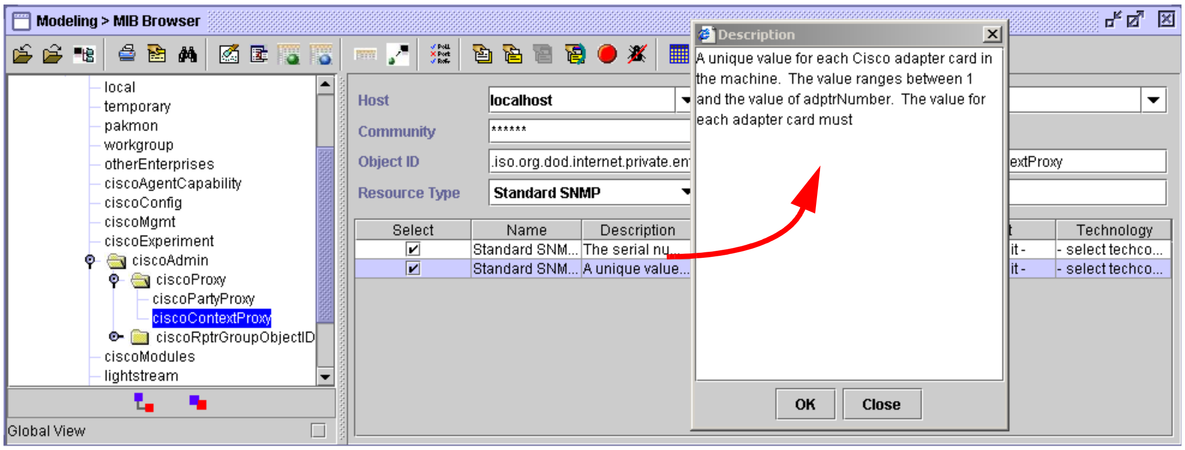
MIB Browsing

Viewing MIB Node Description

MIB Node descriptions can be used to see the details of a MIB attribute. This gives the list of all the vital characteristics of the node such as the Node, OID, Path, MibModule, Syntax, Status, and so on.

To view MIB node description

1. Select a **node** from the MIB tree.
2. Click **Description** cell for a pop-up window providing the description.

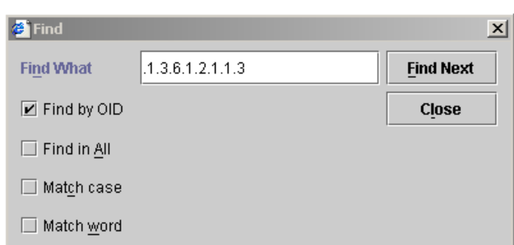


Searching an MIB node

MIB nodes can be searched from the loaded MIBs by name. Partial names can also be used. For example, sysName can be searched using "sysn".

To search a MIB node

1. Click on the **Find Node** toolbar icon.
2. Specify the complete or partial node name.
3. Click **Find**.



Expanding an MIB tree

MIBs are shown in the form of a tree structure. The tree can be expanded to see all the nodes present in the loaded MIBs.

To expand an MIB tree

1 Click **Expand** to expand the MIB tree.

2 Expanded trees can be collapsed by clicking **Collapse**.

SNMP Operations

Configuring SNMP parameters

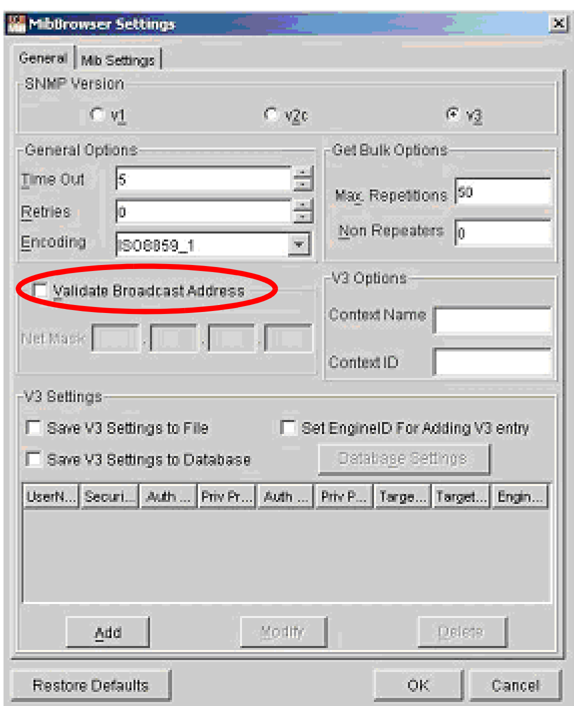
The parameters, such as host, port, and community, can be set in the MIB Browser's main window.

Other SNMP parameters can be configured from a toolbar icon. These parameters include the basic protocol options related to SNMP.

To configure SNMP parameters

1 Click the **snmp parameters** toolbar button.

The following dialog box is displayed:



2 Configure SNMP parameters.

3 Click **Add** to configure security-related parameters for SNMPv3.

4 **Validate Broadcast Address** check box can be used to check the validity of the broadcast address provided.

5 Click **OK**.

Note: Last saved SNMP parameters can be restored using “Restore Defaults” button.

Performing SNMP operations

Get, Get Next, Get Bulk and **SNMP Walk** operations can be performed from the MIB browser. To specify an object to an SNMP agent, both the Object ID (which defines the type of object) and the instance (the specific object of the given type) must be provided.

To specify an Object ID

1 Select a MIB node from the MIB tree.

2 Or type it in the Object ID text field.

To specify the instance

For non-tabular or scalar objects, the instance is 0. For example, *sysDescr* is a scalar object under the system group in RFC1213-MIB and it should be specified as *sysDescr.0* in Object ID field of the MIB Browser. MibBrowser adds it to the selected node while performing the **GET** operation.

Performing SNMP Get

The **GET** operation is performed to get one or more values from the managed objects.

To perform SNMP Get

1 Select the desired node in the **MIB tree**.

2 Click the **Get SNMP Variable**.

This operation gets all objects under the selected MIB object, or the specific object if the MIB node and instance are specified

Note: If the selected node in the MIB tree has child nodes or columnar nodes, the **GET** operation is performed as an SNMP walk, that is, the values for all the nodes under that sub-tree are retrieved.

In case, if the MIB is not loaded in the MibBrowser, the exact numbered OID with full instance (for example 1.3.0 for sysUpTime) should be specified and the **GET** operation should be performed.

Performing SNMP Get Next

This operation is similar to the SNMP **Get** operation, but retrieves the value of the next OID in the tree. This operation is used for traversing the MIB tree.

To perform SNMP Get Next

1 Select a node from the **MIB Tree**.

2 Click **Get Next SNMP Variable**.

Performing SNMP Get Bulk

To retrieve voluminous data from a large table, the **GETBULK** operation is performed. A **GETBULK** request is performed by giving an OID along with two other parameters, namely a Max-Repetitions value and a Non-Repeaters value. The **GETBULK** operation is performed only on SNMPv2c and SNMPv3.

To perform SNMP Get Bulk

1 Select a **MIB node**.

2 Configure the MIB Browser to either SNMPv2c or SNMPv3 as desired.

Note: This can be done through MibBrowser Settings icon on the toolbar.

3 Under the same **MIB Browser Settings** panel, the **Max-Repetitions** field and the **Non-Repeaters** field are enabled. The **Max-Repetitions** value specifies the number of lexicographic successors to be returned for the remaining variables in the variable-bindings list. The default value in this field is *50*. The **Non-Repeaters** value specifies the number of variables in the variable-bindings list for which a single lexicographic successor is to be returned. The default value in this field is 0.

4 Click **Get Bulk SNMP data**.

Performing Table Gets

Table get can be performed from the MIB Browser and the result will be shown in the tabular form.

To perform table get

1 Specify the proper agent host name or IP address in the Host field of MibBrowser.

2 Specify a valid OID or select the OID by traversing through the MIB Tree. The OID should be a valid table OID.

3 Click the View SNMP data table button on the toolbar.

This would invoke the SNMP Table of the specified table OID.

Performing SNMP Walk

A device can be “walked” using MIB browser. SNMP Walk is implemented by using SNMP get bulk. In case of SNMP v1 devices, **GET Next** is used for walking the device.

To perform SNMP walk

1. Select a MIB node from the MIB tree.
2. Click **SNMP Walk** toolbar icon.
3. If MIB browser prompts for selecting SNMP v2c/v3, it means v1 (get next) is being used. For efficient walk, change SNMP version if device supports v2c/v3 and again Click **SNMP Walk** toolbar icon.
4. The device is “walked” starting from the selected MIB node. If the user does not select a MIB node for SNMP walk, the complete device is walked.

Stopping SNMP Walk

SNMP walk can take time to complete. So the user is given the functionality to stop ongoing SNMP Walk operation.

To stop SNMP walk

1. Click the **Stop SNMP query** toolbar icon.

Using MIB Browser Graphs

Real-time plotting of the SNMP data can be done using graphs. The SNMP data to be polled should be an *integer* or *unsigned integer* data type. Typically, the values that are plotted will be of type **Counter, Gauge,** or **Timeticks**.

To plot graphs

1. Specify the proper agent host name or IP address in the host field of the MIB Browser.
2. Specify a valid variable. The variable must be an integer or unsigned integer (Counter, Gauge, TimeTicks). This variable can be entered directly in the variable field or it can be selected through the MIB tree.
3. Select the **Graph Tool** icon.

   The updated Bar graph shows the results of periodically polling the agent for the specified OID.
4. Select the **Log Polled Values** and the results can be logged to a CSV file.
5. Click on the **Export** and select the **Export Type**. Verify the chart can be exported to files as all the available Export Types.

Stopping Ongoing SNMP Operations

Operations that are started by the user can be stopped by the user.

To stop SNMP operations

1 Click **Stop**.

Printing MIB Information

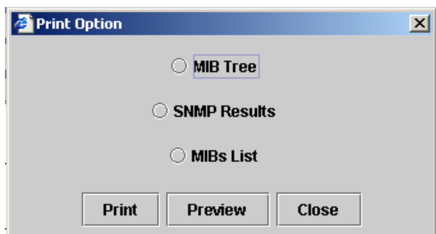
The following MIB information can be printed from the MIB Browser:

- MIB Tree
- SNMP Operations
- MIB List

1 Click the **Print** icon on the toolbar.

2 Select the desired print option.

3 Click **Print**.



Command Line MIB Browser

If the user for some reason is not able to use the MIB browser integrated in the GUI, then command line MIB browsing can be used. The command line browser provides features that are independent of the application's server. These features include:

- Load MIBs from client machine.
- SNMP operation.
- Print features
- Find operation.

Command line MIB browsing can be invoked using **startMibBrowser.bat** or **startMibBrowser.sh**.

Configuring Device Type Mapping through Configuration File

Device type to MIB mapping can also be configured through a configuration file, which is read at server startup.

This file is present in MIB directory specified in the mibbrowser section of the config file. Whenever the user updates a device type mapping through the GUI, this file is updated.

Configuration File Format

<device type>=<comma separated MIB list>

MIB list should contain MIB file name with path relative to MIBs dir.

For example, if Cisco_12000 device type needs to be mapped to RFC1213-MIB and RFC1231-MIB present in common folder inside MIBs dir. Config entry should look like

Cisco_12000=common/RFC1213-MIB: common/RFC1231-MIB

Note: You can use the percentage sign (%) as a wildcard. You can type it before and/or after a variable (such as a parameter or an IP address), or you can type the percentage sign (%) by itself to list all items available for a particular field.

Managing Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) are used to measure the performance network performance of a resource. Because indicators are mathematical expressions, they provide a value that indicates the state of a particular aspect of a resource.

For example, to measure packet loss, you can specify the InCells (from the destination) *minus* OutCells (from the source) *equals* Packets(unit).

To measure the performance and/or QoS for a resource metric you need to look at the indicators, which are also known as a KPIs.

In general, KPIs are mathematical expressions, or calculations, used to measure specific attributes of a performance class. Every performance class has one or more performance indicators.

Indicators take quality metrics and use them in mathematical expressions to derive a value. However, while the value derived by the performance indicator is an important part of quality assessment, quality is not actually assessed at the performance indicator level.

Network performance is assessed at the policy level when you compare performance indicator values against an assigned threshold value that you define to assess quality in a policy.

Building Expressions

When you are building an expression, the following special characters are used as tags. You can type the following special characters into your expression:

Operator	Function
(	Left bracket to enclose sub-formula elements
)	Right bracket to enclose sub-formula elements
+	Add adjacent elements

Operator	Function
-	Subtract adjacent elements
*	Multiply adjacent elements
/	Divide adjacent elements
Max	Get the maximum value and apply
Min	Get the minimum value and apply
Scale	Apply a scaling factor to the result
>	Greater than
<	Less than
!=	Not equal to
==	Is it equal to?
&&	AND
||	OR
<=	Less than or equal to
=>	Greater than or equal to
#	Stands for CautiousAdd operation. (If either operand is -1001 (stands for not available), then the result is -1001 (not available).)
?	Stands for Selection operation. For example, if you put the **?** operator in the following formula, op1op2?, you are saying that if op1>-1001, then the result shall be op1, no matter the value of op2, if op2>-1001, only when op1 <= -1001, then the result shall be op2, otherwise, return -1001.
~	Stands for CautiousSubtract operation. For example, op1op2~, if either op1 or op2 is less than or equal to -1001, return -1001, otherwise, return op1-op2.
$	Stands for Condition Selection operation. For example, op1op2$, if either op1 or op2 is less than or equal to -1001, return -1001; otherwise, return op1.
Clear	Clear the currently displayed formula

- First operand (Main operand) — the main operand can take any value.
- Second operand (Scale operand) — the scale must be a positive value

The output of the scale operation can be determined from the table below:

Output	Condition
-2	Main < - Scale
-1	0 > Main > - Scale
0	Main = 0
1	Scale > Main > 0
2	Main > Scale

The scale operation is very useful for the health calculations.

KPI Definition Process

After defining the quality metrics for a resource you can define the KPIs.You can use the modeling features to map quality metrics to KPIs.

KPIs are related to the following elements:

Element	Description
Analysis Type:	is a flow based analysis associated with tunnel and funnel (that is, have 2 or more endpoints where flow between the endpoints needs to be measured. Single-Point/Cloud are endpoints evaluated singly or as a group).
Performance Class(es):	is a grouping of KPIs.
Formula:	are the mathematical expressions based on one or more quality metrics and/or KPIs.
Quality Metrics:	are the quality metrics that are used to create a KPIs are bound to the technology.
Unit:	specifies the units of expression used.

PerforMAX allows you to create many performance classes based on a single technology code. PerforMAX also has a number of KPIs already defined (that is, they come pre-loaded, out-of-the-box) for many of the included technology codes. Use the features in the KPIs option to manage performance indicators.

KPI Procedures

You can display the list of existing KPIs by accessing the KPIs list. The KPIs list includes the KPI's name, and description. You can perform the following procedures for KPIs:

- Displaying a KPI List
- Adding KPIs
- Editing KPIs
- Viewing KPIs
- Deleting KPIs

Displaying a KPI List

You can display a list of KPIs that meet a specific criteria using the filtering feature.

To display a list of KPIs

1 From the **Modeling** menu, select **KPIs**.

The KPIs default screen is displayed:

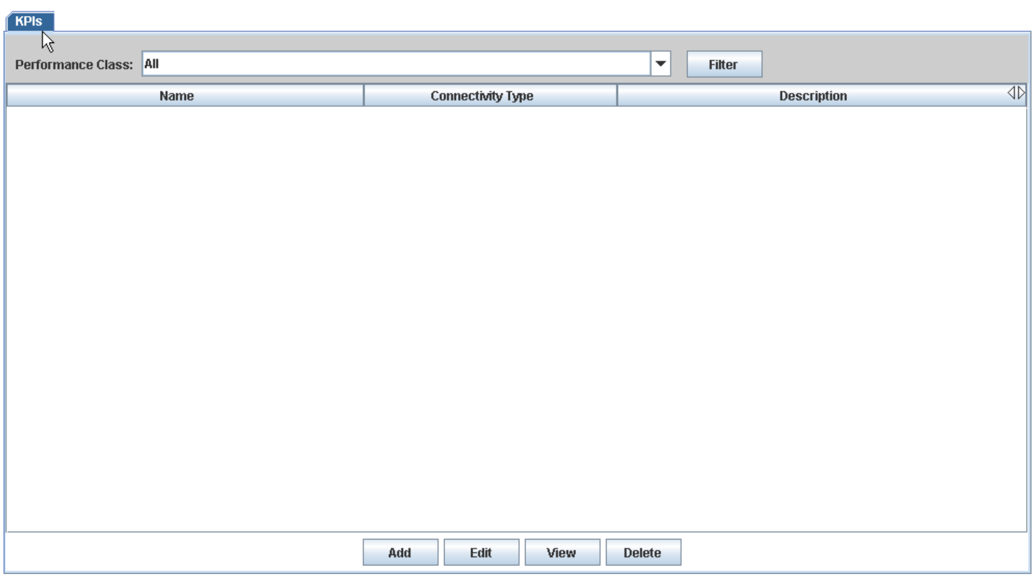


Figure 3-1: KPI Default Screen

2 Select the **Performance Class** from the drop-down list or accept the default (**All**).

3 Click **Filter**.

The KPI list is displayed.

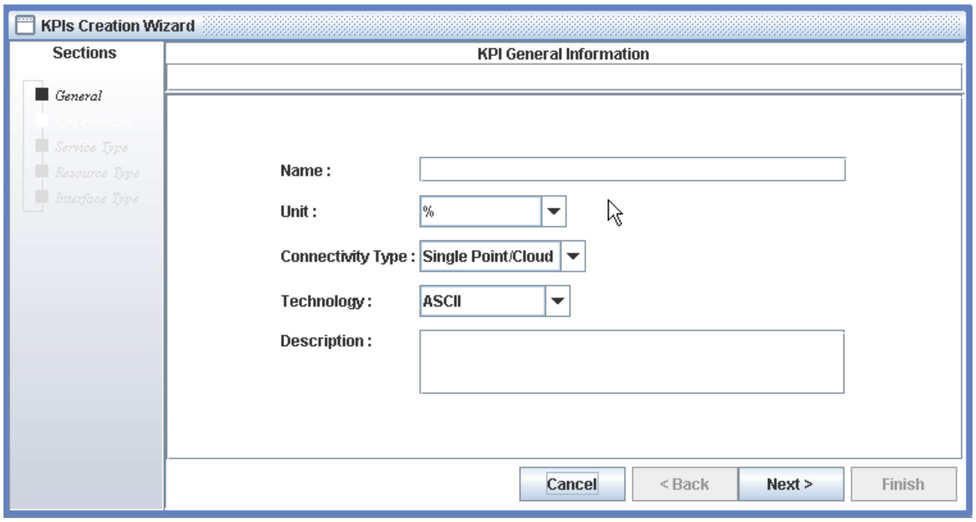
Adding KPIs

After you are certain the appropriate building blocks are in place (see “Building a Model” for detailed information), you can proceed to add KPIs by performing both parts of this two-part procedure.

To add a KPI (General - part A)

1 From the KPI default screen or list, click **Add**.

The KPI General Information screen is displayed:



2 In the **Name** box, type the name of the KPI being created.

3 From the **Unit** list box, choose the unit of the indicator formula.

4 From the **Connectivity Type** list, choose one of the following:

- **Single Point/Cloud**
- **Tunnel/Funnel**

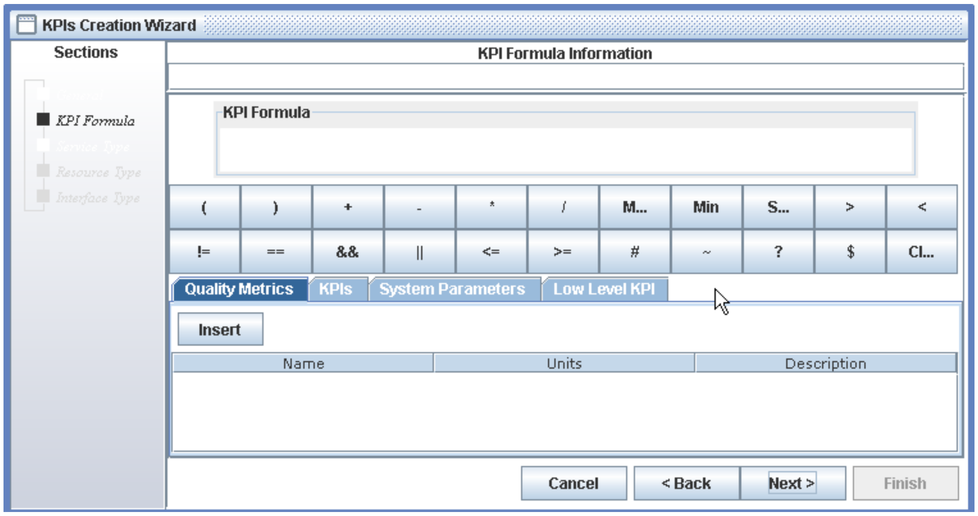
5 From the **Technology** list, choose the technology with which the KPI is to be associated.

6 In the **Description** box, type a description for the KPI.

This text will appear beside the KPI name in the KPI list.

7 Click **Next**.

The KPI Formula Information screen is displayed:



8 Go to "To add a KPI (Creating Indicator Formulas - part B)".

To add a KPI (Creating Indicator Formulas - part B)

1 Do one of the following:

If you want to insert a...	then...
Quality Metric operand	see "To insert a quality metric operand"
Indicator operand	see "To insert an indicator operand"
System Parameter operand	see "To insert a system parameters operand"
Low Level KPI	see "To insert a low level KPI"

To insert a quality metric operand

a From the **KPI Creation Wizard**, select the **Quality Metrics** tab.

b Select a Quality Metric from the list.

c Click **Insert**.

d Click the appropriate **Operator** button. You can switch among the available tabs before selecting **Finish**, inserting any number of desired operands and operators.

e Click **Next**.

Note: PerforMAX expressions require a minimum of two operands.

To insert an indicator operand

a Click the **KPIs** tab.

b Select an indicator from the **KPIs** list.

c Click **Insert**.

d Click the appropriate **Operator** button.

e Click **Next**.

To insert a system parameters operand

a Click the **System Parameters** tab.

b Click one of the following option buttons:

- **Collection Interval** — takes the data from the CG parameters.
- **Collection Time** — causes the system parameter to be calculated at the collection time.

c Click **Insert**.

d Click the appropriate **Operator** buttons.

e Click **Next**.

To insert a low level KPI

a Click the **Low Level KPI** tab.

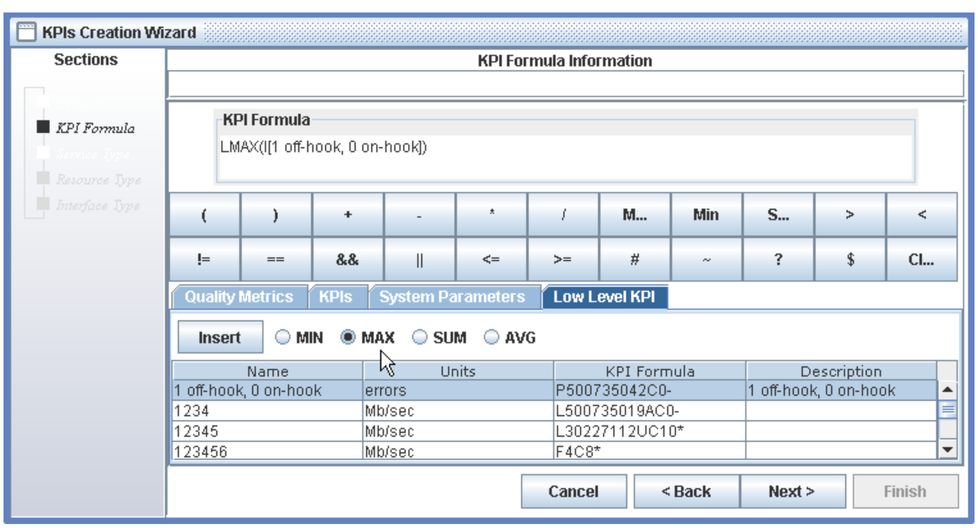
b Select an indicator from the **KPIs** list.

c Select the appropriate **Operator** button.

- MIN
- MAX
- SUM
- AVG

d Click **Insert**.

A screen similar to the following is displayed:



e Click **Next**.

The next three wizard screens are optional. To associate the KPI with a Performance Class, Resource Type, or Interface Type, click **Next** to the desired wizard screen.

Note: The **Finish** button is active through the next three wizard screens. The user may click **Finish** at any time during the final three wizard screens.

To create a new low level KPI (from existing KPI)

a From the KPI Formula Information screen, insert the appropriate operand after the existing KPI Formula. For example, you might insert a plus (+) sign.

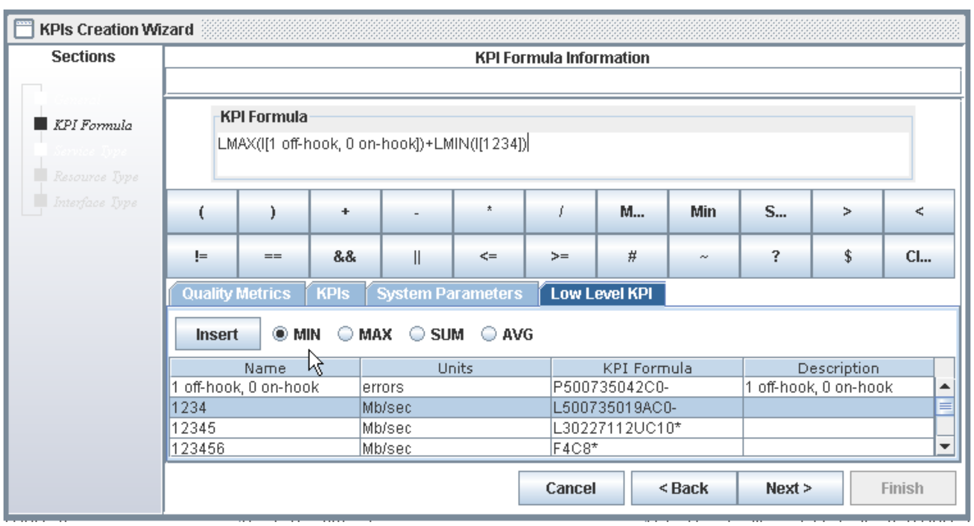
b Select an indicator from the **KPIs** list.

c Select the appropriate **Operator** button.

- MIN
- MAX
- SUM
- AVG

d Click **Insert**.

A screen similar to the following is displayed:



The new formula is added to the existing KPI.

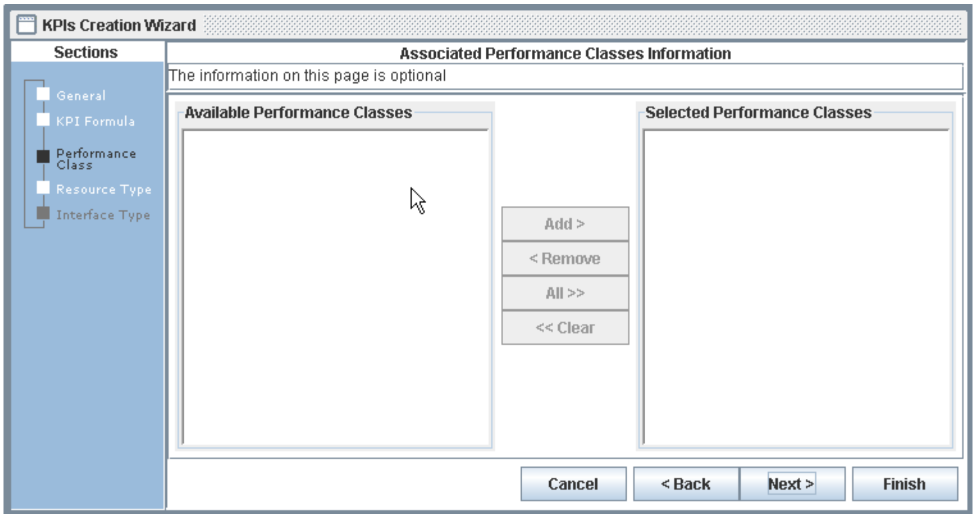
For all of the following procedures, the wizard screens are optional.

To associate a KPI with a performance class or resource/interface type

1. To associate the KPI with a Performance Class, Resource Type, or Interface Type, click **Next** to navigate to the desired screen.

   Note: The **Finish** button is active throughout the following wizard screens. The user may click **Finish** at any time during the existing wizard screens.

   A screen similar to the following is displayed:



2. Select a **Performance Class, Resource** and or **Interface Type** from the following screens and click **Add** to move the selection from the **Available** to the **Selected** pane.
3. Click **Next**.
4. When you are finished adding the information, click **Finish**.

   You are returned to the KPI list.

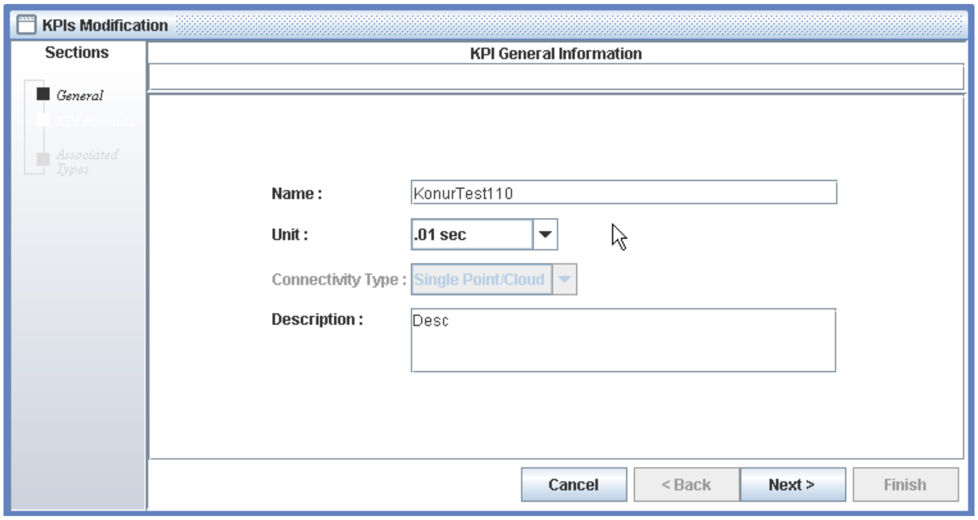
Editing KPIs

You can edit existing KPIs to your specific needs.

To edit a KPI

1. From the KPIs list, select a specific KPI and click **Edit**.

   A screen similar to the following is displayed:



2 From the **Name** field, type in a new name.

3 From the **Unit** drop-down list, select a new unit.

4 From the **Description** field, type in a new description.

Note: These are the only fields that you can edit.

5 Click **Next**.

The KPI Information screen is displayed.

Note: You cannot edit any of these fields.

6 Click **Next**.

The Associated Types Information screen is displayed.

Note: You cannot edit any of these fields.

7 Click **Finish**.

Note: You may or may not be able to delete or edit a KPI because it may have dependencies on other records.

Viewing KPIs

After generating a KPI list, you can view the details of a KPI record in read-only format.

To view a KPI

1 From the KPI list, select, or highlight, the row corresponding to the KPI to be viewed.

2 Click **View**.

The KPI General Information screen is displayed in read-only format.

3 Click **Next** to view the *KPI Formula and Associated Types Information screens* in their respective order.

4 Click **Cancel** to return to the *KPI list*.

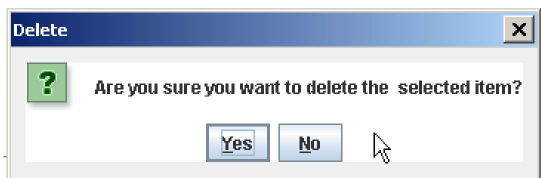
Deleting KPIs

You can delete existing KPIs from the KPI list that are no longer used or valid.

To delete a KPI

1 From the KPIs list, select a specific KPI and click **Delete**.

The following confirmation message is displayed:



2 Click **Yes** to delete the KPI or click **No** to return to the KPI list.

Note: You may or may not be able to delete or edit a KPI because it may have dependencies on other records.

Managing Performance Classes

A Performance Class specifies the performance metrics (Indicators) and associates these metrics with a technology code.

Performance Classes contain one or more Indicators. In addition, an Indicator can be reused in multiple Performance Classes.

Connections are defined based upon Performance Classes.

When you create a performance class, you must specify a name, process code, analysis type, and description.

In addition, you also need to define the indicators defining the performance classes overall performance attributes (for example, availability, errored seconds, MTTR, and so on).

When you purchase a technology code from Trendium, certain default performance class are added to the database and displayed in the Performance Classes list.

For more information about which performance classes are added for each technology code, contact Trendium Customer Service.

Performance Class Procedures

You can perform the following procedures on performance classes:

- Displaying a Performance Class List
- Adding a Performance Class
- Editing a Performance Class
- Viewing a Performance Class Record
- Deleting a Performance Class

Displaying a Performance Class List

You can display a list of performance classes based on its technology.

To display a performance class list

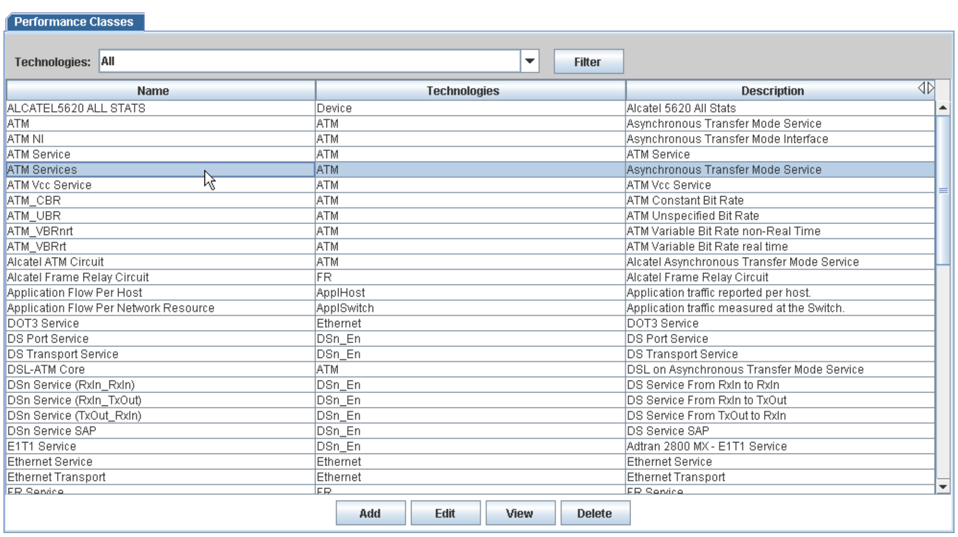
1 From the **Modeling** menu, select **Performance Classes**.

The default screen is displayed.

2 Select a technology from the **Technologies** drop-down list or accept the default (**All**).

3 Click **Filter**.

A screen similar to the following is displayed.



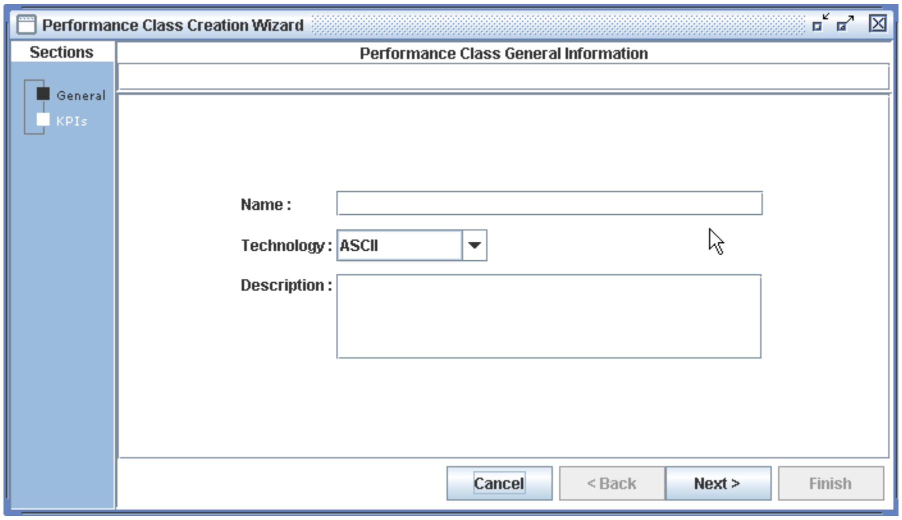
Adding a Performance Class

You can add a new performance class to the database by performing the following steps.

To add a performance class

1 From the default or generated list screen, click **Add**.

The following screen is displayed:



2 In the **Name** field, enter a descriptive name.

3 From the **Technology** drop-down list, choose a technology code.

You can change technology and select **Get** to display different lists of KPIs but you cannot mix KPI of different technologies in the same performance class.

4 In the **Description** field, type a description of the technology code.

5 Click **Next**.

6 In the **KPIs** column, select a KPI(s) and click **Add**.

7 Click **Finish**.

You are returned to the default list.

Editing a Performance Class

You can change the name, description and associates KPI of an existing performance class.

To edit a performance class

1 From the default list, select the performance class to be edited.

2 Click **Edit**.

The Performance Class Modification screen is displayed.

3 Type new information (if necessary) in any of the following fields:

- **Name**
- **Description**

4 Click **Next**.

5 In the KPIs screen, you can add or remove KPIs from a performance class.

If a performance class has indicators that you cannot remove because they form part of a dependency, then the indicators will be ghosted in a list at the bottom of the **Selected Indicators** list (see Figure 3-2 below).

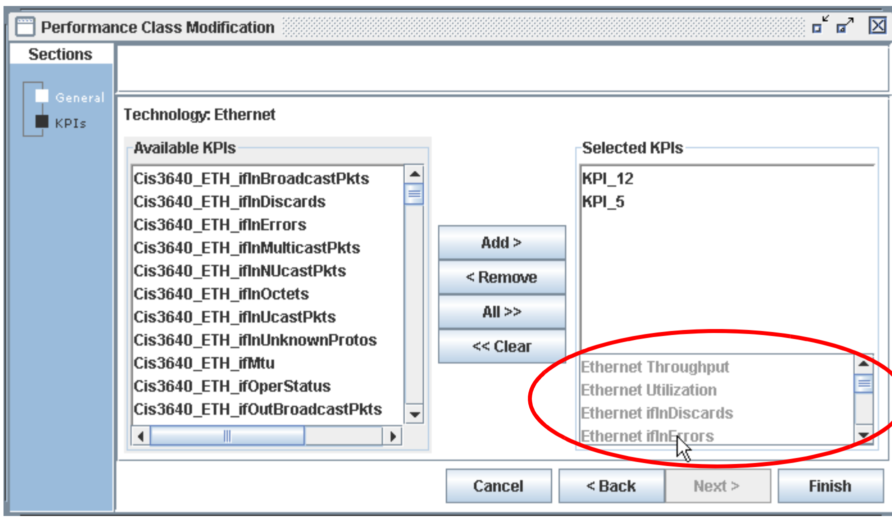


Figure 3-2: Uneditable Indicators

6 Click **Finish**.

You are returned to the default list.

Viewing a Performance Class Record

You can view the details of a performance class record after generating a list. This option allows you to view the record in a read-only format.

To view a performance class record

1 From the default list, select, or highlight, the row corresponding to the record that you want to view.

2 Click **View**.

3 Use the **Next** and **Back** buttons to navigate through the screens.

4 Click **Cancel** to close the screen.

You are returned to the default list.

Deleting a Performance Class

You can delete a performance class record provided that it not referenced in any other components.

To delete a performance class

1. From the default list, select, or highlight, the row corresponding to the record that you want to view.
2. Click **Delete**.

   A confirmation message asks "Are you sure you want to delete the selected item?"
3. Click **Yes** to confirm the deletion or **No** to cancel the deletion process.

Data Collection Templates

4

The following documentation provides information and procedures about data collection templates.

- Managing Action Templates
- Managing Policy Templates
- Managing Connection Templates
- Managing Performance Templates

In PerforMAX you can create and use the following types of templates:

- Action Templates - templates that let you associate actions, such as e-mailing, paging, scripts, provisioning, and alarms, with events.
- Policy Templates - templates that let you associate conditional rules with metrics from your performance indicators.
- Connection Templates - templates that let you associate actions with policies.
- Performance Templates - templates that let you customize condition threshold values.

You must create both the relevant action and policy templates before you can create Connection templates. Connection templates, which are instantiated as Connection Groups (CGs), are a required component.

Understanding the Template Workflow

Before you can create connection offerings, which represent what you sell to a customer, you must create the relevant Action, Policy, and Connection templates.

Templates are generic rules upon which CGs are based. Basically, a template is any set of rules that specifies how PerforMAX will respond to an event, condition, and so on. These rules are reusable, meaning that you can define them once and reference them repeatedly. PerforMAX supports the following types of templates:

- Action
- Policy
- Performance
- Connection

Because templates allow you to reuse information, they can help you reduce the amount of work associated with the creation process.

When you reference templates, they are instantiated. In other words, PerforMAX creates a copy of the template you are referencing at the location from which you are referencing the template.

As a result, you can make changes to this copy of the template without the changes affecting the original template. Likewise, any changes that you make to the original template will not affect the instantiated templates.

Generally, because templates are typically referenced as part of other templates and components, templates have dependencies.

Action templates are built with the assumption that they will be referenced in Policy templates. Policy templates are built so that they can be referenced in Connection templates and so on.

Basically, as you move through the template creation process you move from Action templates to Policy templates (where you incorporate or “gather” action templates) and then to Connection templates (where you incorporate the policy templates).

Connection Groups (CGs) are based on Connection templates. In particular, Connection templates are comprised of one or more policy and action templates. Connection templates use Policy and Action templates as building blocks, allowing you to “mix and match” templates to create a variety of offerings.

Managing Action Templates

Action templates are generic rules that instruct PerforMAX to perform an action such as sending an e-mail, initiating a pager or Netcool notification, or running a script. Here are some examples of the potential use of these actions:

- Send an e-mail or pager notification to system administrators when system utilization reaches a certain threshold
- Notify a network fault management system when certain system parameters are exceeded
- Run a shell script on the PerforMAX server when certain system parameters are exceeded

Action templates can be associated with one or more Policy templates. For more information on incorporating action templates within Policy templates, see "Managing Policy Templates".

PerforMAX allows you to create, modify, and view Action templates. When creating an Action template, you customize the template's name, description, and action type.

The parameter choices available in the Action Type Parameters are based on the action type selected in the General screen.

After you create an Action template, it appears in the Action Templates list.

In addition, you can delete action templates directly from the Action Templates list, provided they have not been referenced elsewhere.

SNMP Traps

The SNMP trap action handler is configured the same way as the other action handlers; that is, it is configured through the action template wizard and the creation wizard.

When a violation occurs, an SNMP trap is sent to a specific destination host or broadcast to the network using the SNMP network protocol.

PerforMAX has 25 fields of SNMP violation data (for example, violationDevice, violationInterface, and so on). In general, most users will be interested in violation data that identifies customers, devices, interfaces, states the severity; and provides a description.

You can see the SNMP violations by viewing the data with the MIB browser. See "Using the MIB Browser" in "Modeling" for more information.

Note: The SNMP trap handler is not started automatically when you install PerforMAX. For information about starting the SNMP trap handler, see the *PerforMAX System Administrator Guide*.

Action Template Procedures

You can perform the following procedures on action templates:

- Displaying an Action Template List
- Adding Action Template Records
- Editing Action Template Records
- Viewing Action Template Records
- Deleting Action Template Records
- Updating Action Parameters

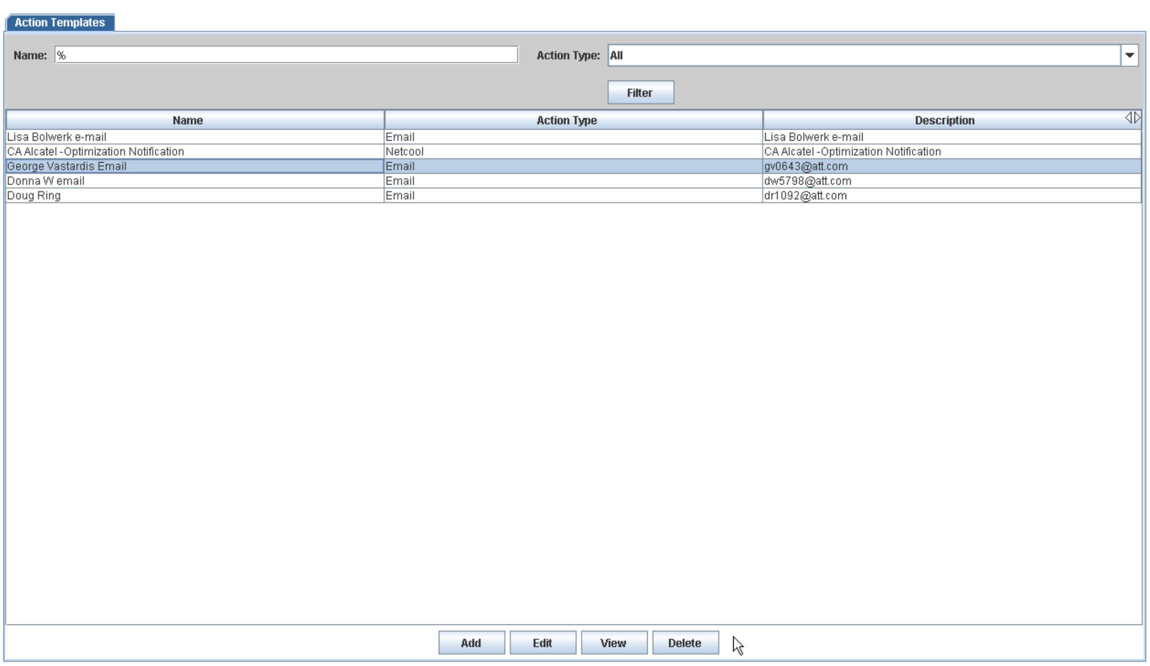
Displaying an Action Template List

You can display a list of Action templates that meet a specific criteria by using the filtering feature.

To display an action template list

1. From the **Catalog** menu, select **Actions**.
2. In the **Name** field, enter the Action template name or, to list all templates, accept the default (**%**).
3. Select the **Action Type** from the drop-down list or accept the default (**All**).
4. Click **Filter**.

A screen similar to the following is displayed:



Adding Action Template Records

You can add an Action template record, which includes specifying an action type and entering the appropriate parameters for the template. See “Using Action Tokens” for more information.

To add an action template record

1. Click **Add**.
2. Enter an action template name in the **Name** field.
3. Select an **Action Type** from the drop-down list.
4. In the **Description** field, enter additional comments, and click **Next**.
5. Enter valid parameter information.

Using Action Tokens

Tokens are defined and handled by the Action handlers based on the Violation Structure the Policy Manager is going to provide in case of violation. Following are the currently supported tokens:

$VIOLATION_ID
$VIOLATION_DETAIL
$POLICY_NAME$POLICY_ID
$POLICY_DESCRIPTION
$SLA_NAME
SLA_IDCUSTOMER_NAME
$CUSTOMER_ID
$CUSTOMER_ACCOUNT
$CONTRACT_NAME
$CONTRACT_ID
$OFFERING_NAME
$OFFERING_ID
$CAUSE$TIME
$SDCOUNT$VIOLATIONCOUNT
$SEVERITY

How to Use Action Tokens

The following example will employ the use of an e-mail action. Detailed information for violated Collection Groups can be used in the e-mail action template.

In the following example, an e-mail action is as follows:

[To=support@myprovider.com][Subject=Violation of SLA $SLA_NAME]
[Body Text= I'm very sorry to tell you that your SLA: $SLA_NAME has been violated. Here are the results we have got from the system $VIOLATION_DETAIL].

Note: Double quotes may not be used as a delimiter and the use of them immediately before or after a token, results in the token **not** being processed correctly. For example, a token entered as **"$VIOLATION_DETAIL** would not be processed.

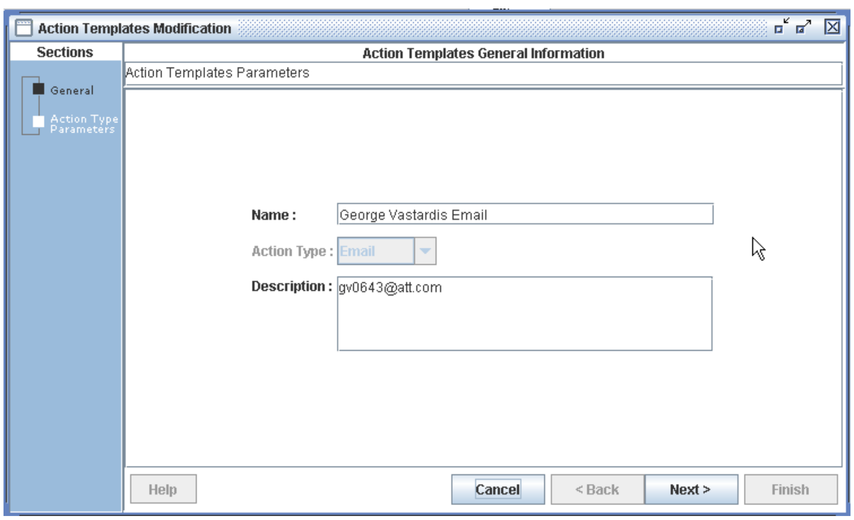
To use action tokens in an e-mail template

1. From the **Catalog** menu, select **Actions**.
2. From the Action Type drop-down list, select **Email**. Click **Filter** to generate a list of Email Action Templates.

   A list displays with Email Action Types only.

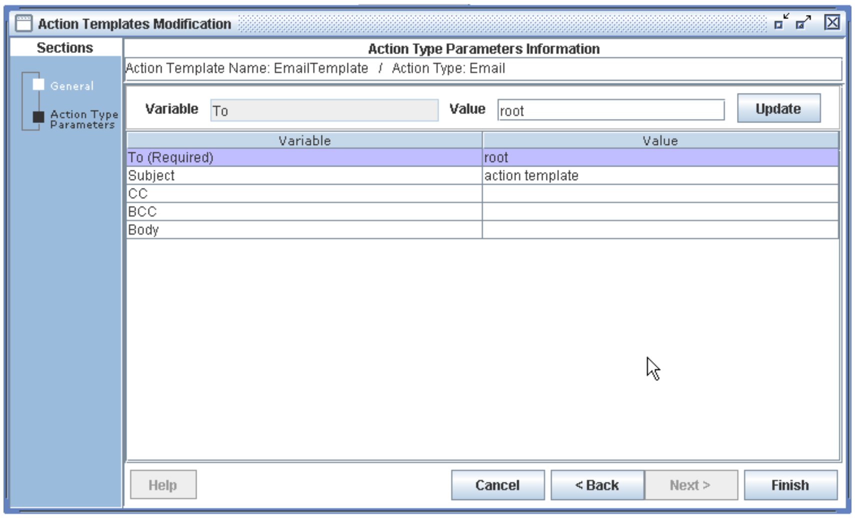
3. Select a template from the list and click **Edit**.

   The following screen is displayed:

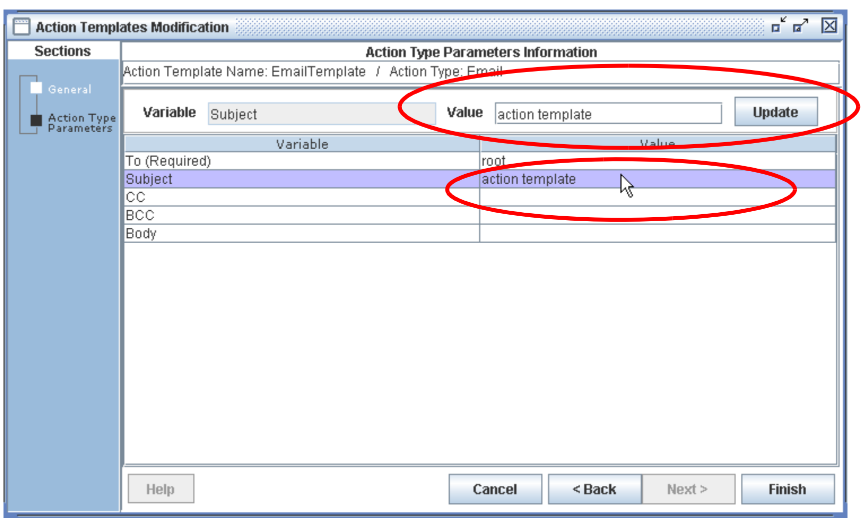


4. Click **Next**.

   The following screen is displayed:



5 Select the **Subject** row.



6 Click inside the **Value** field.

7 Type the token name $SLA_NAME.

8 Click **Update** (the **Subject/Value** row updates with $SLA_NAME).

9 Select the **Body** row.

10 Click inside the **Value** field.

11 Type the token name *"I'm very sorry to tell you that your SLA: $SLA_NAME has been violated. Here are the results we have got from the system $VIOLATION_DETAIL]."*

12 Click **Update** (the **Body**/Value row updates with the *"I'm very sorry to..."*).

13 Click **Finish**.

Shell Action Templates

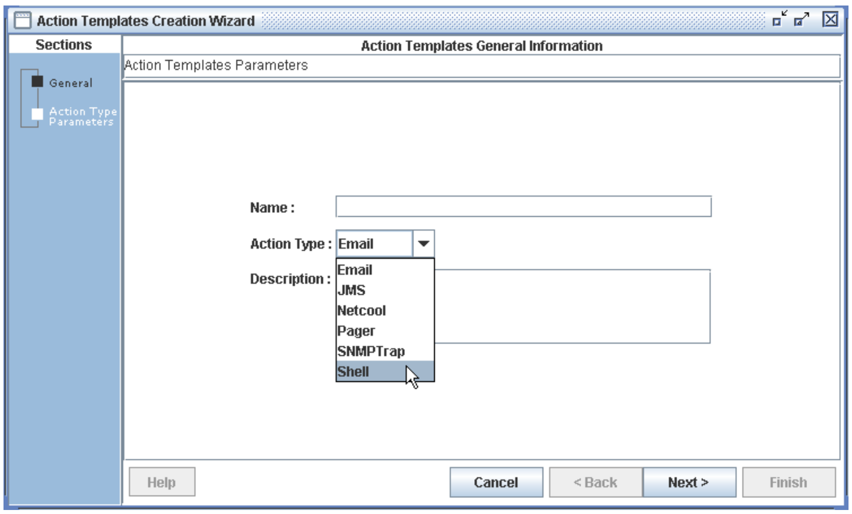
Following is an example of using a shell action template to run a shell script for violated policies.

To add a shell action template record

1 From the **Catalog** menu, select **Actions**.

2 Click **Add**.

The following screen depicts selecting the *Shell Action Template.*

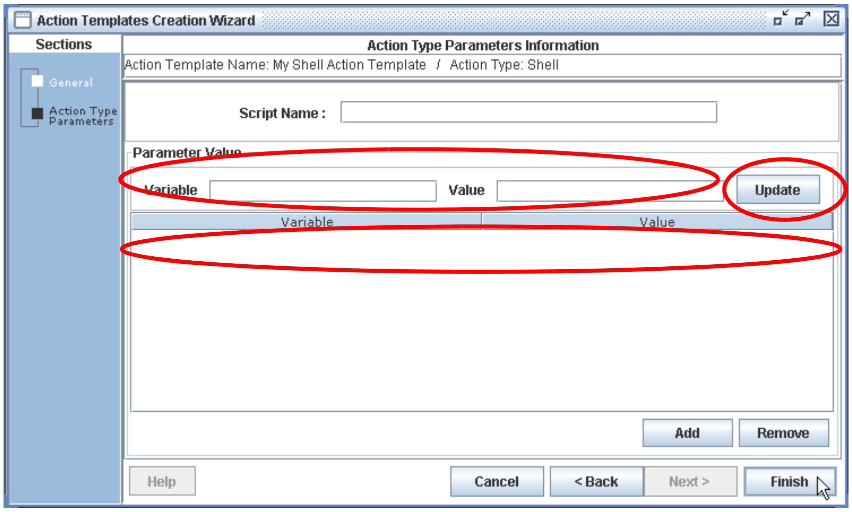


3 Enter an action template name in the **Name** field.

4 Select **Shell** from the **Action Type** drop-down list.

5 In the **Description** field, enter additional comments, and click **Next**.

A screen similar to the following displays:



6 Enter parameter information by:

a. Click **Add.**

b. In the **Variable** field, type a Parameter name.

c. In the **Value** field, type a value for the Parameter and click **Update,** or

d. Select a **Variable** name and modify the **Value** and click **Update** when "editing" a parameter value(s).

7 When you have completed adding/editing the Action Template, click **Finish**.

Editing Action Template Records

You can edit existing action template records after generating an action template list. The edit feature allows you to make modifications to certain action template fields.

To edit an action template record

1 Select, or highlight, the row corresponding to the action template to be edited.

2 Click **Edit**.

3 Click **Next** and **Back** to navigate through the display screen and make the appropriate edits.

Note: To change the action parameters, see "Updating Action Parameters".

4 Click **Finish** to save your edits.

Viewing Action Template Records

You can view the details of an Action template in read-only format.

To view an action template record

1. Select, or highlight, the row corresponding to the action template to be displayed.
2. Click **View**.
3. Click **Next** and **Back** to navigate through the display screen.
4. Click **Cancel** to exit the screen.

Deleting Action Template Records

You can remove, or delete, an action template record from the database.

To delete an action template record

1. Select, or highlight, the row corresponding to the action template to be deleted.
2. Click **Delete**.

   A confirmation message asks "Are you sure to delete the selected item?"
3. Click **Yes** to confirm the deletion.

Updating Action Parameters

Use this procedure to update action parameters while adding or editing action template records.

To update e-mail action type parameters

1. Generate a list of e-mail action types.
2. Select an e-mail parameter row.
3. Click **Edit**.

   The Action Templates General Information screen is displayed.
4. Click **Next**.

   The Action Type Parameters Information screen is displayed.
5. Select a variable to be updated.

   Note: The Variable box at the top left shows the row selected
6. Enter the appropriate information in the Value box and click **Update** to add it to the record.

   Note: You can specify the following e-mail fields: To (Required), Subject, CC, BCC, and Body.
7. Click **Finish**.

To update network fault management action type parameters

1 Generate a list of NFM action types.

2 Select an **e-NFM** parameter row.

3 Click **Edit**.

The Action Template Modification screen is displayed.

4 Click **Next**.

The Action Type Parameters Information screen is displayed.

5 Select a variable to be updated.

Note: The Variable box at the top left shows the row selected.

6 Select an appropriate value from the pull-down list and click **Update** to add it to the record.

Note: You can specify the following variables: Severity and Consolidate on Customer, Service, Device, or Interface.

7 Click **Finish**.

To update pager action type parameters

1 Generate a list of pager action types.

2 Select a **pager** parameter row.

3 Click **Edit**.

The Action Template Modification screen is displayed.

4 Click **Next.**

The Action Type Parameters Information screen is displayed.

5 Select a variable to be updated.

Note: The Variable box at the top left shows the row selected.

6 Select an appropriate value from the pull-down list (or enter the value manually, where appropriate) and click **Update** to add it to the record.

Note: You can specify the following Pager fields: Pager Number (Required), Call Back Number, Dial Option, Outside Line, and Device.

7 Click **Finish**.

To update shell action type parameters

1 Generate a list of shell action types.

2 Select a **shell** parameter row.

3 Click **Edit**.

The Action Template Modification screen is displayed.

4 Click **Next**.

The Action Type Parameters Information screen is displayed.

5 Select a variable to be updated.

Note: The Variable fields display the row selected.

6 Update the text in the Variable box.

7 Enter the value in the Value box for this variable.

8 Click **Update** to change the Variable/Value pair in the selected row.

9 Repeat steps 2- 8 until you have entered all Variable/Value pairs for the shell script.

10 Click **Finish**.

Managing Policy Templates

Policy templates are an essential aspect of managing network performance since they let you virtually represent the business rules associated with the performance objectives in PerforMAX. In other words, policy templates create a series of conditions or rules for events.

For instance, if a certain condition is met, such as the throughput rises above a certain level, PerforMAX will page your system administrator automatically. The action, paging your system administrator, derives from an action template that you have already established. The policy templates merely define the conditions.

As the following figure illustrates, action templates are not only built before you create policy templates, but they also require you to reference policy templates. The creation of policy templates may contain one or more action templates or no action templates at all.

.

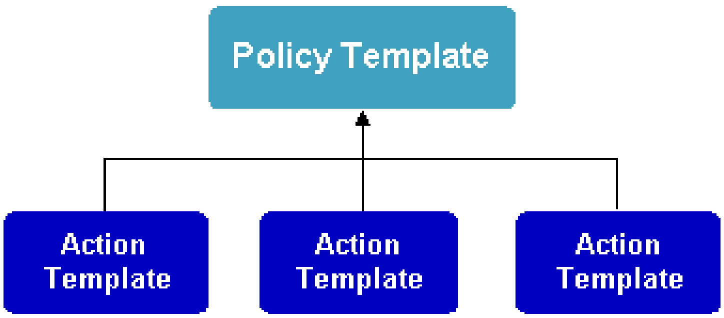


Figure 4-1: Policy Template Workflow

When you create policies, you are defining acceptable levels of service quality. Policies take the data collected and assess it in relation to the conditions that you have set that define quality metrics.

Here are some examples of the ways you might want to use policies:

- **Priority classes based on frequency of events**
  For example, the front office may create a policy to escalate the priority of an alert if performance degradation occurs over a specified period of time. This is accomplished by creating a policy that includes a performance condition, occurrence over a certain time frame, and alert priority.

When you create a policy template, you must first specify a performance class (and its corresponding connection type) and effective time. The policy template's conditions can consist of any of the KPIs supported by the associated performance class. PerforMAX supports the following:

- A one to many association of action templates to conditions
- Conditions to action templates
- Rules to policy templates

PerforMAX lets you create, modify, and view policy templates. When creating a policy template, you must enter a name and description as well as associate a performance class and connection type.

Then, you must define when the policy template will be active. This includes specifying the effective time's period, range of reoccurrence, and recurrence pattern for that policy template. This information determines when the policy template is active.

For example, you can specify that the template will become active after the first thirty days of service (effective period) between 12:00 AM and 11:59:59 PM (range of recurrence) every weekday (recurrence pattern).

You can also associate billing penalties with policy templates. For example, you might want to have a penalty appear on the bills of any customers who use excess service.

When you include a billing penalty in a policy template, all customers who are associated with the offering in which the policy template was used will be penalized automatically.

Finally, you can create rules by associating conditions (for example, If Availability > 99.9%) with action templates (for example, e-mail, script, and so on).

Policy Template Procedures

You can perform the following procedures on policy templates:

- Displaying a Policy Template List
- Adding Policy Template Records
- Editing Policy Template Records
- Viewing Policy Template Records
- Deleting Policy Template Records

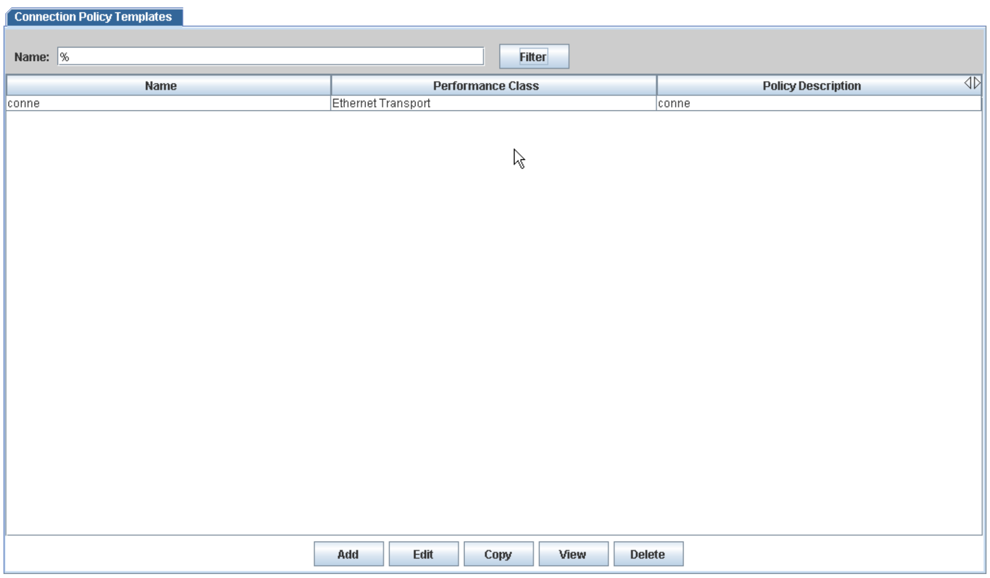
Displaying a Policy Template List

You can display a list of connection or performance policy templates based on the policy template name and policy type.

To display a connection policy template list

1. From the **Catalog** menu, select **Conn. Policy Templates**.
2. In the **Name** box, enter the policy template name or (to list all templates), accept the default (**%**).
3. Click **Filter**.

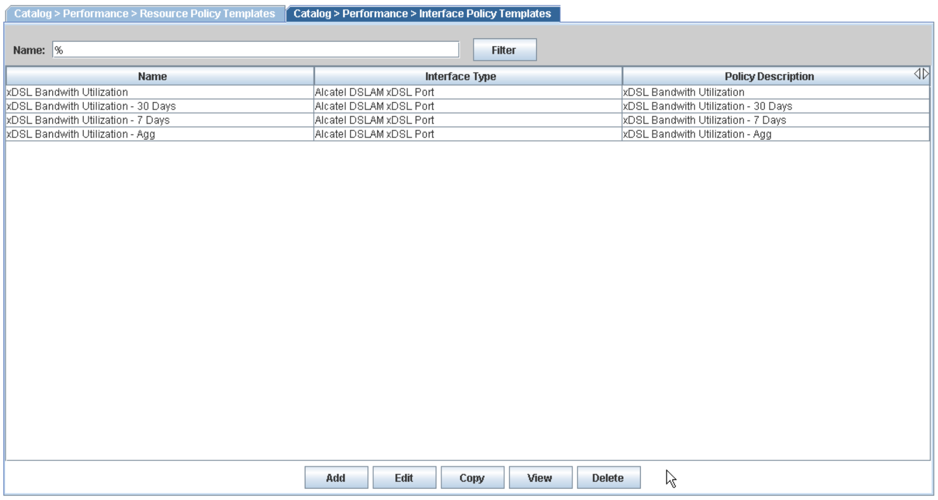
A screen similar to the following is displayed:



To display a performance policy template list

1. From the **Catalog** menu, select **Perf. Policy Templates**.
2. Select the **Resource** or **Interface Policy Templates** tab.
3. In the **Name** box, enter the policy template name or, to list all templates, enter a percentage sign (**%**).
4. Click **Filter**.

A screen similar to the following is displayed:



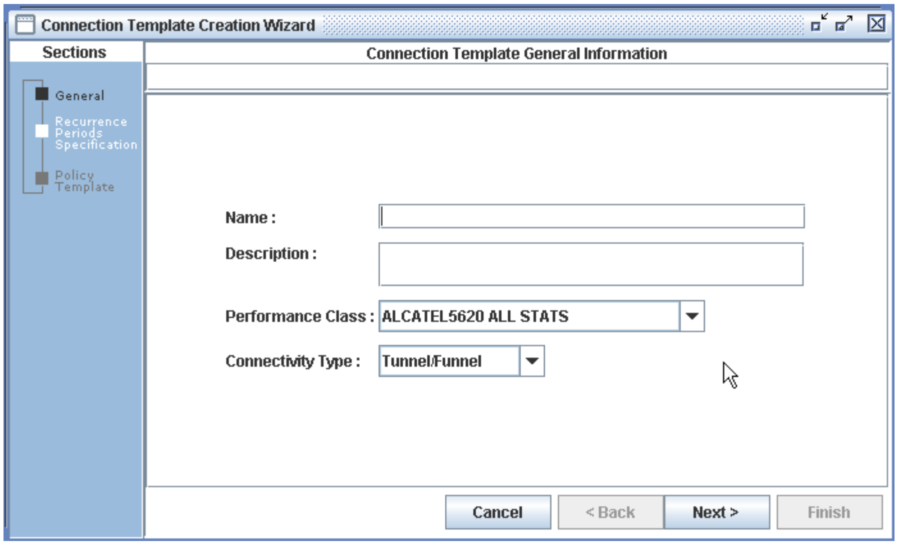
Adding Policy Template Records

You can add a connection or performance policy template record to the database. This includes associating a performance class, connection type, effective period and policy rules. Policy template rules include associating a performance condition with threshold value and an action template.

To add a policy template record, general (part A)

1. Go to the Connection or Performance Policy Templates screen, depending on which type of policy template you want to add.
2. Click **Add**.

A screen similar to the following is displayed:



3 Enter a template name in the **Name** field.

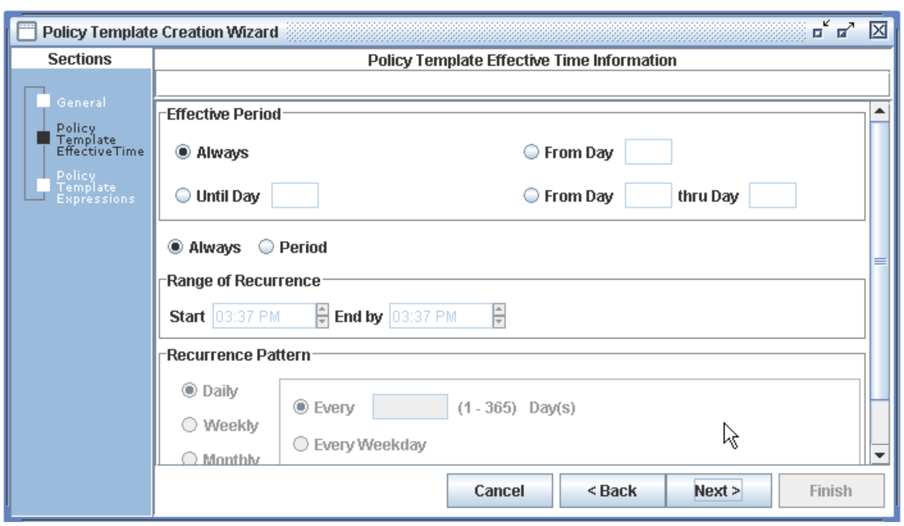
4 In the **Description** field, enter the description.

5 Select a performance class from the **Performance Class** drop-down list.

6 Select a connectivity type from the **Connectivity Type** drop-down list.

7 Click **Next**.

A screen similar to the following is displayed:



8 Go to "To specify policy template effective time (part B)".

To specify policy template effective time (part B)

1 Under **Effective Period**, select the policy's effective period (**Always**, **Until Day ___**, **From Day ___**, or **From Day ___ Thru Day**).

2 Above the **Range of Recurrence** box, select one of the radio buttons: **Always** or **Period**.

If you select...	then go to...
Always	"To specify policy template effective time (part B)" step 3.
Period	"Setting Time Periods"

Setting Time Periods

a In the **Range of Recurrence** box, select a **Start** and **End by** time.

b In the **Recurrence Pattern** box, select one of the radio buttons to specify when policy recurrence is: **Daily**, **Weekly**, **Monthly**, or **Yearly**.

If you select...	then also specify...
Daily	• **Every ___ of days**. Enter the number of days to pass before resource monitoring occurs again, **OR** • **Every Weekday.** Use this selection to specify that resource monitoring will occur every day except weekends and holidays.
Weekly	• **Recur Every ___ (1-52) Weeks on**. Enter the number of weeks to pass before resource monitoring occurs again, **AND** • **Sunday - Saturday**. Select day(s) of the week on which resource monitoring is to occur.
Monthly	• **Day**. Enter the day number (day of the month for a one month period) or the appropriate day number) if a period longer than one month is selected), **OR** • Select the **First - Last** drop-down list box to specify the day of the week and if it is to occur **Every (1-12) Month**.
Yearly	• Select the month from the drop-down list box (**January - December**) and enter the day of the month resource monitoring is to occur, **OR** • Select the **First - Last** drop-down list box to specify the day of the week and the month it is to occur (**January - December**).

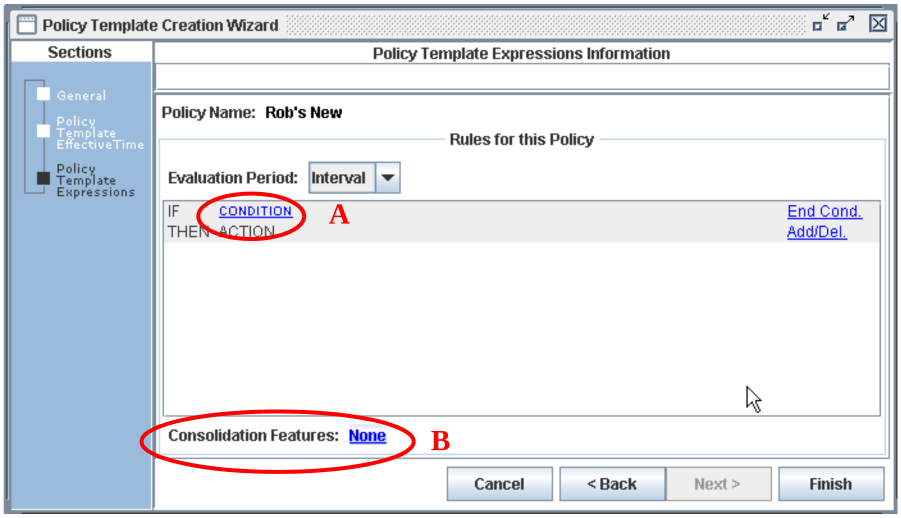
3 Click **Next**. Go to "To add policy template expressions (part C)".

To add policy template expressions (part C)

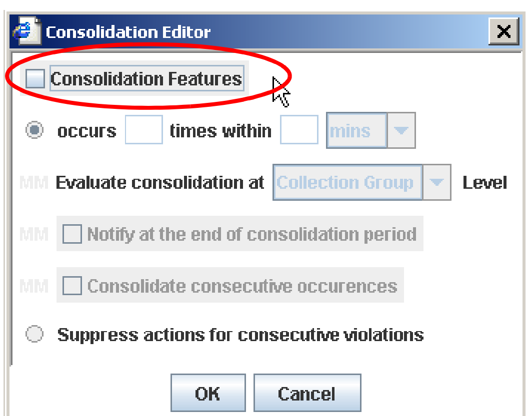
1 Create a rule:

a. Create a condition:

1) Click **Condition** to display the Condition Editor (A in the following graphic).



2) Create the appropriate condition components by selecting a time period, Quality of Service metric, operator, and threshold value.
3) You can consolidate alerts generated by the rule by selecting the **Consolidation Features** link (B in the above graphic).
4) The **Consolidation Editor** dialog box is displayed. Select the appropriate consolidation options (if necessary).

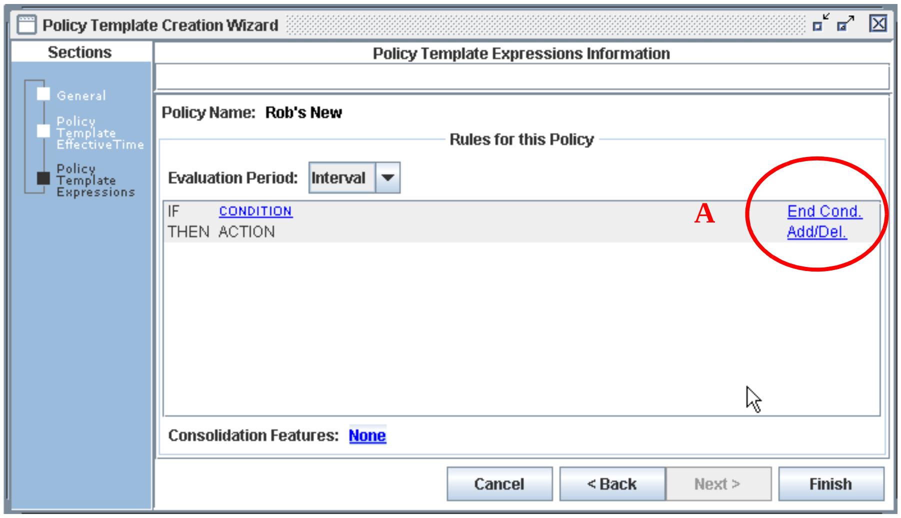


5) Click the **Consolidation Features** check box.
6) Enter a value for **occurs _ times within**.
7) Select a value from the time period drop-down list.
8) If “**occurs *n* times within *n* mins**” is selected, continue with selecting the **Evaluate consolidation at** the **Collection Group** or **Interface** level using the drop-down list.
9) Select/clear the **Notify at the end of the consolidation period** check box.
10) Select/clear the **Consolidate consecutive occurrences** check box, or
11) If **for n% of all Services** is selected, enter the desired percentage, or

12) Select the **Suppress actions for consecutive violations** option button.

13) Click **OK** to close the Condition Editor.

14) Click **End Cond** to display selections.



15) Select **AND** or **OR** to add another condition to the rule, **End Cond.** to end the condition, or **Delete Cond.** to delete the selected condition (A in the above graphic).

b. Associate an action with the condition.

1) Click **Add/Del** to display the Action Editor.

2) Under **Available Actions**, select the action template(s) to associate with and click **Add**.

•**Remove -** removes the highlighted selection from the **Selected Actions** dialog.

•**All -** adds all **Available Actions** to the **Selected Actions** dialog.

•**Clear -** clears all **Selected Actions**

3) Click **OK**.

2 After creating the rules to include in the template, click **Finish**.

Note: Use the previous procedures to edit and/or add conditions to the Policy Template.

Editing Policy Template Records

You can edit existing policy template records after generating a policy template list. The edit feature allows you to make modifications to certain fields.

To edit a policy template record

1 Select, or highlight, the row corresponding to the policy template to be edited.

2 Click **Edit**.

3 Click **Next** and **Back** to navigate through the display screen and make the appropriate edits.

4 Click **Finish** to save your edits.

Viewing Policy Template Records

You can view the details of a policy template record after generating a policy template list. This features lets you view the record in read-only format.

To view a policy template record

1. Select, or highlight, the row corresponding to the policy template to be displayed.
2. Click **View**.
3. Click **Next** and **Back** to navigate through the information screen.
4. Click **Cancel** to exit the screen.

Deleting Policy Template Records

You can remove, or delete, a policy template record from the database.

To delete a policy template record

1. Select, or highlight, the row corresponding to the policy template to be deleted.
2. Click **Delete**.

   A confirmation message appears asking "Are you sure you want to delete the selected item?"
3. Click **Yes** to confirm the deletion.

Condition Editor

You can customize conditions by using the Condition Editor. A condition consists of a time interval (for example, daily, monthly, and so on), KPI (for example, VCC CDR, Availability, and so on), and an operator (for example, greater than, less than, equal to, and so on). For example, a condition could consist of the following syntax:

```
If Daily Ethernet Throughput is less than 95%
```

In addition, the Condition Editor supports consolidation of faults so that the Network Operations Center is not bombarded with multiple alerts generated by the same condition. For example, you can include multiple occurrences of events within a specified time period as illustrated by the following syntax:

```
If Daily Ethernet Throughput is less than 95 and this occurs 3 times within the
next 5 minutes
```

Creating Multiple Conditions

You can also manage conditions through the use of the mathematical operators AND and OR. The result is a multi-condition rule that increases or decreases the sensitivity of the rule. For example, using AND to create a multi-condition rule increases the rule's sensitivity to performance degradation as AND requires both conditions to be true for the condition to be true as illustrated in the following syntax:

```
If Daily Ethernet Throughput is less than 95
AND
If Daily Ethernet Availability is less than 95
```

Conversely, OR decreases a rule's performance degradation sensitivity as it only requires one of the conditions to be true as illustrated in the following syntax:

```
If Daily Ethernet Throughput is less than 95
OR
If Daily Ethernet Availability is less than 95
```

Consolidation also enables you to create different levels of sensitivity to performance degradation. For example, you can create variations of the same condition to create a performance degradation hierarchy based on the relative importance of a subscriber. For example, the following condition:

```
If Daily Ethernet Throughput is less than 95 and this occurs 2 times within the
next 3 minutes
```

is more sensitive than a condition that states:

```
If Daily Ethernet Throughput is less than 95 and this occurs 2 times within the
next 10 minutes
```

By including a more stringent consolidation condition, the policy becomes more sensitive to performance degradation.

The Condition Editor also supports consolidation based on the following:

- time (that is, only send alert at end of consolidation period)
- consecutive occurrences (that is, do not send consecutive alerts for the same rule)
- percentage of total connections affected (for example, this rule occurred for 75% of all active connections)

In addition, the Condition Editor supports consolidation at the collection group and connection levels (evaluate the condition for each collection group or connection, respectively).

After creating a condition, you must select an action to complete the rule. The Condition Editor allows you to associate one or more actions with a condition.

An action instructs PerforMAX to perform a specific behavior if the associated condition is met (for example, e-mail bob_smith@yourdomain.com, run copy.sh script, and so on).

After selecting the appropriate action(s), the rule is complete. A policy template supports multiple rules as illustrated in the following syntax:

```
IF Daily VCC Availability is less than 99.99%

AND

IF Daily Utilization is greater than 90% 5 times within a five minute
period
```

```
THEN
E-Mail bob.smith@yourdomain.com
IF Monthly Utilization is greater than 95% for 30% of all interfaces
OR
IF Monthly VCC Availability is less than or equal to 98
AND
IF Monthly Utilization is greater than 90% for 30% of all interfaces
THEN
Page 555.555.5555
```

Summary

Once added to PerforMAX, the policy template is displayed in the Policy Templates tab. In addition, PerforMAX integrates performance management with fault management enabling you to correlate resource faults (for example, application, network, server) with domains and connections and vice versa. You can add, display, and delete policy templates; you cannot edit them.

Managing Connection Templates

PerforMAX lets you create a variety of connection templates. Connection templates allow you to customize condition threshold values. These templates are comprised of one or more policy templates. When completed, connection templates are the basis upon which CGs (Connection Groups) are created: connection templates are instantiated every time you create aCG.

When you reference a connection template, the template is instantiated. That is, the information contained in the template is reproduced as part of the agreement.

You can, therefore, alter the policies associated with a specific offering without altering the original template.

Consequently, you can reference a connection template that doesn’t have the exact policies that you need to save time and then make changes to the instantiated version when you create the new agreement.

As a service provider, you can create connection templates that encompass policy templates and include parameter ranges. The policy templates also specify actions to be taken if it does not meet specific objectives.

While creating a connection template, you must also specify the policy templates which include the KPIs on which the policies are built. You can also add more KPIs to be calculated for the template.

PerforMAX lets you create and view connection templates. You must specify the template's name, description, type, and connection type. You also need to specify the following:

- Effective time — when the policy is active
- Exclusion time — when the policy is inactive
- Maintenance time — when the policy is inactive due to service maintenance

In addition, for each time, you define a range of reoccurrence (for example, between 9:00 AM and 5:00 PM) and a recurrence pattern (for example, every weekday).

After you have defined these parameters, you must specify the policy template(s) to associate with the connection template. Connection templates can only include policy templates based on the same performance class. As a result, you will not be able to select templates for other performance classes.

You can see all the policy rules associated with the selected policy templates in the Policy Templates Condition Thresholds screen. You can customize these conditions by modifying the threshold value(s) for the condition(s) within each rule.

Connection Template Procedures

You can perform the following procedures on connection templates:

- Displaying a Connection Template List
- Adding Connection Template Records
- Editing Connection Template Records
- Viewing a Connection Template
- Copying Connection Template Records
- Deleting Connection Template Records

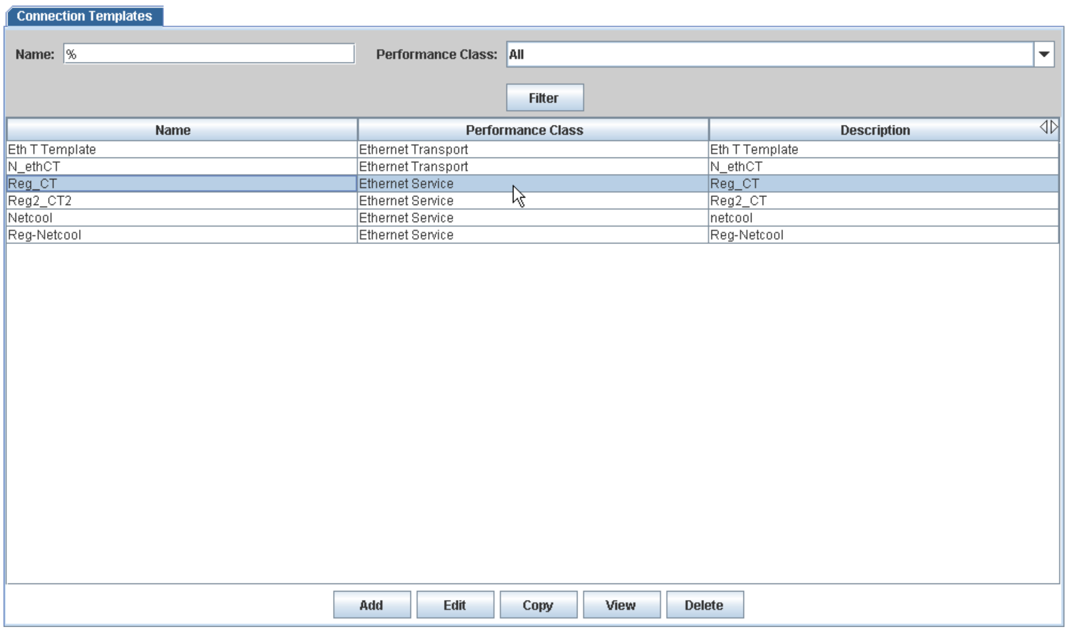
Displaying a Connection Template List

You can display a list of connection templates that meet a specific criteria using the filtering feature.

To display a connection template list

1. From the **Catalog** menu, select **Connection Templates**.
2. In the **Name** box, enter the connection template name or (to list all templates) accept the default (**%**).
3. Select the **Performance Class** from the drop-down list.
4. Click **Filter**.

A screen similar to the following is displayed:



Adding Connection Template Records

You can add a connection template record to the database provided that you follow all procedures and sub-procedures. You can also modify a referenced policy template's threshold value(s) to further customize the connection template.

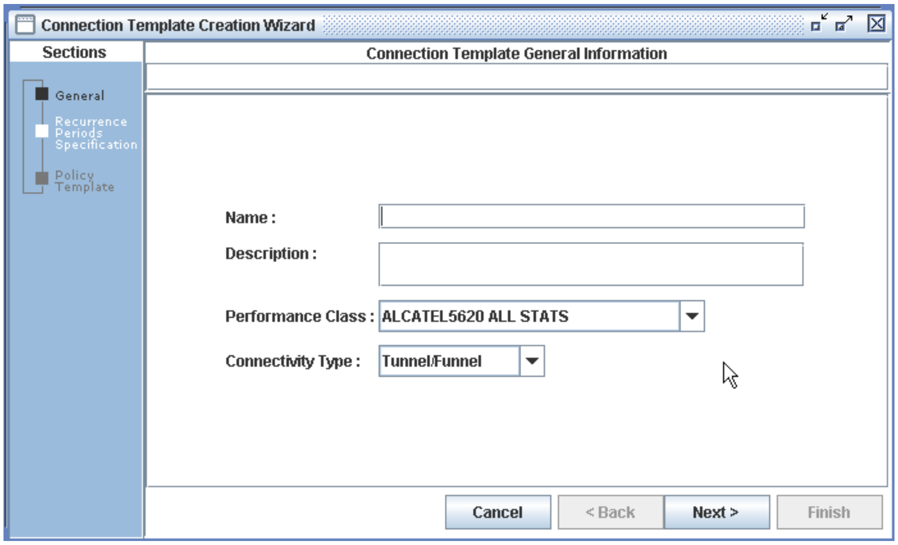
To add a connection template record (part A)

1 From the **Catalog** menu, select **Connection Templates**.

The default screen is displayed.

2 Click **Add**.

The Connection Template General Information screen is displayed:



3 Enter a template name in the **Name** field.

4 In the **Description** field, type a description of the template.

5 Select a **Performance Class** from the drop-down list.

6 Select a **Connectivity Type** (if applicable) from the drop-down list.

7 Click **Next**. Go to “To specify an effective time (part B)”.

The Recurrence Periods Specification Information screen is displayed.

To specify an effective time (part B)

1 Click the **Effective Time** tab.

2 Click one of the following option buttons:

- **Always** — indicates the template will always be effective.
- **Never** — indicates the template will never be effective.
- **Period** — indicates the template will only be effective during the specified period.

3 Do one of the following:

If you selected the...	then...
Always option button	go to step 7
Never option button	go to step 7
Period option button	go to step 4

4 In the **Range of Recurrence** section, type the time at which the policy will become effective in the **Start** field.

5 In the **End by** field, type the time at which the policy will stop being effective.

6 In **Recurrence Pattern** section, select one of the following option buttons:

- **Daily** — sets data collection to occur on a daily basis.

  If you selected Daily, select one of the following:

  - **Every** — specifies that data collection will occur every day in the week including weekends.
  - **Every Weekday** — specifies that data collection will only occur from Monday to Friday.

- **Weekly** — sets data collection to occur on a weekly basis.

  If you selected Weekly, do the following:

  1) Type a value in the **Recur Every** text box to represent the number of weeks to elapse before the connection template becomes active again.
  2) Select the day of the week the connection template is to become active.

- **Monthly** — sets data collection to occur on a monthly basis.

  If you selected Monthly, select one of the following option buttons:

  - **Day**
  - **The**

- **Yearly** — sets the data collection to occur on a yearly basis.

  If you selected Yearly, select one of the following option buttons:

  - **Every**
  - **The**

7 Do one of the following:

If you want to...	then...
specify a time that you do not want the policy to run	go to "To specify an exclusion time (part C)".
specify a time that you do not want the policy to run because of maintenance	go to "To specify a maintenance time (part D)".
Finish	click **Finish**.

To specify an exclusion time (part C)

1 Click the **Exclusion Time** tab.

2 Repeat steps 2 to 6 from the "To specify an effective time (part B)" sub procedure.

3 Do one of the following:

If you...	then...
want to specify a time that the connection template will not be available because you are performing network maintenance	go to "To specify a maintenance time (part D)".

If you...	then...
do not want to specify a maintenance time	click **Next**, and go to “To add a policy template (part E)”.

4 To proceed, continue on to “To specify a maintenance time (part D)”.

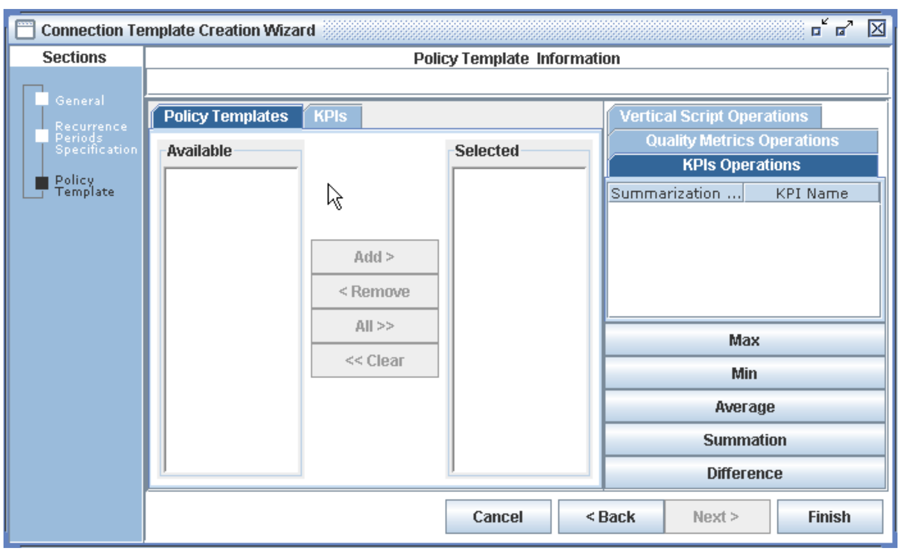
To specify a maintenance time (part D)

1 Click the **Maintenance Time** tab.

2 Repeat steps 2 to 6 from the “To specify an exclusion time (part C)” sub procedure.

3 Click **Next**, and go to “To add a policy template (part E)”.

A screen similar to the following is displayed:



To add a policy template (part E)

1 Click the **Policy Templates** tab.

2 In the **Available Policy Templates** column, select the policy template, and click **Add**.

The policy template that you selected will appear in the Selected column.

3 Repeat step 2 to include additional policy templates.

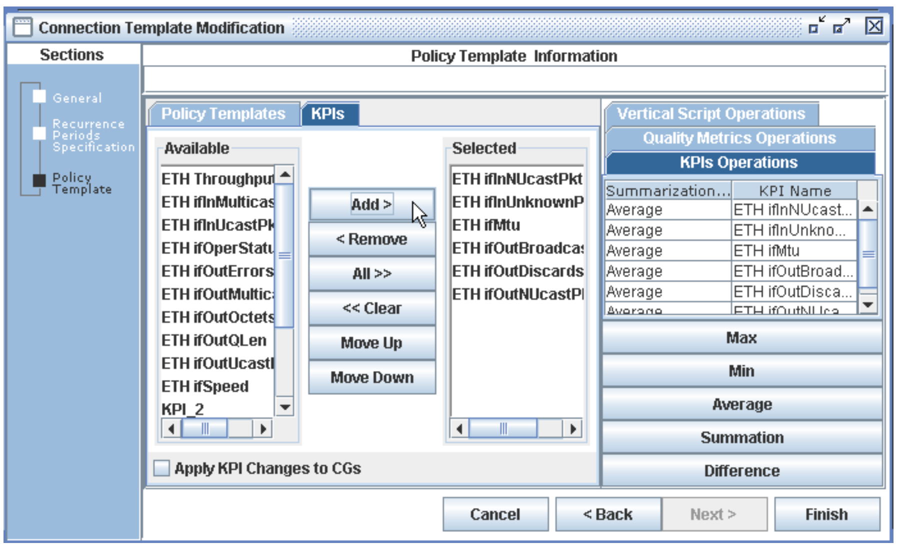
4 Do one of the following:

If...	then...
you want to add a KPIs independently of its associated policy to the connection template	go to "To add a Key Performance Indicator (KPI)".
you do not want to add any additional KPIs to the connection template	go to step 6.

To add a Key Performance Indicator (KPI)

a. Navigate to the Policy Template Information screen.

A screen similar to the following is displayed:



b. Click the **KPIs** tab.
c. In the **Available** list, select the KPI to add to the connection template, and click **Add**.
d. Repeat step c to include additional KPIs.

- Click **All** to move all KPIs from the Available list to the Selected list.
- Click **Clear** to clear all KPIs in the Selected list.
- Click **Remove** to remove a single KPI in the Selected list.
- Click **Move Up** to move a KPI up in the Selected list.
- Click **Move Down** to move a KPI down in the Selected list.

5 To edit **KPI Operations, Quality Metric Operations,** or the **Vertical Script** aggregation methods**:**

a. Click the appropriate tab, on the right side of the window
b. For each KPI displayed on the tab, select the KPI in the window and select an aggregation method. by clicking the appropriate aggregation button.

c. Repeat step to modify each desired KPI.

Note: KPI Operations is the summarization over time. “How do you want the interval data summarized at the end of the day and the month?”

Note: Quality Metric Operations is the summarization over points. For clouds and funnels, (only the source side), data is collected from multiple places but needs to be summarized into one value for the interval. This provides, “how do you want the multipoints summarized per interval?”

6 Click **Next**

The Policy Templates Conditions Thresholds wizard is displayed.

To change the threshold values in a condition

1 If necessary, modify threshold values of associated conditions. If you do not want to modify any threshold values, go to step 2.

a. Select the Policy Template to be modified

b. Click the condition portion of a rule.

c. In the threshold box, type the appropriate value.

d. Click **OK**.

e. Repeat step a. to step c. for all of the threshold values to be modified.

2 Do one of the following:

If...	then...
you want to apply the threshold changes to individual CGs	go to “To apply KPI changes to a Connection Group”.
you want to apply the changes to this connection template	click **Finish**. The procedure is complete.

Editing Connection Template Records

You can edit a connection template record using the following procedures. You can also modify associated threshold value(s) to further customize the policy.

To edit a connection template record

1 Select a connection template from the list and click **Edit**.

The Connection Template General Information screen is displayed.

2 Edit the **Name** and/or **Description** fields, if desired.

3 Click **Next**.

The Recurrence Periods Specification Information screen is displayed.

4 Do one of the following:

If...	then...
you want to edit effective times	go to "To specify an effective time (part B)".
you want to edit exclusion times	go to "To specify an exclusion time (part C)".
you want to edit maintenance time	go to "To specify a maintenance time (part D)".

5 After reviewing or editing the contents of the Recurrence Periods Specification screen, click **Next**.

6 The Policy Template Information screen displays. Do one of the following:

If...	then...
you want to add policy templates	go to "To add a policy template (part E)".
you want to edit a KPI list	go to "To edit a KPI list".
you want to apply KPI changes to a CG	go to "To apply KPI changes to a Connection Group".
you do not make changes	go to step 7.

7 Click **Next**.

The Policy Templates Conditions Thresholds Information screen is displayed.

8 Go to the next procedure.

To change the threshold values in a condition

1 If necessary, modify threshold values of associated conditions. If you do not want to modify any threshold values, go to step 2.

a. Click the condition portion of a rule.
b. In the threshold box, type the appropriate value.
c. Click **OK**.

d. Repeat step a. to step c. for all of the threshold values to be modified.

2 Do one of the following:

If...	then...
you want to apply the threshold changes to individual CGs	go to the next procedure.
you want to apply the changes to this connection template	click **Finish**. The procedure is complete.

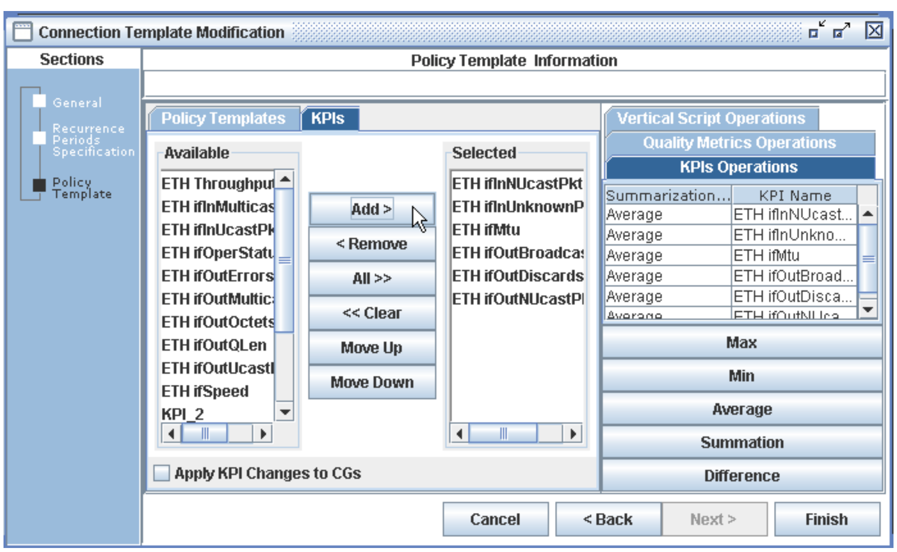
To edit a KPI list

1 Select a connection template from the list and click **Edit**.

The Connection Template General Information screen is displayed.

2 Click **Next** until you get to the Policy Template Information screen. Click the **KPIs** tab.

The KPI list for the Connection Template is displayed:



3 Select the Available KPIs and click **Add** to add them to the Selected KPIs list.

- Click **All** to move all KPIs from the Available list to the Selected list.
- Click **Clear** to clear all KPIs in the Selected list.
- Click **Remove** to remove a single KPI in the Selected list.
- Click **Move Up** to move a KPI up in the Selected list.
- Click **Move Down** to move a KPI down in the Selected list.

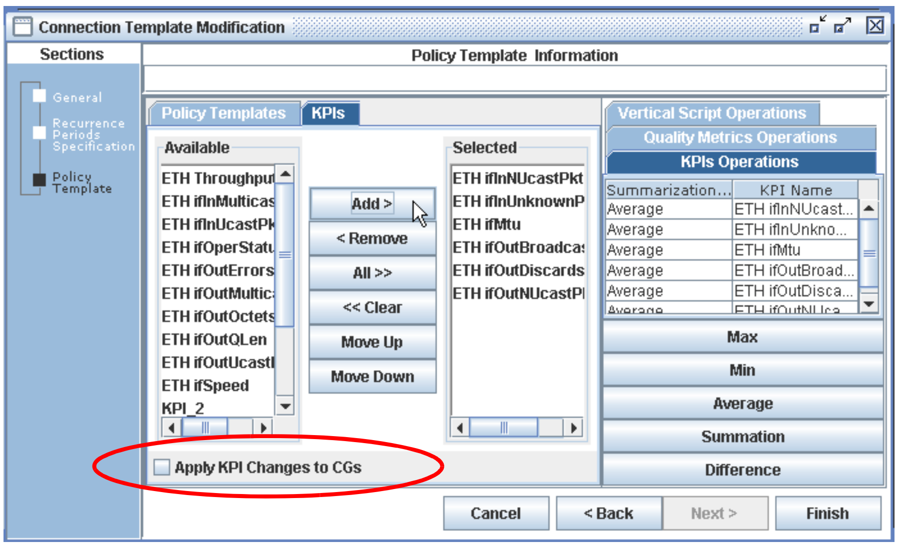
4 After you have selected the KPIs to modify, click **Next**.

5 Click **Finish** to complete the changes.

To apply KPI changes to a Connection Group

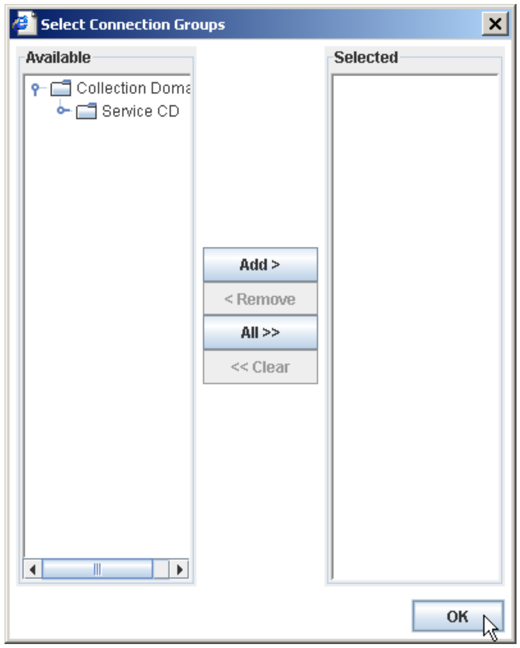
1 From the Policy Template Information screen, click the **KPIs** tab.

The following screen displays:



2 Select the **Apply KPI Changes to CGs** check box.

The Select CG dialog box is displayed:



3 Expand the Available CGs pane on the left and select the CG titles that you want to add the KPI changes to.

4 Click **Add >**. The CG titles are now selected. These CGs will have the KPI changes added to them.

- Click **All** to move all CGs from the Available list to the Selected list.
- Click **Clear** to clear all values in the Selected list.
- Select an CG in the Selected CGs pane and click **Remove** to remove a single CG.

5 When you have selected the CGs to modify, click **OK**.

6 Click **Finish** to complete the changes.

Add a New Policy Template to a Connection Template

1 Add a new policy template. See "Adding Policy Template Records" for detailed information.

2 Select **Catalog** > **Connection Templates**. Click **Filter**.

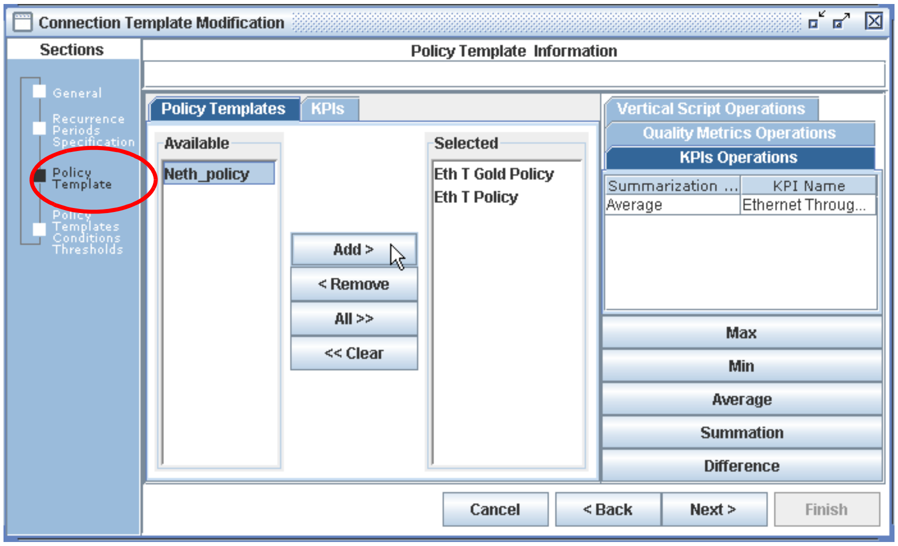
The Connection Template list is displayed.

3 Select a connection template from the list and click **Edit**.

The Connection Template General Information screen is displayed.

4 Click **Next** until you get to the Policy Template Information screen.

A screen similar to the following is displayed:



5 Select the **Policy Templates** tab and select a policy template from the **Available** pane. Click **Add**.

6 Click **Next** and click **Finish**.

Viewing a Connection Template

You can view the details of a connection template in read-only format after generating a template list.

To view a connection template

1 Select, or highlight, the row corresponding to the template to be viewed.

2 Click **View**.

3 Click **Next** and **Back** to navigate through the screens.

4 Click **Cancel** to exit the screens.

Copying Connection Template Records

Instead of creating all of your templates “from scratch,” you can copy an existing connection template and then make changes to it to form your new template.

To copy a connection template record

1 Select, or highlight, the row corresponding to the template to be copied.

2 Click **Copy**.

3 Change the **Name** and/or **Description** fields.

4 Click **Next**.

5 Make all other changes, as needed.

6 Click **Finish** to save as a new connection template.

Deleting Connection Template Records

You can delete a connection template from the database, provided you have not already referenced the template elsewhere.

To delete a connection template record

1 Select, or highlight, the row corresponding to the template to be deleted.

2 Click **Delete**.

A confirmation message appears asking “Are you sure to delete the selected item?”

3 Click **Yes** to confirm the deletion or **No** to cancel the deletion process.

Uneven Collection Cycles

Uneven collection occurs when a collection is triggered for a Connection with a collection cycle that is a multiple of the collection cycle supported by at least one endpoint (a.k.a. Service Access Point (SAP)).

See Appendix B, “Uneven Collection Cycles” for detailed information.

Managing Performance Templates

PerforMAX lets you create a variety of performance templates. Performance templates allow you to customize condition threshold values.

While creating a performance template, you must also specify the policy templates which include the KPIs on which the policies are built. You can also add more KPIs to be calculated for the performance template.

PerforMAX lets you create and view performance templates and perform template related tasks, including:

- Displaying a Performance Template List
- Adding/Editing/Viewing/Copying/Deleting Performance Template Records

Performance Template Procedures

You can perform the following procedures on performance templates:

- Displaying a Performance Template List
- Adding/Editing/Viewing/Copying/Deleting Performance Template Records

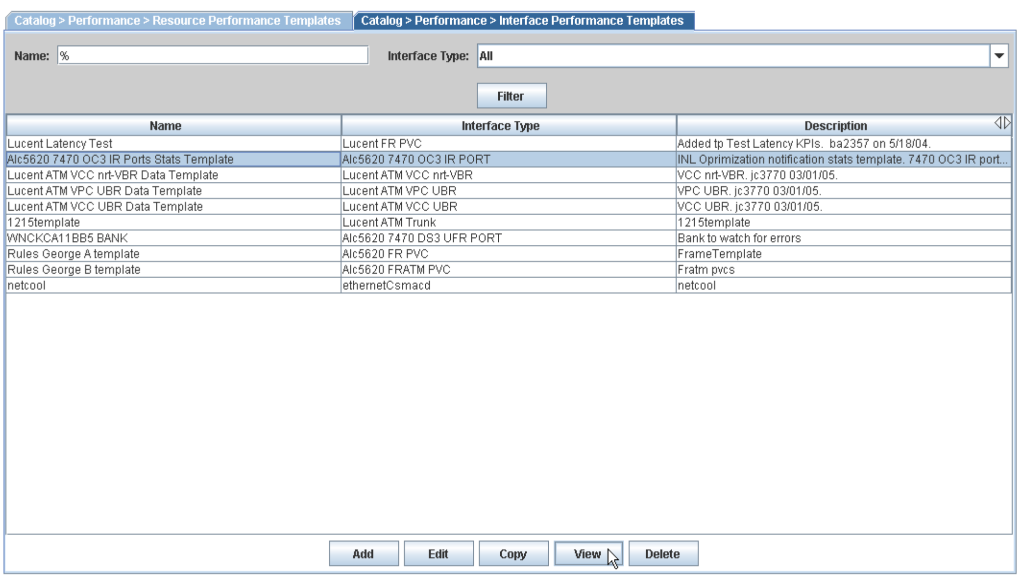
Displaying a Performance Template List

You can display a list of performance templates meeting specific criteria using the filtering feature.

To display a performance template list

1. From the **Catalog** menu, select **Performance Templates**.
2. Select the **Catalog > Performance Resource/Interface > Performance Templates** tab.
3. In the **Name** box, enter the performance template name or, to list all templates, enter a percentage sign (**%**).
4. Select the **Resource** or **Interface Type** from the drop-down list.
5. Click **Filter**.

A screen similar to the following is displayed:



Adding/Editing/Viewing/Copying/Deleting Performance Template Records

You can add, edit, view, copy or delete performance template records using the same procedures as when working with connection template records. In Performance Templates, you use the Add, Edit, Copy, View and Delete buttons at the bottom of the screen.

See the following topics for detailed information.

- Adding Connection Template Records
- Editing Connection Template Records

- Viewing a Connection Template
- Copying Connection Template Records
- Deleting Connection Template Records

Resources Monitoring

5

The following documentation provides information and procedures for monitoring resources.

The topics in this documentation include the following:

What is Resource Performance?

After your resources (for example, devices, support systems, interfaces) are registered with PerforMAX, you can then monitor them at the physical level. This allows you to monitor device and interface performance independent of the services they support.

PerforMAX lets you monitor your resources in the following ways:

- You can generate a report for a specific period in time
- You can monitor them dynamically in real-time (so that you can see events as they occur)

When you choose to monitor in “real time,” PerforMAX displays all of the data points on the report in the reporting interval.

For example, if you have 15 minutes set as your reporting interval, then PerforMAX will display all of the data points that are collected in that period.

If you only want to see an average of the data points for a 15 minute period, select 15 minutes as the collection interval in a real-time report.

The data points have labels associated with them. Essentially, the real-time option shows time-series data points rather than aggregated auto-collection intervals.

Evaluating Resources

PerforMAX’s functionality lets you evaluate the performance of your resources. You can evaluate performance by cross-referencing and combining information specific to certain types of services.

For example, you can analyze services in relation to other similar services, in relation to the other services you are selling to a customer, or in relation to other services you are using.

Collection Domains

A logical grouping that provides ways of hierarchically combining collection groups. You can combine different collection groups in one collection domain.

In turn, these collection domains can subsequently be combined to form a domain. By creating domains and collection domains you can view the statistics in various ways in your reports.

See "Collection Domain Procedures" for more information.

Collection Domain Procedures

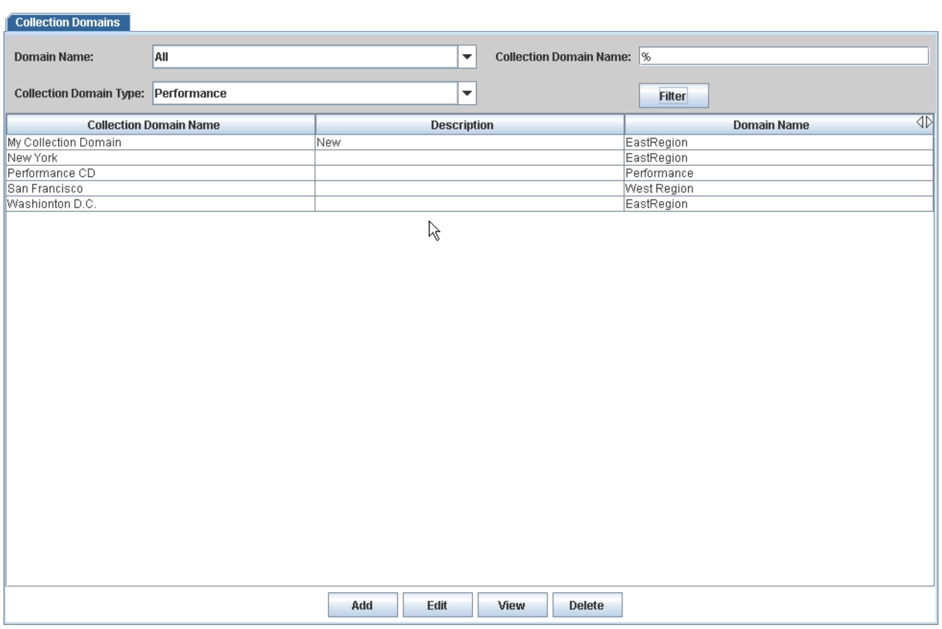
Displaying Collection Domains

You can display a list of collection domains that meet specific criteria by using the filtering feature.

To display a collection domains list

1. From the **Domains** menu, select **Collection Domains**.
2. Select your specific criteria or accept the defaults and click **Filter**.

A screen similar to the following displays:



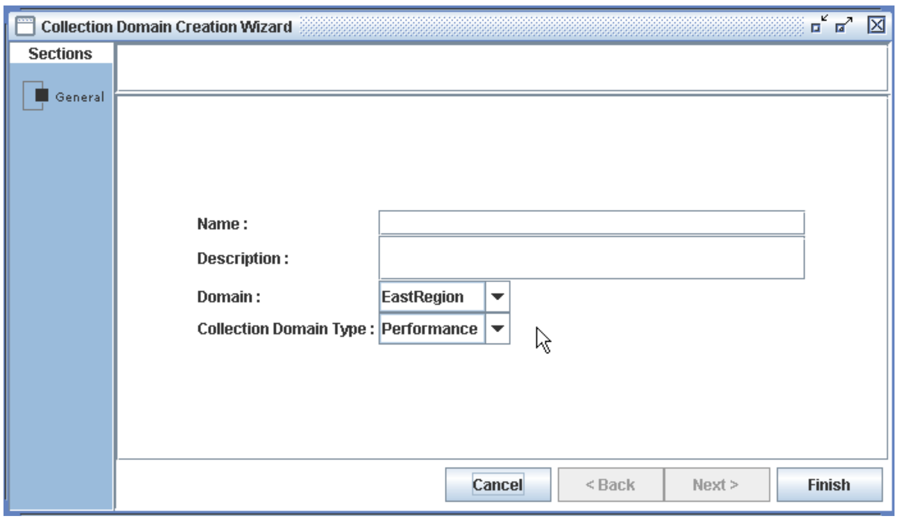
Adding a Collection Domain

You can add a collection domain to the database. This provides PerforMAX with the information necessary to collect and evaluate specific performance data associated with a new domain.

To add a collection domain

1 From the Collection Domains default screen, click **Add**.

The following screen displays:



2 Enter a **Name** (domain) and **Description**.

3 Select a **Domain** from the drop-down list.

4 Select a **Collection Domain Type** (**Performance** or **Connection**) from the drop-down list.

5 Click **Finish**.

Editing a Collection Domain

You can edit a collection domain in PerforMAX.

To edit a collection domain

1 Select the **Collection Domain** from the list that you want to edit.

2 Click **Edit**. The user can then modify the **Description** field only.

3 Click **Finish**.

Viewing a Collection Domain

You can view a collection domain’s information in read-only format.

To view a collection domain

1 Select the **Collection Domain** to view.

2 Click **View**.

The Collection Domain Display screen displays the information in read-only format.

3 Click **Cancel** to return to the **Collection Domains** list.

Deleting a Collection Domain

You can delete a collection domain from the PerforMAX database.

To delete a collection domain

1 Select the **Collection Domain** from the list that you want to delete.

2 Click **Delete**.

A delete confirmation message is displayed:

3 To continue with the **Delete** operation, click **Yes**. Click **No** to stop the deletion process.

Collection Groups

Collection groups are based on devices or interfaces. Collection groups are a set of rules that specify the collection of device data based on resource type, collection cycle, resources, resource interfaces, and performance indicators.

See “Resource and Interface Collection Groups” and “Collection Group Procedures” for more information.

Resource and Interface Collection Groups

There are two types of collection groups, and you can gather real-time information for both of these types of collection groups:

- **Interface Collection Groups** — collect data which is comprised of storage statistics (for example, storage size, storage used, and so on).
- **Resource Collection Groups** — collect data which is comprised of system parameters (for example, system up time, number of users, and so on).

You can view the list of existing resource and resource interface collection groups by accessing the Resource Collection Groups and Resource Interface Collection Groups lists, respectively.

The Resource Collection Group list displays the name, resource type, and collection interval for each collection group.

The Resource Interface Collection Group list displays the name, resource type, interface type, and collection interval for each collection group.

In addition, you can add, view, and delete resource and resource interface collection groups using the Collection Group wizard.

You can specify collection group names as well as specify the resource type and collection cycle for the collection group.

You use the Resource Interface Collection Group wizard to manage the creation, viewing, and modification of resource interface collection groups.

While the Resource Interface Collection Group wizard is similar to the Resource Collection Group wizard, there are some notable differences, as discussed below.

You can also specify a collection group name, resource type, resource interface type, and collection cycle for the collection group. In addition, you can specify the resource interface to include in the collection policy.

Finally, you need to specify the calculations performed on the data the collection policy gathers.

Note: Concurrent edits of an Interface Collection Group (adding the same interface from 2 or more GUI wizards) will cause duplicate entries of the interface in the Collection Group.

Collection Group Procedures

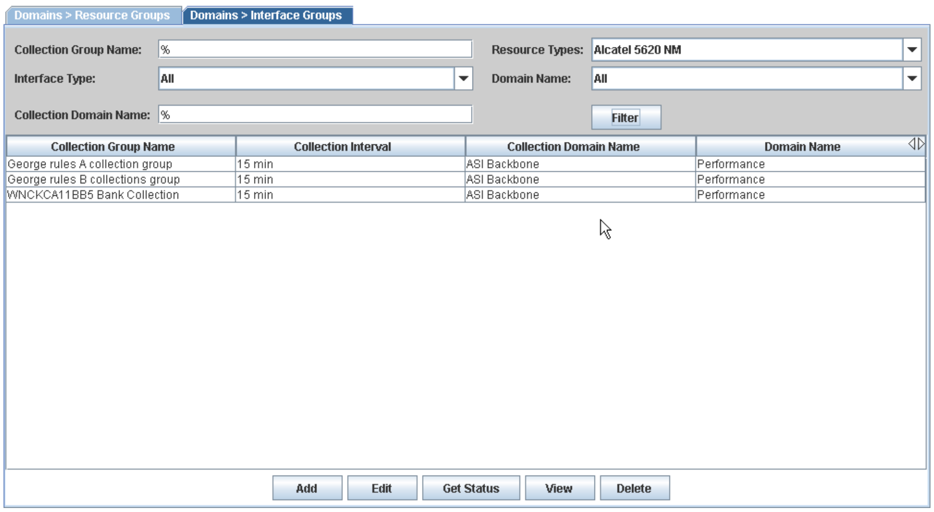
Displaying a Collection Group List

You can display a list of collection groups that meet specific criteria using the filtering feature. Resources that are to be monitored are organized as collection groups.

To display a collection group list

1. From the **Domains** menu, select **Collection Groups**.
2. Select the **Domains > Resource** or **Interface Groups** tab for the category of the collection group that you want displayed.
3. Select your specific criteria or accept the defaults and click **Filter**.

A screen similar to the following displays:



Adding a Collection Group

You can add a collection group to the database. This provides PerforMAX with the information necessary to collect and evaluate specific performance data associated with a new set of devices or interfaces.

To add a collection group (part A - general)

1 From the Collection Groups default screen (or list), select the **Domains** > **Resource** or **Interface Groups** tab.

2 Click **Add**.

The Resource/Interface Group Information screen displays.

3 Type a collection group name in the **Collection Group Name** field.

Note: The following special characters may not be used to specify a Collection Group Name; (~, !, @, #, $, %, ^, &, *, -, +, =).

4 Select the resource type from the **Resource Types** drop-down list.

a. If creating an interface collection group, ensure that the **Domains** > **Interface Groups** tab is selected and select the interface type from the **Interface Type** drop-down list.

Note: When adding an Interface Group, not all resource types are displayed. Create performance templates for additional resource types to allow resource type selection.

5 Select a performance template from the **Performance Template** drop-down list.

6 Select a collection cycle from the **Collection Cycle** drop-down list.

7 Select a domain from the **Domain** drop-down list.

8 Select a collection domain from the **Collection Domain** drop-down list.

9 Click **Next**. Go to “To add a collection group (part B - recurrence periods).”

To add a collection group (part B - recurrence periods)

1 On the **Effective Time** tab, select one of the radio buttons: **Always**, **Never**, or **Period**.

If you select...	then go to...
Always	"To add a collection group (part B - recurrence periods)" step 4.
Never	"To add a collection group (part B - recurrence periods)" step 4.
Period	"Setting time periods."

Setting time periods

a In the **Range of Recurrence** field, select a **Start** and **End by** time.

b In the **Recurrence Pattern** field, select one of the radio buttons to specify when resource monitoring will occur: **Daily**, **Weekly**, **Monthly**, or **Yearly**.

If you select...	then also specify...
Daily	• **Every ___ of days**. type the number of days to pass before resource monitoring occurs again, **OR** • **Every Weekday.** Use this selection to specify that resource monitoring will occur every day except weekends and holidays.
Weekly	• **Recur Every ___ (1-52) Weeks on**. type the number of weeks to pass before resource monitoring occurs again, **AND** • **Sunday - Saturday**. Select day(s) of the week on which resource monitoring is to occur.
Monthly	• **Day**. type the day number (day of the month for a one month period) or the appropriate day number) if a period longer than one month is selected), **OR** • Select the **First - Last** drop-down list to specify the day of the week and if it is to occur **Every (1-12) Month**.
Yearly	• Select the month from the drop-down list (**January - December**) and type the day of the month resource monitoring is to occur, **OR** • Select the **First - Last** drop-down list to specify the day of the week and the month it is to occur (**January - December**).

2 If necessary, select the **Exclusion Time Tab** and repeat the settings in step b.

3 If necessary, select the **Maintenance Time Tab** and repeat the settings in step b.

4 Click **Next**. Go to "To add a collection group (part C - resource interfaces)".

To add a collection group (part C - resource interfaces)

1 Add resources to the collection group:

   a. In the **Resource Name** field, type the resource name or, to list all resources, accept the default (**%**).

2 Click **Get**.

3 Under the **Available Resources/Interfaces** pane, select the resource(s) to include in the collection group and click **Add**. The resource is added to the **Selected Resources/Interfaces** pane.

4 Click **Next**. At the Rules for this Policy screen, add any rules or conditions for this collection group. Click **Finish**.

Editing a Collection Group

You can edit the details of a collection group record after generating a collection group list. The edit feature allows you to edit the data fields used to create a Collection Group record.

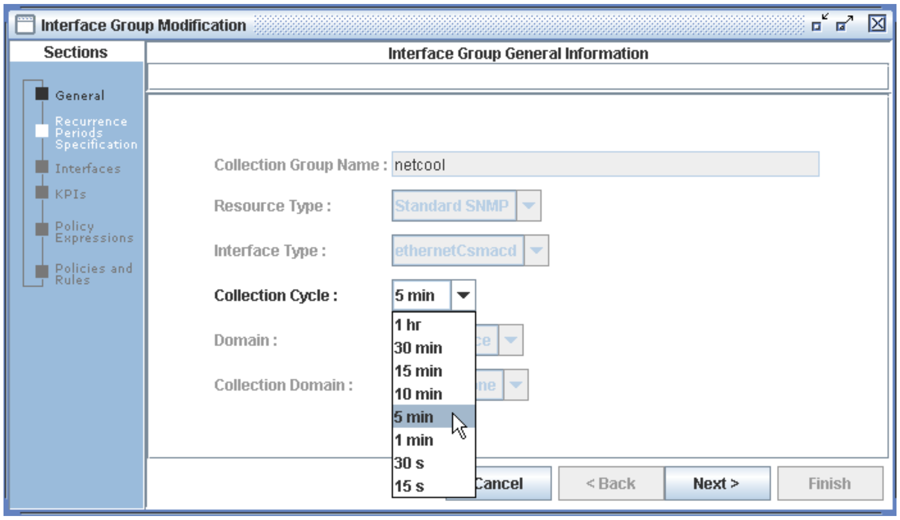
To edit a collection group

1 Select the **Domains** > **Resource** or **Interface Groups** tab.

2 Select the row corresponding to the resource interface to be edited.

3 Click **Edit**.

A screen similar to the following displays:



4 Select a new collection cycle interval from the **Collection Cycle** drop-down list.

Note: You can edit the Collection Cycle (interval) time from the *Collection Group Modification screen*.

5 Click **Next** (click **Back** to navigate to the previous screen).

6 Click **Cancel** to exit the wizard.

7 Click **Finish** after your changes are made.

Viewing a Collection Group

You can view the details of a collection group after generating a collection group list. You can view the collection group in read-only format.

To view a collection group

1 Select the **Domains > Resource** or **Interface Groups** tab.

2 Select, or highlight, the row corresponding to the resource/interface to be viewed.

3 Click **View**.

4 Click **Next** and **Back** to navigate through the wizard.

5 Click **Cancel** to exit the wizard.

Deleting a Collection Group

You can delete a collection group from the database.

To delete a collection group

1 Select, or highlight, the row corresponding to the collection group to be deleted.

2 Click **Delete**.

3 Click **OK** to confirm the deletion.

Performance Evaluation Feature

The Performance Evaluation feature provides you with extensive end-to-end real-time performance reporting capabilities.

This feature is different from the PerforMAX Reporting feature, which provides a historical snapshot or report of an event; that is, the event is frozen in time.

There are two different types of performance evaluations:

- **Interface Collection Groups** consist of a set of resource interfaces that may be from one or multiple resources that are all monitored for a common set of resource metrics.
- **Resource Collection Groups** consist of a set of resources that are all monitored for a common set of resource metrics.

These performance evaluations produce the following types of charts:

- Conformance
- Network Performance

You can specify the date range, collection interval, and format of the reports. The following report formats are supported:

- Tabular text
- Line chart
- Two-dimensional bar chart
- Three-dimensional bar chart

In addition, you can have multiple indicators charted on the same graph. The indicators can be multiply selected and charted on the same graph providing that they are for the same connection or the same policy.

When you want to chart multiple indicators, you simply select both indicators and click **Submit**.

The real-time scale can be modified by right-clicking and dragging to adjust the zoom range.

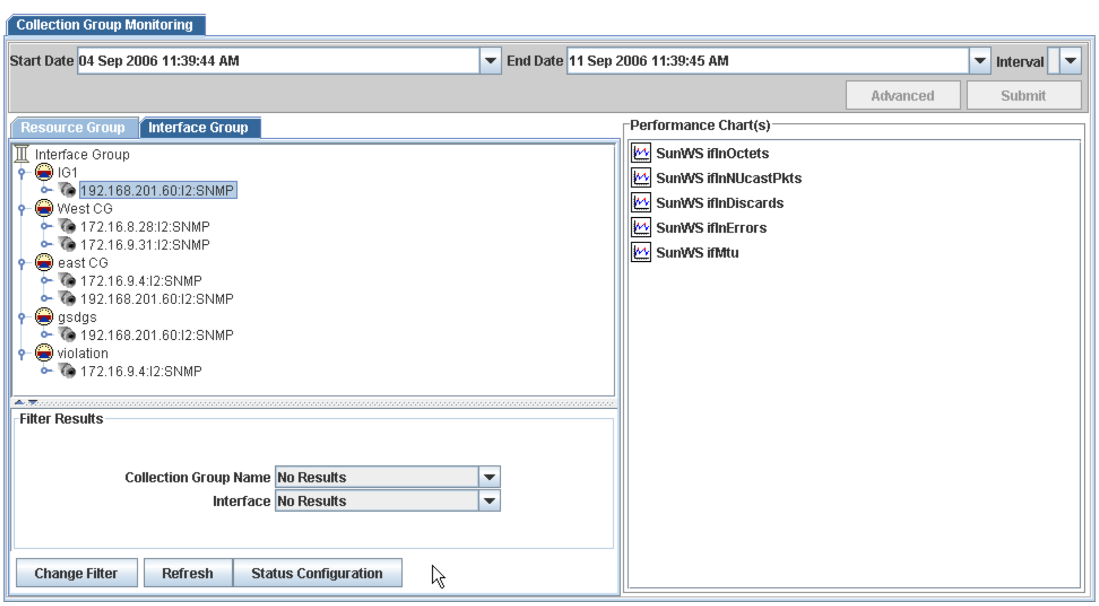
Generating a Performance Chart

Use the following procedure to generate a performance chart of your collection groups.

To generate a performance chart

1 From the **Monitoring** menu, select **Collection Group Monit**.

A screen similar to the following is displayed:



2 Select the **Resource** or **Interface Group** tab which corresponds to the category of collection group for which a chart is to be generated.

3 Navigate through the collection group tree and select the resource/interface for which a chart is to be generated.

4 Select an item from the **Performance Chart(s)** pane for which a chart is to be generated.

Note: To select multiple resource/interface types, hold the **Ctrl** button on your keyboard.

5 Set the **Start** and **End Date** for the chart:

a. Select a start date from the **Start Date** drop-down list.
Set the Year and Month using the controls above the calendar; set the Time using the controls below the calendar.

b. Repeat step a. to select an **End Date**.
OR

a. Click **Advanced**.

b. Select date range based on the last day, hour, minute, and so on or last January, February, and so on.

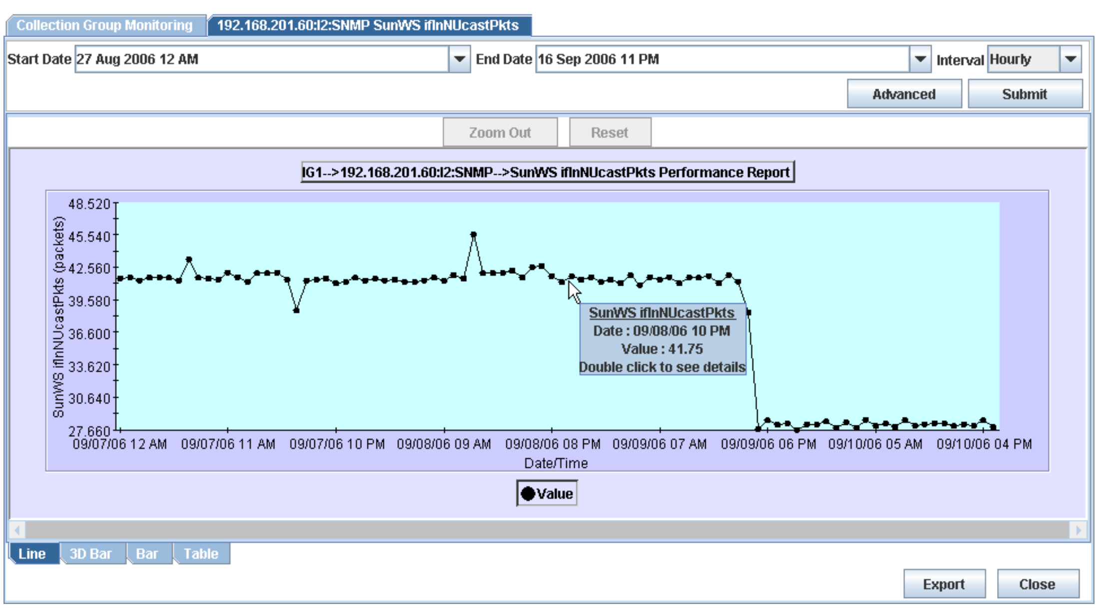
c. Click **OK**.

6 Select a date/time interval from the **Interval** drop-down list:

Note: If you select the **Real Time** option, the chart will continually update based on the collection cycle setting.

7 Click **Submit** to generate the chart.

A chart similar to the following is displayed:

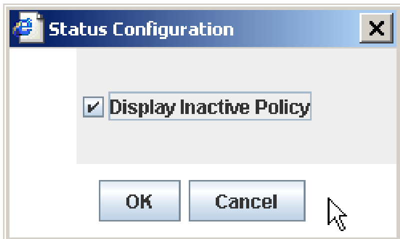


Optionally you can:

To change status configuration

8 Click **Status Configuration** to select the display options based on status.

A dialog box similar to the following is displayed:



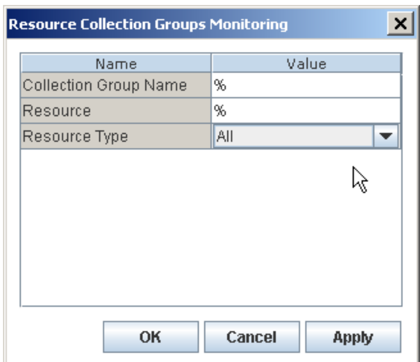
9 Accept the default(s) or clear/check the check box(es) to change the information to be displayed. Click **OK**. The item tree (left pane) is refreshed with the newly configured status.

and/or

To find a specific element in the tree

10 Click **Change Filter**.

A dialog box similar to the following is displayed:



11 Type the new filter criteria for the fields, or accept the default (**%**) for all.

12 Select the new **Type,** from the drop-down list for the desired filter criteria.

13 Click **Apply** and **OK** to display the filtered information.

Working with Performance Evaluation Charts

PerforMAX's performance evaluation includes features that control the display of information including changing the chart type (for example, bar, line, table, and so on) and increasing/decreasing the magnification of the chart.

In addition, you have the ability to save the chart as a PDF, Post Script, JPEG, CSV, or HP PCL file.

To change the display format of a chart

You can change the display format of the generated charts network performance data (for example, you can display a report in Line, 3D Bar, Bar or Table format).

1. After generating a chart, select the tab corresponding to the format type that you want the chart to display as. You can select:
    - Line
    - 3D Bar
    - Bar
    - Table

To save the chart

1. Click **Export**.
2. Select the option button next to the type of file to be generated. You can select:
    - Adobe PDF
    - CSV
    - Post Script
    - HP PCL
    - JPEG Image
3. Click **OK**.
4. Navigate to the location where you want to save the file to and click **Save**.

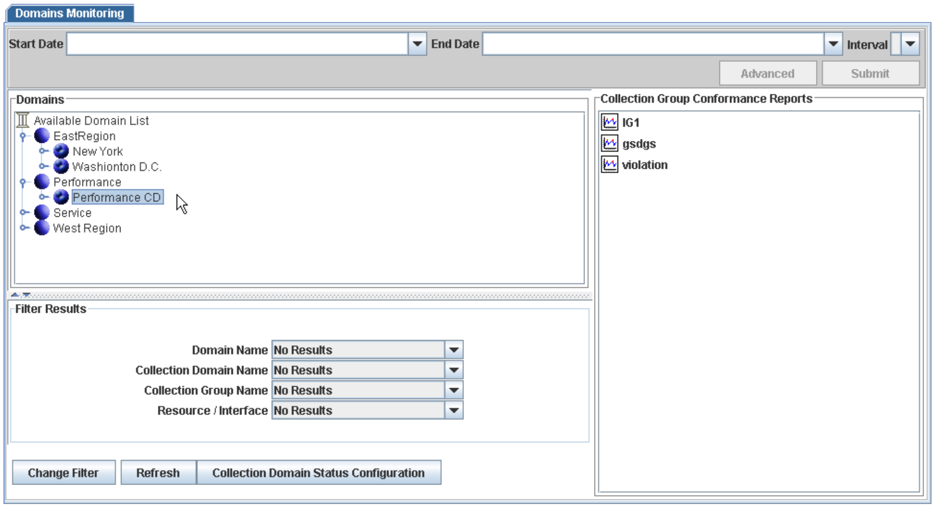
Domain Monitoring

Use the following procedure to generate a collection domain performance evaluation chart.

To generate a collection domain performance evaluation chart

1. From the **Monitoring** menu, select **Domain Monitoring**.

    A screen similar to the following displays:



2 Navigate through the Domains tree and select the resource/interface for which a chart is to be generated for.

Note: To select multiple domains or interfaces, hold the **Ctrl** button on your keyboard.

3 Select an item from the **Reports** pane for which a chart is to be generated.

4 Set the start and end date for the chart:

a. Select a start date from the **Start Date** drop-down list.
Set the Year and Month using the controls above the calendar; set the Time using the controls below the calendar.

b. Repeat step a. to select an **End Date**.
OR

a. Click **Advanced**.

b. Select date range based on the last day, hour, minute, and so on or last January, February, and so on.

c. Click **OK**.

5 Select a date/time interval from the **Interval** drop-down list:

Note: If you select the **Real Time** option, the chart will continually update based on the collection cycle setting.

6 Click **Submit** to generate the report.

Optionally you can:

To change collection domain status configuration

This feature works in the same manner throughout the PerforMAX application. See “To change status configuration” for detailed information.

and/or

To find a specific element in the tree

This feature works in the same manner throughout the PerforMAX application. See "To find a specific element in the tree" for detailed information.

Connections Monitoring

6

The following documentation provides information and procedures for connections monitoring.

The topics in this documentation include the following:

- Connection Assessment
- Specifying the Connections to be Collected
- Connections Monitoring Charts

Connection Assessment

Connection Assessment is the comparison of a connection's pre-defined performance thresholds with the actual service received. In other words, connection assessment compares pre-defined minimum/maximum values for specific aspects of a connection (for example, daily data loss ratio > 0.5) against the actual connection information collected by PerfoMAX indicators.

Therefore, for each KPI within a connection, Connection Assessment compares connection quality against specific thresholds. As a result, you can compare a combined measurement against pre-defined connection objectives to determine overall connection quality in real time.

If violations occur, Connection Assessment can create internal system notifications so that you can prioritize corrective actions based on their connection impact.

Benefits

Connection Assessment gives you a comprehensive understanding of the connection quality derived from traffic and operational performance at the infrastructure layer(s). This comprehensive understanding of connection quality allows you to analyze and target infrastructure issues that affect connection quality and thus improve customer satisfaction and also optimize network utilization.

Connection Assessment also lets you assess connection capacity, allowing for the sale of under-utilized capacity with advance knowledge of traffic trends and patterns on packet-switched and statistically multiplexed networks. You can readily understand traffic patterns — time-of-day, day-of-week, or other trends — and then make smart decisions on sales and business priorities.

Specifying the Connections to be Collected

You can use PerfoMAX to collect connection templates.

You can display a list of connection templates that meet specific criteria using the filtering feature. The connection template filtering process includes entering a template name, scanning all connection group records in the database, and displaying only those records meeting the specified criteria.

You can add a connection template to the database. This allows you to collect data and assess the connection's performance.

To create a connection template record, you must select an offering, connection specification status, start and end dates, and a resolution window. In addition, you must create agreement(s) consisting of connection(s) based on specific collection cycle(s).

You can edit connection templates after generating a connection group specification list. The edit feature allows you to make modifications to certain template fields.

Connection Groups Procedures

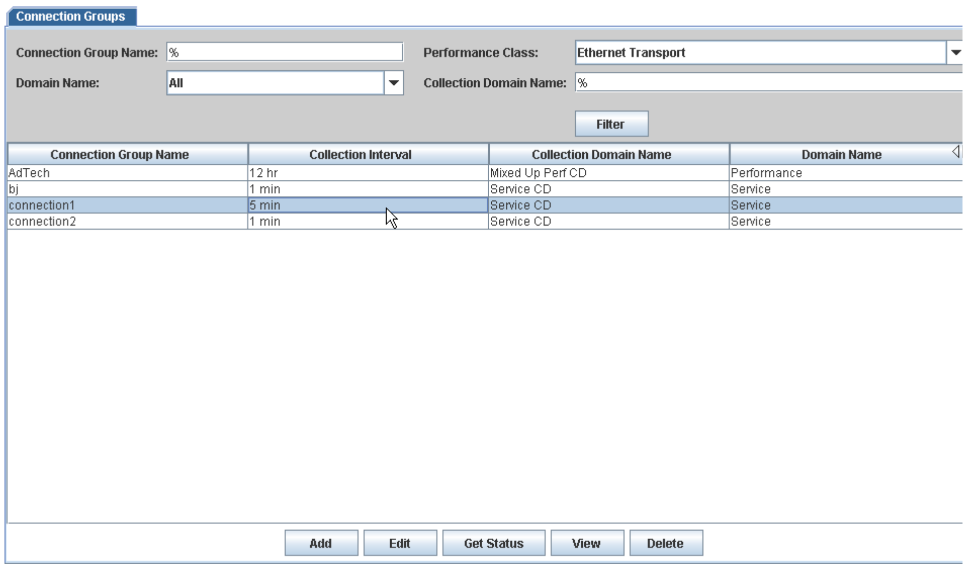
Displaying Connection Groups

You can display a connection groups list that meets specific criteria by using the filtering feature.

To display connection groups

1. From the **Domains** menu, select **Connection Groups**.
2. In the **Connection Group Name** field, type the name, or to list all collection domains (regardless of the associated offering), accept the default (**%**).
3. Select the **Performance Class** from the drop-down list.
4. In the **Collection Domain Name** field, type the domain name or accept the default (**%**).
5. Select the **Domain Name** from the drop-down list or accept the default (**All)**.
6. Click **Filter**.

   A screen similar to the following is displayed:



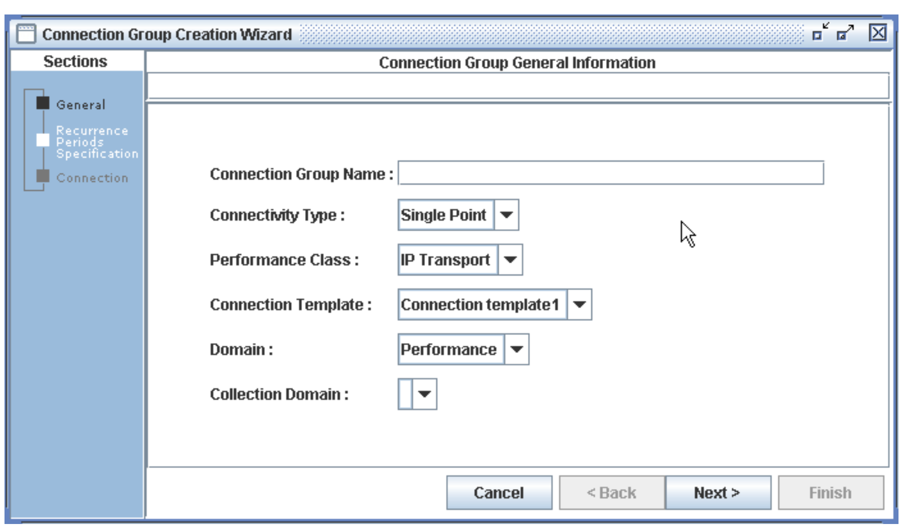
Adding a Connection Groups Record

You can add a connection groups record to the database. This allows you to collect data and assess the connection's performance.

To add a connection groups record, general (part A)

1 From the **Connection Groups** default screen, click **Add**.

A screen similar to the following is displayed:



2 Type a name in the **Connection Group Name** field.

3 Select a **Connectivity Type** from the drop-down list.

4 Select a **Performance Class** from the drop-down list.

5 Select a **Connection Template** from the drop-down list.

6 Select a **Domain** from the drop-down list.

7 Select a **Collection Domain** from the drop-down list.

8 Click **Next**.

The Recurrence Periods Specification Information screen is displayed.

To specify recurrence periods (part B)

1 On the **Effective Time** tab, select one of the option buttons:.

If you select...	then...
Always or **Never**	click **Next**.
Period	go to "Setting time periods."

Setting time periods

a In the **Range of Recurrence** box, select a **Start** and **End by** time.

b In the **Recurrence Pattern** box, select one of the option buttons to specify when specification collections will occur:.

If you select...	then also specify...
Daily	**Every ___ of days**. type the number of days to pass before resource monitoring occurs again, **OR** **Every Weekday.** Use this selection to specify that resource monitoring occurs every day except weekends and holidays.
Weekly	**Recur Every ___ (1-52) Weeks on**. type the number of weeks to pass before resource monitoring occurs again, **AND** **Sunday - Saturday**. Select days of the week on which resource monitoring occurs.
Monthly	**Day**. type the day number (day of the month for a one month period) or the appropriate day number if a period longer than one month is selected, **OR** Select the **First - Day** drop-down list box to specify the day of the week and if it is to occur **Every (1-12) Month**.

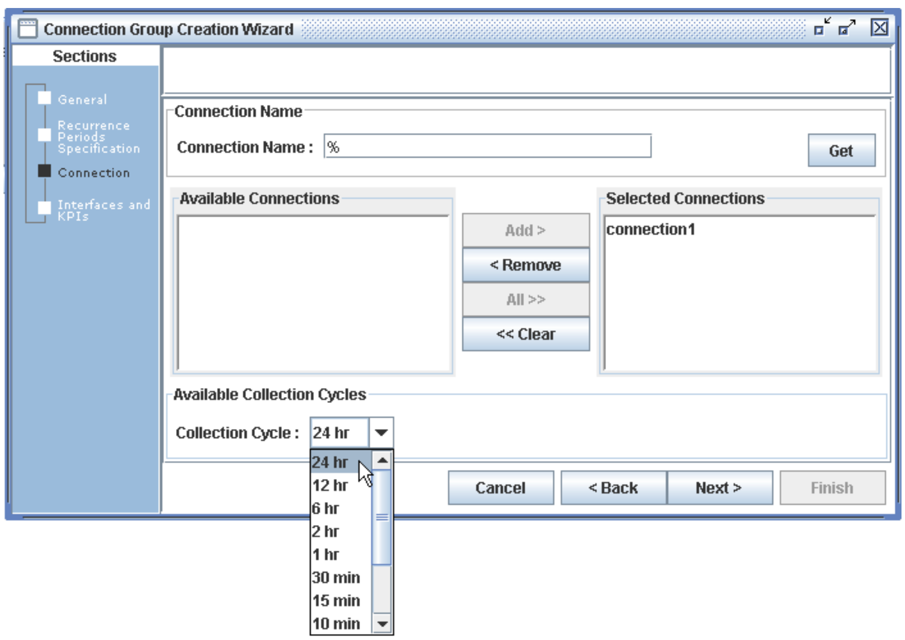
If you select...	then also specify...
Yearly	Select the month from the drop-down list box (**January - December**) and type the day of the month resource monitoring is to occur, **OR** Select the **First - Day** drop-down list box to specify the day of the week and the month it is to occur (**January - December**).

2 If necessary, click the **Exclusion Time Tab**.

3 If necessary, select the **Maintenance Time Tab**.

4 Click **Next**.

A screen similar to the following is displayed:



To retrieve a connection (part C)

1 In the **Connection Name** field, type the name or, to list all connections, accept the default (**%**).

a. Click **Get**.

b. In the **Available Connections** panel, select the connections that you want to include and click **Add >**.

Note: To select multiple connections at once, press the **Ctrl** key on your keyboard as you make the selections.

Note: Connections can only be reused if they are used in the same Connection Template, not the instantiation of the template but the actual template.

2. From the **Collection Cycle** drop-down list, choose the collection cycle for the connection.
3. Click **Next**.

   The Collected Group Indicators screen is displayed.
4. Click **Finish**.

Editing a Connection Groups Record

The edit feature allows you to make modifications to connection groups records.

To edit a connection groups record

1. From the Connection Groups list select, or highlight, the row corresponding to the record to be edited.
2. Click **Edit**.
3. Edit the fields that are available and click **Next** each time you complete a section.

   Note: From this mode, you can edit the list of all available policies for the selected connection template when the connection group was originally created.

4. Click **Next** and **Back** to navigate through the screens.
5. When finished, click **Finish**.

Note: You can edit the Collection Cycle (interval) time from the *Connection Group Modification screen*.

Viewing a Connection Groups Record

The view feature allows you to view connection groups records in read-only format.

To view a connection groups record

1. From the list select, or highlight the row corresponding to the record to be viewed.
2. Click **View**.
3. Click **Next** and **Back** to navigate through the wizard.
4. Click **Cancel** to exit the wizard.

Deleting a Connection Groups Record

You can delete a connection groups record from the database.

To delete a connection groups record

1. From the list select, or highlight the row corresponding to the record to be deleted.
2. Click **Delete**.

   A confirmation message appears asking "Are you sure you want to delete the selected item?"

3. Click **Yes** to confirm the deletion.

Connections Monitoring Charts

Connections Monitoring Charts provide a comprehensive understanding of the connection performance derived from traffic and operational data at the infrastructure layer(s).

This knowledge allows you to analyze and target infrastructure issues that affect service quality and thus improve customer satisfaction and also optimize network utilization.

Connections Monitoring Charts also enables you to assess capacity, allowing for the sale of under-utilized capacity with advance knowledge of traffic trends and patterns on packet-switched and statistically multiplexed networks.

You can readily understand traffic patterns — time-of-day, day-of-week, or other trends — and then make smart decisions for sales and business priorities.

You must first define the Connection group specifications to start Connections Monitoring.

A Connection group specification is a set of rules, which are defined using templates, which are applied to the KPIs (analyzed raw data) collected from Connections.

Connection groups may be setup in the following ways.

- Adding Resources
- Verifying Discovered Resource Interfaces
- Creating Connections
- Creating Action Templates
- Creating Policy Templates
- Creating Connection Templates

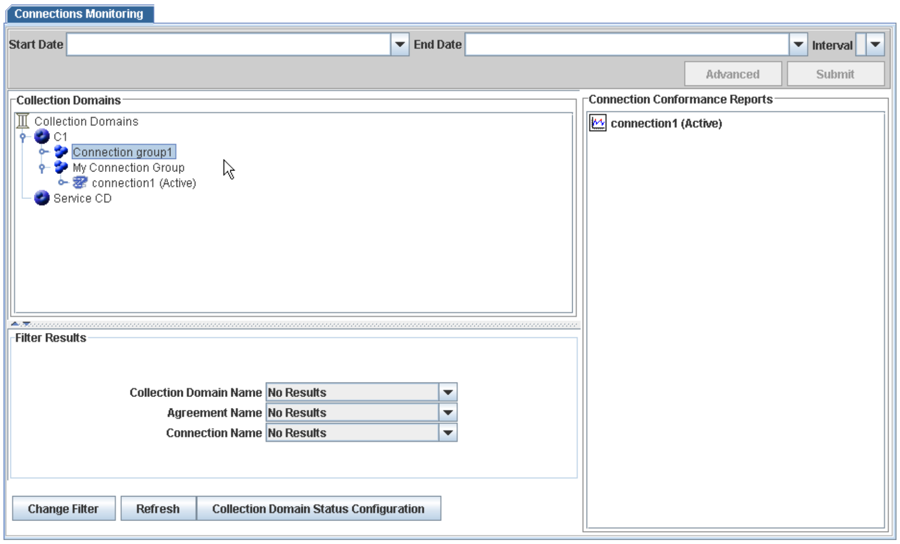
Generating a Performance Chart

Use the following procedure to generate a performance chart of your connections.

To generate a performance chart

1 From the **Monitoring** menu, select **Connections Monitoring**.

A screen similar to the following is displayed:



2 Navigate through the domains tree (left hand pane) and select the component to monitor. You can select various aspects of a connection group domain including policies, connections, and connection templates.

Note: To select multiple components at once, press the **Ctrl** key on your keyboard as you make the selections.

3 Select a chart from the right side pane for which a report (Collection Domain/Connection Group/Connection) is to be generated.

4 Set the **Start** and **End Date** for the report:

a. Select a start date from the **Start Date** drop-down list.
Set the Year and Month using the controls above the calendar; set the Time using the controls below the calendar.

b. Repeat step a. to select an **End Date**.
OR

a. Click **Advanced**.

b. Select date range based on the last day, hour, minute, and so on or last January, February, and so on.

c. Click **OK**.

5 Select a reporting interval from the **Interval** drop-down list:

Note: If you select the **Real Time** option, the chart will continually update based on the collection cycle setting.

Note: When the reporting interval is changed the Start Date/End Date will automatically adjust to the granularity established by the Interval setting. For example:

Interval	Start/End Date Format
Hourly	01 Jan 2003 10PM

Interval	Start/End Date Format
Daily	01 Jan 2003
Weekly	Wed 01 Jan 2003
Monthly	Jan 2003
Quarterly	Jan 2003
Yearly	2003

6 Click **Submit** to generate the report.

A Connection Groups Monitoring Chart is displayed.

This feature works in the same manner throughout the PerforMAX application. See “To generate a performance chart” in “Resources Monitoring” for detailed information.

Optionally you can:

To change status configuration

This feature works in the same manner throughout the PerforMAX application. See “To change status configuration” in “Resources Monitoring” for detailed information.

and/or

To find a specific element in the tree

This feature works in the same manner throughout the PerforMAX application. See “To find a specific element in the tree” in “Resources Monitoring” for detailed information.

Working with Performance Evaluation Charts

PerforMAX’s performance evaluation includes features that control the display of information including changing the chart type (for example, bar, line, table, and so on) and increasing/decreasing the magnification of the chart. In addition, you have the ability to save the chart as a PDF, Post Script, JPEG, CSV, or HP PCL file.

This feature works in the same manner throughout the PerforMAX application. See “Working with Performance Evaluation Charts” in “Resources Monitoring” for detailed information.

Alerts Monitoring

7

The following documentation provides information and procedures about managing alerts.

The following topics are included in this documentation:

- Managing Alerts
- Alert Monitoring Procedures
- Displaying an Alert Management List
- Displaying Alert Management Records
- Displaying Real-Time Charts

Managing Alerts

PerforMAX has a feature that lets you find alerts. The Alert Manager polls the policies looking for any connections associated with a specific policy that have been violated.

You can display a list of alerts occurring within a specific time period using the application's filtering feature. The alert management filtering process includes selecting a start and end date during which the alerts occurred and specifying the type of alert (historical or continuous).

After you select a specific filtering process, all alert records in the database are scanned, and only those records meeting the specified criteria are displayed.

You can display the details of an alert management record after generating an alert management list. The alert view feature provides read access to the data fields used to create an alert management record.

The following table illustrates the severity of alerts, according to their color:

Table 7-1: Alert Color Coding

INCIDENT SEVERITY	ELEMENT COLOR
critical	red
major	orange
minor	yellow
warning	cyan
indeterminate	grey
cleared	green
none	blue

Alert Monitoring Procedures

Displaying an Alert Management List

You can display a list of alerts occurring within a specific time period using the filtering feature.

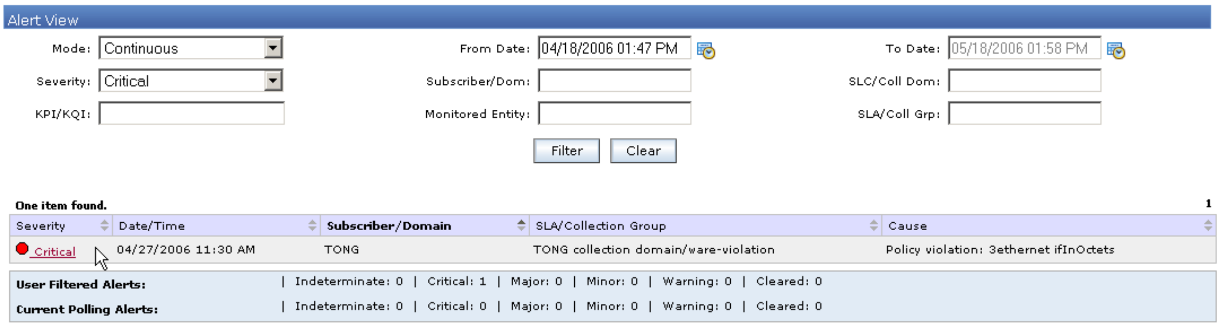
To display an alert management list

1. From the **Monitoring** menu, select **Alert Viewer**.
2. From the Alert View screen, Click **Filter**. To further define any filtering criteria:
   a. Click the **From Date** drop-down list.
   b. Set the year and month using the controls above the calendar; set the time using the controls below the calendar.
   c. Click to select the date from the calendar.
   d. Repeat steps a.-c. for **To Date**.
3. Select an alert severity type.
   a. Click the **Severity** drop-down list and select a degree of severity. For example, Critical.
   b. Click to **Mode** drop-down list and select the alert viewing mode. For example, Major.
4. Click **Filter**.

Note: The **Mode** drop-down list provides the following Alert options:

- **Current Polling Window** (default) **-** provides Alert view of pre-configured polling window period
- **Continuous -** provide the Start Date and click **View**
- **Historical -** provides Alert views of a specific period (Start Date-End Date)
  - In Continuous mode, the auto-refresh feature is enabled. The auto-refresh feature allows the list of alerts in the screen to be updated automatically without having to click the **Filter** button.
  - In Historical mode there is no dynamic update of the screen; the information remains static, and there is no suspend/resume control.

The following screen is displayed:



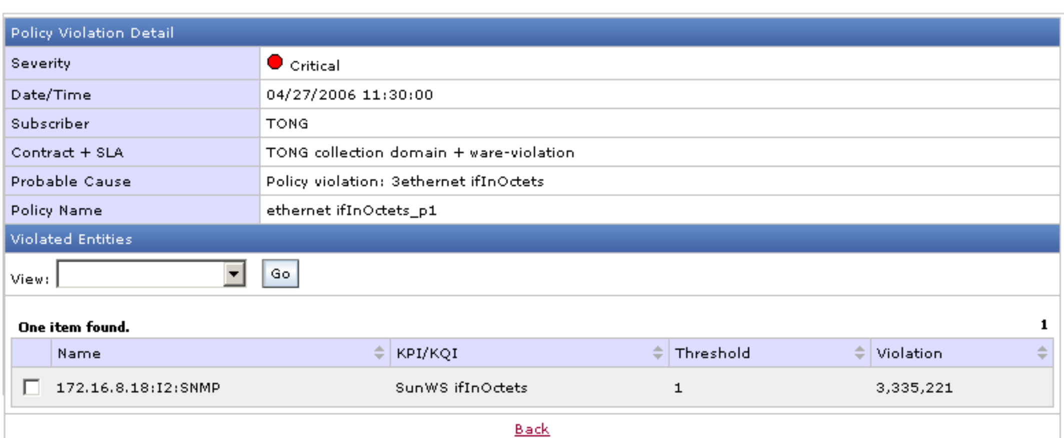
Displaying Alert Management Records

You can display the details of an management record after generating an alert management list. The view feature provides read access to the data fields used to create an alert management record.

To display an alert management record

1 Click a specific violated entity from the list.

A screen similar to the following is displayed:



The Policy Violation details are displayed including the:

- Severity
- Date/Time
- domains

- Probable Cause
- Policy Name

In addition, the specific Violated Entities are shown which include:

- Name
- KPI/KQI
- Threshold
- Violation

From here you can perform the following tasks:

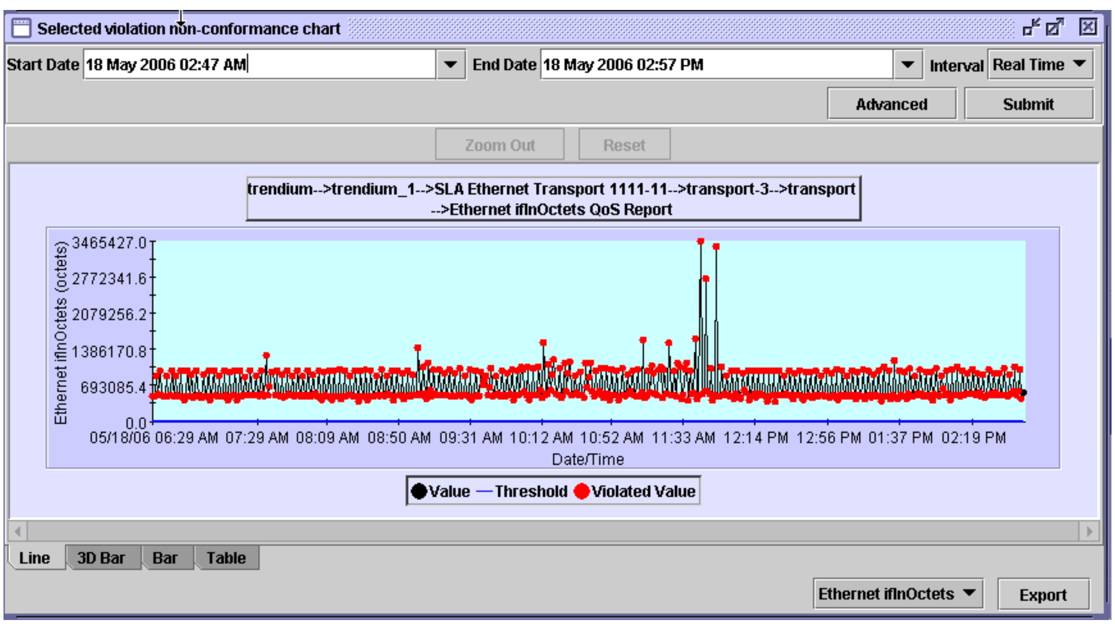
- Displaying Real-Time Charts

Displaying Real-Time Charts

To display real-time charts

1 From the Policy Violation Detail screen, select the desired **Violation** from the list. From the **View** drop-down list select **Real-Time Charts** and click **Go**.

A screen similar to the following is displayed:



See “Performance Evaluation Feature” for more information.

Conformance View

8

The following documentation provides information and procedures for Conformance View.

Some of the topics in this documentation include the following:

Conformance View provides a graphical representation of the various entities that exist in PerforMAX and how they relate to each other.

Conformance View displays the domains, collection domains, collection groups and elements that are in PerforMAX.

Domains, Collection Domains and Collection Groups are known as Nodes and are specifically displayed in the Information Tree window.

Elements are the Connections and the Resources/Interfaces in the Collections Groups and are specifically displayed in the Viewing window.

Understanding Conformance View

Use the Conformance View feature to show which domains are conforming to their policies in:

- Collection Groups
- Connection Groups

Conformance View gathers all the definitions for all elements in your network. After gathering all the data, it displays a view of your network using the tree layout, automatically sizing the view to fit in the current viewing window. See "Conformance View Layout" for detailed information.

Conformance View listens for violation events received by the client at set intervals and updates the alert and status information to maintain an up-to-date indication of the customer's state. See "Alerts Monitoring" for more information.

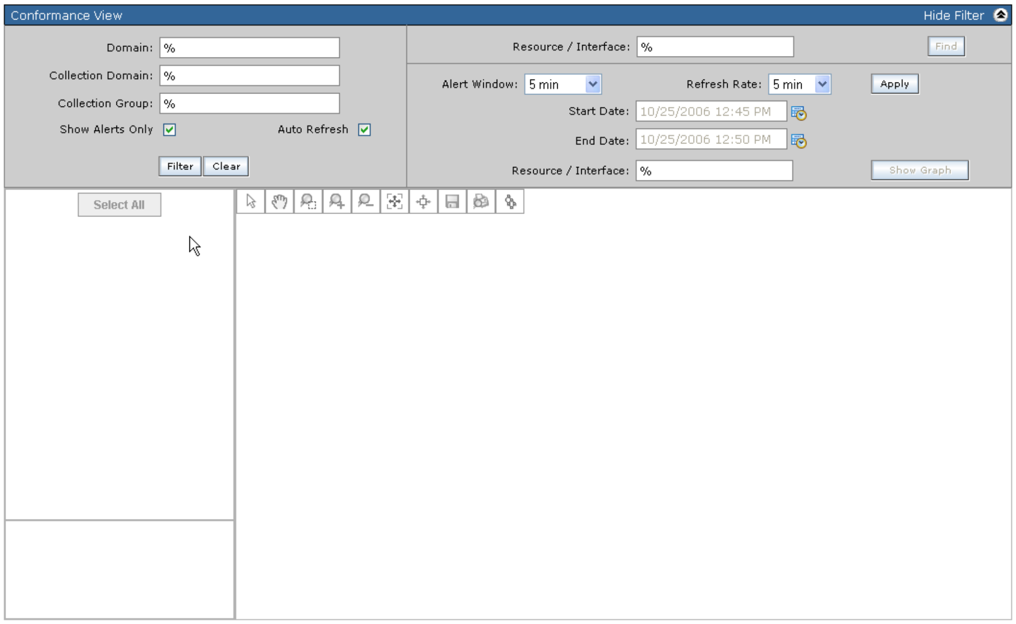


Figure 8-1: Conformance View Default Screen

Conformance View lets you perform the following tasks:

- Display domains.
- Display elements
- Navigate the view using the Viewing or Overview window. The Overview View window is a thumbnail image of the view, and it allows you to zoom in and out and move the focal point quickly.
- Select Domain(s) and expand them to see their associated Collection Domains and Collection Groups.
- Show alert and status information using the **Show Alerts Only** feature. This feature restricts the display to show only non-conforming domains, collection domains, collection groups and/or elements.
- Automatically refresh the screen to show new data.
- Change the focus point of a view to a specific element.
- Follow links and automatically pan to the next element in the link.

Conformance View Layout

The following figure shows a sample screen of the Conformance View layout. The default screen occupies the entire viewing area in the browser window (or work area).

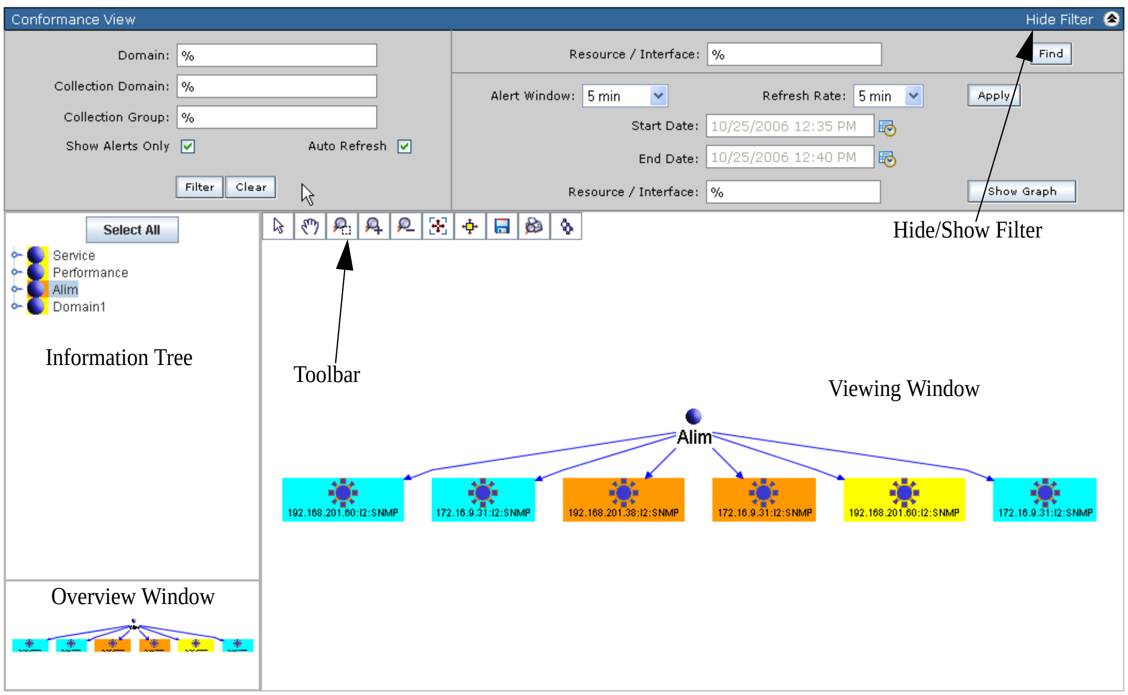


Figure 8-2: Conformance View Work Area

The following windows/toolbar make up the work area in Conformance View.

- Information Tree
- Overview Window
- Viewing Window
- Conformance View Toolbar
- Hide/Show Filter

Information Tree

The Information Tree is the primary area for displaying your system's nodes (Domains, Collection Domains and Collection Groups. You can modify which domain's are displayed using the **Filter** tool.

Click the **Select All** button to select all the domains in the tree.

Overview Window

The Overview window allows you to control the focus of the view using a thumbnail image of the element view. You can zoom in and out and move the focal point.

The image in the Overview window represents your current conformance view. The blue zoom window on the navigator represents your viewing window (the area on the screen used to display the conformance view). When you move or resize the zoom window, you change your viewing window correspondingly.

See "Using the Overview Window" for more information.

Viewing Window

The Viewing window is the primary area for displaying your system's elements. You can navigate and perform various tasks while in this window.

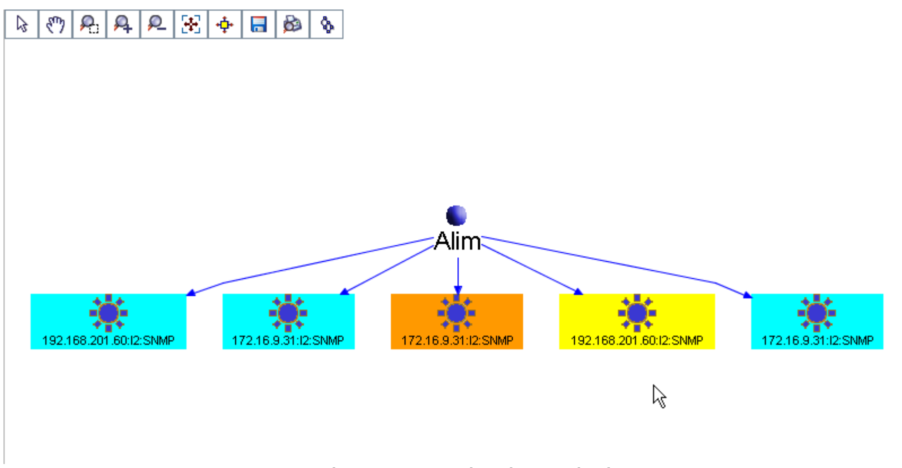


Figure 8-3: Viewing Window

The Viewing window allows you to focus on an element and examine details or to get a broader view of the status of all elements in the view.

You can zoom in on the view to focus on more detail or zoom out to get a broader view and see more elements in the view. See "Using the View Scale Tools" for detailed information.

Hide/Show Filter

The Hide/Show Filter button lets you hide or show the filtering option fields. Click Hide to hide the fields and show to redisplay them.

Conformance View Toolbar

You can navigate and perform tasks within the viewing window using the Conformance View toolbar.

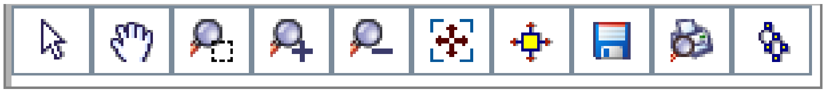


Figure 8-4: Conformance View Toolbar

Various tools allow you to perform different task in the view window. For example, some tools allow you to change the magnification of the element you are viewing. You can zoom in to show more detail or zoom out to see more elements and more of the entire element view.

Other tools allow you to change the contents of the view and to work with those elements in the view. While others allow you to save or print the displayed graph.

See "Toolbar Details" for more information.

Toolbar Details

The Conformance View toolbar lets you perform various tasks and navigate within your viewing window.

The following functions can be performed:

Table 8-1:Conformance View Toolbar

Name	Description
Select	Selects an specific node(s) or element(s) in the view window
Pan	Pan to drag canvas in any direction for navigational purposes
Zoom	Marquee zoom to select a rectangle for zooming into
Zoom In	Zooms in on specific node or element
Zoom Out	Zooms out on specific node or element
Fit in Page	Fit into canvas
Link Navigation	Move from element to element automatically
Save as Image	Saves displayed graph as image file
Print	Prints displayed graph
Hierarchical Layout	Displays graph in hierarchical view

Navigating within the View

Conformance View provides several methods for you to navigate your view of the domains:

- Changing the focal point by selecting a specific element
- Selectively zooming and panning to change your view
- Automatically following links from one element to the next one

If the entire image is not visible in the viewing window, you can navigate to bring another area of the image into view.

Using the Overview Window

The Overview window allows you to control the focus of the view using a thumbnail image of the domain view. You can zoom in and out and move the focal point.

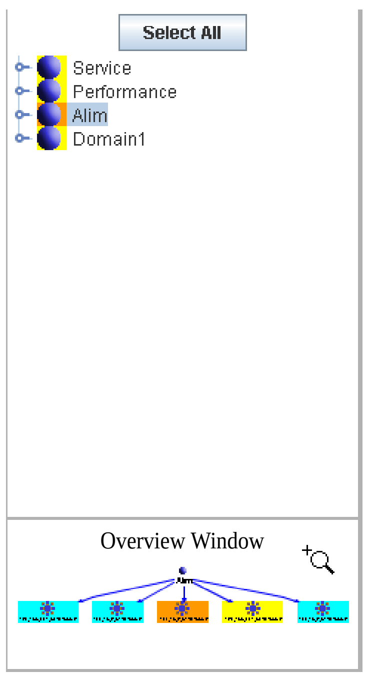


Figure 8-5: Overview Window

The image in the Overview window represents your current conformance view. The blue bordered zoom window on the navigator represents your viewing window (the area on the screen used to display the conformance view). When you move or resize the zoom window, you change your viewing window accordingly.

To zoom in or out on an area

1. Click the **Select** button and in the Overview window select a specific node of the graph by dragging a corner of the navigator window to the size you want.
2. Click outside the blue-zoom window (your mouse pointer turns to a hand, which you use to pan over the area you want to view).

To focus on a new area

1. Use your mouse to select a new area and drag a corner of the navigator window to the size you want.

Using the View Scale Tools

Using the Zoom Tool

The **Zoom** tool allows you to quickly zoom in on a specific area of the viewing window. The zoom magnifies the selected area and moves it to the center of the viewing window.

To use the zoom tool

1. Click the **Zoom** button.
2. Click in the viewing window and drag to create the marquee around the area you want to view.

Using the Zoom In/Out Tool

The **Zoom In/Out** tool allows you to quickly zoom in or zoom out on objects in your viewing window. When you zoom in, you increase the scale, thereby showing more detail. When you zoom out, you reduce the scale, thereby showing more of the entire view.

To use the zoom in/out tool

1. Click the **Zoom In/Out** button to zoom in or out of the respective graph or node.

Using the Fit in Page Tool

The **Fit in Page** tool allows you to quickly view all elements. The **Fit in Page** tool zooms out so that all elements in your view are centered and visible in the viewing window.

To use the fit in page tool

1. Click the **Fit in Page** button.

All elements are restored to fit within the current viewing window.

Using the Pan Tool

The **Pan** tool allows you to quickly move the domain view within the viewing window.

To use the pan tool

1. Click the **Pan** button.
2. Click anywhere in the viewing window and drag it to the new position.

Using Link Navigation

The Link Navigation tool allows you to follow links from one element to another within the current elementview.

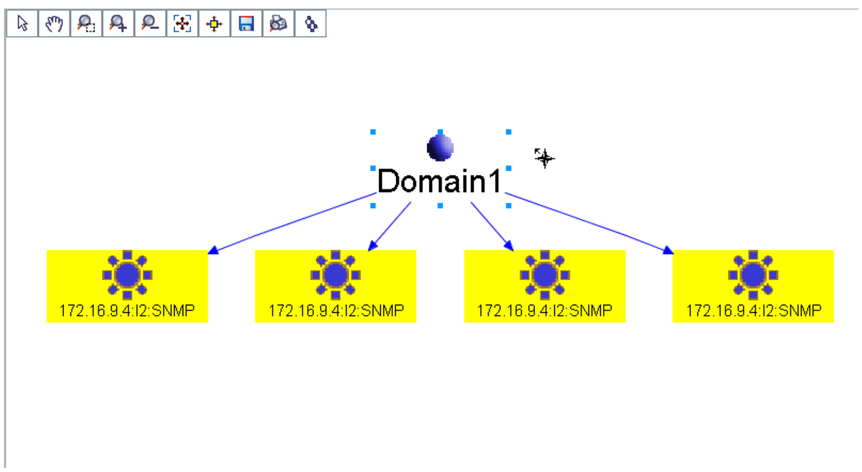


Figure 8-6: Link Navigation View

These links do not represent the physical connection from one resource to another. They simply represent the logical connection between the elements displayed in Conformance View.

- Clicking the right-mouse button on an element moves the selection from one link at that element to the next link at the same element.
- Clicking the left-mouse button on an element moves the selection from one end of the selected link to the other end of the same link.
- Clicking the left-mouse button on any link connected to the node will move to the end of the same link.

Conformance View Filtering Controls

You can limit the number of elements displayed in the view using the filtering controls. Elements are the Domains, Collection Domains, Collection Groups and/or Resource/ Interfaces that you retrieve from the database. Choose a specific filtering criteria to show only those elements that you want to display. See "Display Conformance View" for more information.

The following filtering controls can be applied:

Table 8-1:Conformance View Filtering Controls

Name	Description	Default Value
Domain	Limits domain data to value supplied by user	% = all
Collection Domain	Limits the dataset to only the Collection Domains that match the specified value	% = all
Collection Group	Limits the dataset to only the Collection Groups specified	% = all
Show Alerts Only	Filter criteria to display a subset of domains, collection domains, collection groups and/or elemensts based on existence of violation	Selected
Auto Refresh	Updates the data in the Information Tree, Overview and Viewing windows based on selected filter criteria. n	Selected
Filter	Initiates search for populating the Information Tree. When the search starts, the View is reset to empty	N/A
Clear	Resets Domain, Collection Domain, Collection Group, to empty	N/A
Resource/Interface (Find)	Automatically selects and pans to first specific node in Viewing window	% = all
Alert Window	Controls time duration for Start and End Date. View only mode if the Auto Refresh check box is cleared.	Based on configuration file
Refresh Rate	Controls the frequency of the auto refresh. View only mode if the Auto Refresh check box is cleared.	Based on configuration file
Start Date	The beginning point in time to prepare data for graphical representation. View only mode if the Auto Refresh check box is selected.	Server-Time minus polling window in format: MM/DD/ YYYY HH:MM HR
End Date	The end point in time to prepare data for graphical representation. View only mode if the Auto Refresh check box is selected.	Server-Time in format: MM/ DD/YYYY HH:MM [AM|PM]
Resource/Interface (Show Graph)	Limits the dataset to only the resources/interfaces that match the specified value	% = all

Table 8-1:Conformance View Filtering Controls

Name	Description	Default Value
Show Graph	Displays graph of elements for selected domains from tree list	N/A

The filtering tool has text fields in which you can enter search criteria (that is, words or sequences of words). The filtering tool filters nodes based on their name.

When the filter finds an node that matches a search term, it becomes part of the set of nodes that appear in Conformance View. Elements that do not meet the search criteria are not shown.

The filtering tool has wildcard matching using the "**%**" character as the wildcard. The filtering fields default to **%**. When searching, the wildcard matches any sequence of characters.

For example, typing "%" as the only character in the filter's search fields returns all nodes; this is equivalent to having no filtering.

To filter on the specific node names, enter one or more alpha characters into the appropriate field. Terms with spaces must be surrounded with quotation marks. An example set of search terms for a single filter operation might be: "My Customer," "Joe's Customer," CustomerX, or D%1%.

If any of these terms matches a node name, that node is included in the filtered set. For example, the filter criteria of **D%1%** contains two instances of the wildcard and might match such names as: Doe 100, Dodson 10, and Devlin 102.

Using Conformance View

The following procedures detail how to use Conformance View.

After launching Conformance View you can:

- Display Domain, Collection Domain, Collection Groups
- Display Elements under a Domain
- Display Elements under Multiple Domains
- Display Summary Information for an Element
- Display a List of Indicators for an Element
- Display Performance Evaluation Chart
- Display an Alerts List for an Element
- Display Alert Details for an Alert
- Display a Chart for an Alert Detail

Conformance View Procedures

Display Conformance View

To display conformance view

1 From the **Views** menu, select **Conformance View**.

The following screen displays:

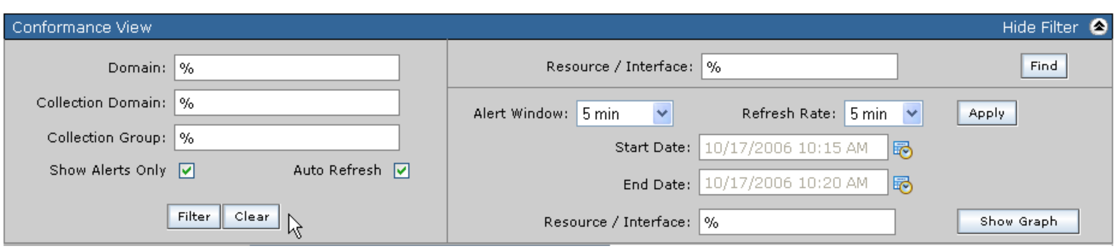


Figure 8-7: Filtering screen

Filtering options are located in the top panel of Conformance View, as seen above.

The left side pane controls the data that is displayed in the Information Tree and is part of the filtering controls.

The right side pane controls the data that is displayed in the Viewing window. You can set criteria on this side to refine how to view the data.

From here you can generate a list of domains using specific criteria.

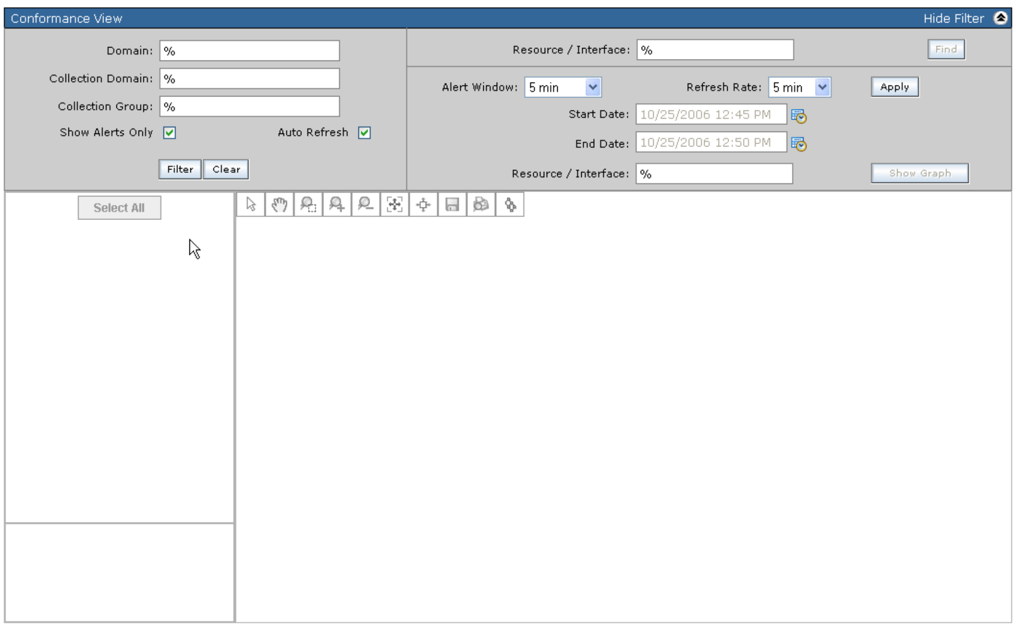
Display Domain, Collection Domain, Collection Groups

From Conformance View you can display Domain, Collection Domain, and Collection Groups. Use this feature to generate a list to see which collection groups and elements might have alert violations.

To display domain, collection domain, collection groups

1 From the dashboard, select **Views** > **Conformance View**.

The following screen displays:

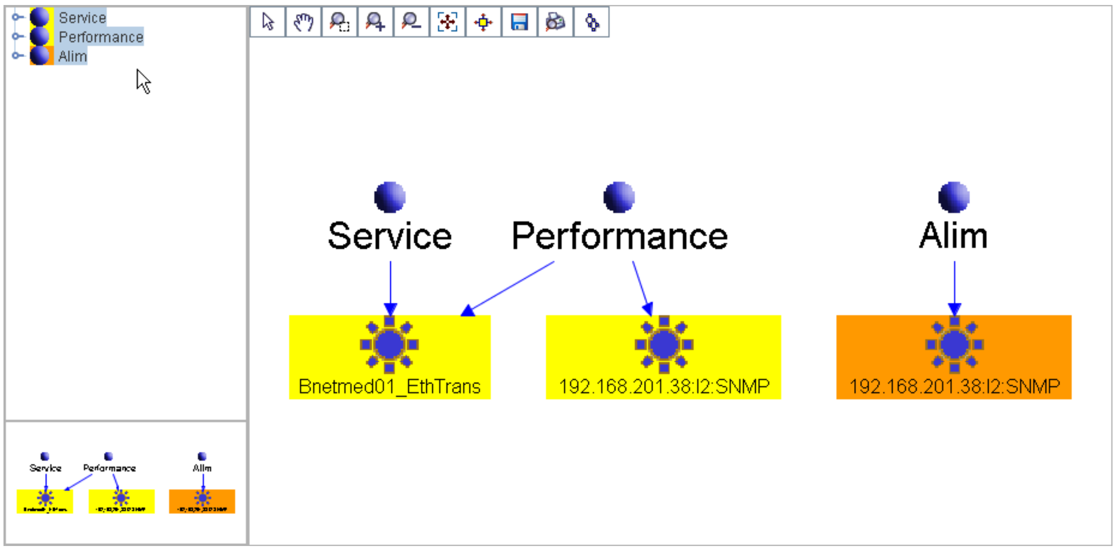


By default the Conformance View main screen shows:

- The filter attributes for **Domain**, **Collection Domain**, and **Collection Group**, as **%** (for all).
- The **Start Date** as the current server time and the **End Date** as five minutes later. These dates are only relevant for violations. Default is view-only mode.
- The **Show Alerts Only** check box is selected.
- The **Auto Refresh** check box is selected.
- The **Alert Window** and **Refresh Rate** check boxes default to config. file settings.
- The Information Tree, Overview and Viewing window are empty.
- The **Select All** button is ghosted.

2 From the Conformance View screen accept all defaults, or enter specific filtering criteria within a specific date range to display a list of domains. Enter additional filtering criteria as necessary and click **Filter**.

A screen similar to the following is displayed:



The Information Tree window displays all the domains in the system that meet your filtering criteria.

The Overview and Viewing window displays the top level node(s) and element(s) of the domains.

You can expand items in the Information Tree window by clicking on the nodes. See “Expanding Nodes” for more information.

The tree structure has the following hierarchy:

Domain > Collection Domain > Collection Groups

Expanding Nodes

You can expand the view of each to show the underlying elements associated with it.

To view the elements within a domain

1 From the Information Tree window, select a specific domain node.

2 Click the node.

The Information Tree displays the elements directly associated with that element.

3 You can then expand or collapse each element by clicking on it.

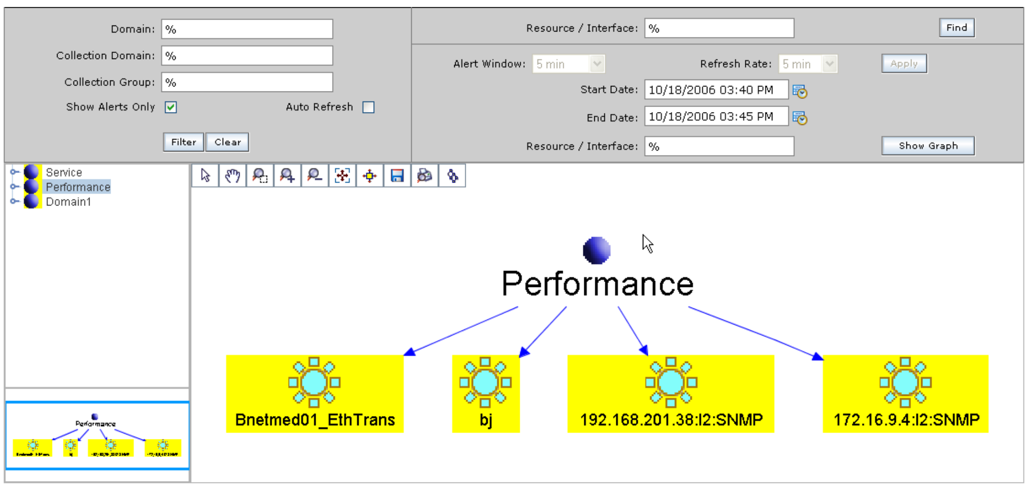
Display Elements under a Domain

From Conformance View you can display elements under a selected Domain. Use this feature to view in detail nodes with conformance issues.

To display elements under a domain

1 Select a specific node from the list in the Information Tree window and click **Show Graph**.

A screen similar to the following is displayed:



The graph is displayed in the View window.

By default the **Show Alerts Only** check box is selected and all violating elements are displayed in the graph. The elements that have a violation (regardless of the violation being associated with the selected domain or not) are displayed with a different color.

The following rules apply when determining the violations of an elements (only applies if the **Auto Refresh** check box is cleared):

- If user provides both an **End Date** and a **Start Date**, then the system finds all violations for the element that exists in the specific date/time window for all policies (for a specific policy and expression, if multiple violations exist, the most recent one of those violations is shown).
- Otherwise, if user provides only an **End Date** then the system finds all violations for the element that exists in the time window where the start date is zero (0) and the end date is the user specified end date, for all policies (for a specific policy and expression, if multiple violations exist, the most recent one of those violations is shown).
- Lastly, of all such violations, the highest severity violation is determined. The display color of the element is set to the standard severity color for that severity level.

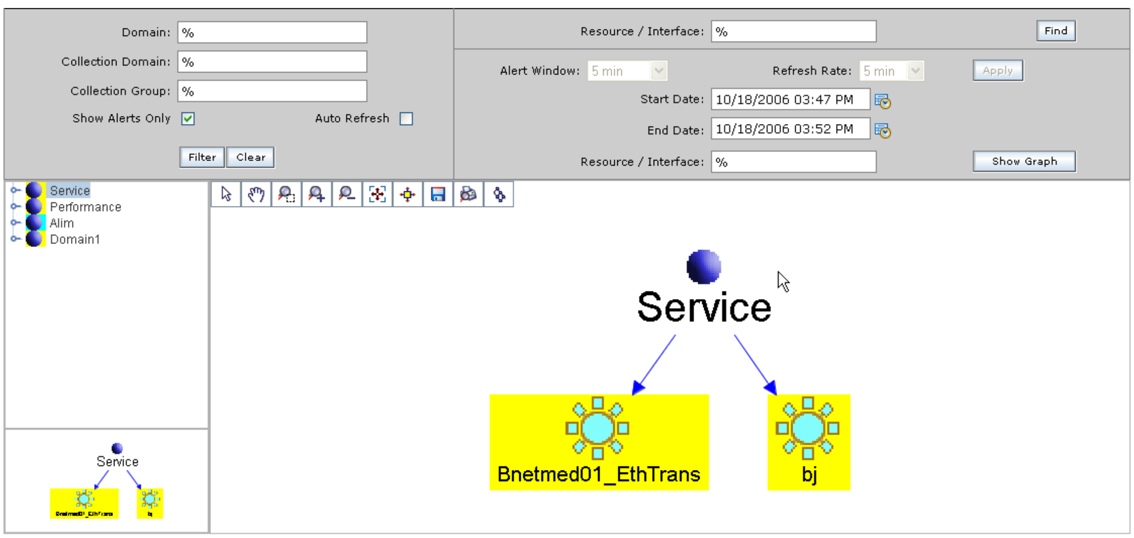
To explain this by an example, let’s assume that an element is used in two policies, POL1 and POL2.

In the specified time interval, the element has three violations for (POL1, KPI1) that occurred in the sequence: Critical, Severe, Clear. In addition, the element has one violation for (POL2, KPI2) in the specified time interval, Major.

Of the three violations for (POL1, KPI1) we pick up the most recent, Clear. Then, there are two violations to compare in terms of severity level: Clear, contributed by (POL1, KPI1), and Major, contributed by (POL2, KPI2). Since Major is a higher severity level than Clear, the highest severity violation for the element is Major.

2 To continue to show only the elements with violations, then: keep the **Show Alerts Only** check box selected and the View window is refreshed with a restricted set of nodes that show only the elements with violations.

A screen similar to the following is displayed:



When the **Show Alerts Only** check box is selected, the following rules are observed:

- Any element that has a violation is displayed.
- For any element, move up the hierarchy following the parent-child relationships starting from the element. This means that all elements in the path are also displayed, even if they do not have violations (in other words, any direct or indirect parent of a violating element is also displayed).
- All other elements are not displayed.

From here, you can also clear the **Show Alerts Only** check box and the View window is refreshed with all the elements (with and without violations) displayed .

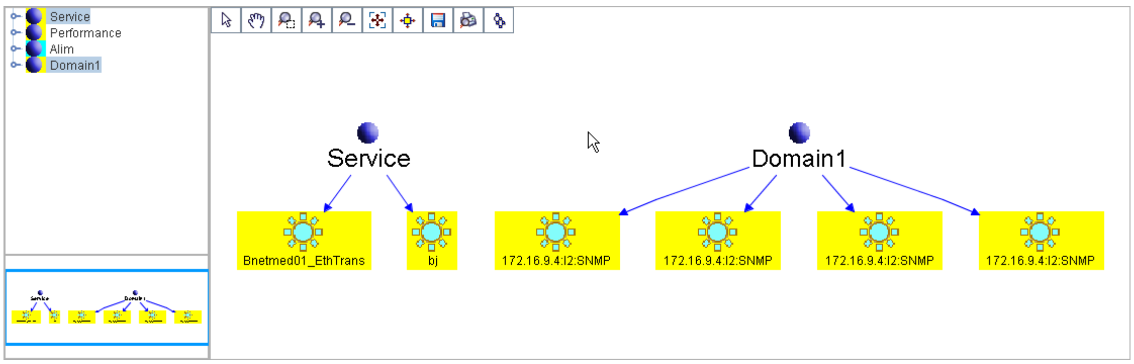
Display Elements under Multiple Domains

From Conformance View you can display elements under multiple Subscribers/ Domains. Use this feature to view in detail multiple subscribers with conformance issues.

To display elements under multiple domains

1 Select multiple domains (**Ctrl** and click mouse) from the list in the Information Tree window and click **Show Graph**.

A screen similar to the following is displayed:



The element graph is displayed in the View window. As the top level nodes, element nodes are also displayed. Under each element node, the associated nodes are displayed

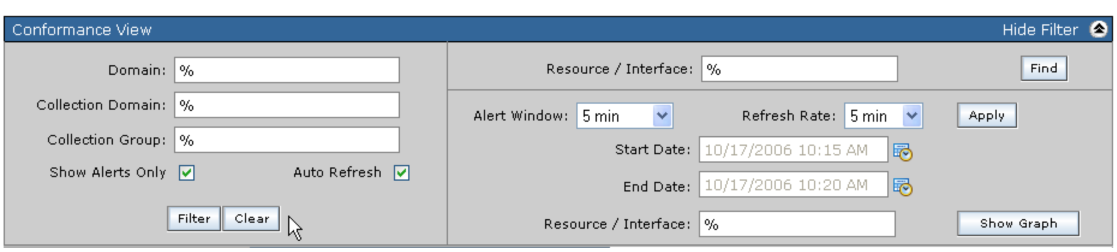
2 To show only multiple element with violations see step 2 in “To display elements under a domain”.

Display Violations in Auto Refresh

From Conformance View you can set the graphs in the View window to automatically refresh by using the Auto Refresh feature. This is similar to the Current Polling Window feature in Alert View.

To display violations in auto refresh

1 After generating an element graph(s) in the View window, leave the **Auto Refresh** check box selected.



When using Auto Refresh, the **End Date** (the current server time), and the data in the View window is refreshed at periodic intervals based on the data in the current polling window and the element information as selected by user.

If the **Show Alerts Only** check box is selected, then the elements displayed in the View window will be restricted to only violating elements and their parents as previously described.

Context Menu

Conformance View's right-click functionality (**Context Menu**) includes the following options:

- **Summary -** Displays summary information for the specified element. The information includes the name, performance class, connectivity type of the element.
- **Indicators -** Displays the collected indicators for a specific element. This list any violations and their severity, name, type, domain and policy. From here you can select a violation and generate a QoS chart.
- **Alerts -** Displays the alert violations for the specified element. The information includes the alert severity, date/time, domain, collection group, and cause of the alert.

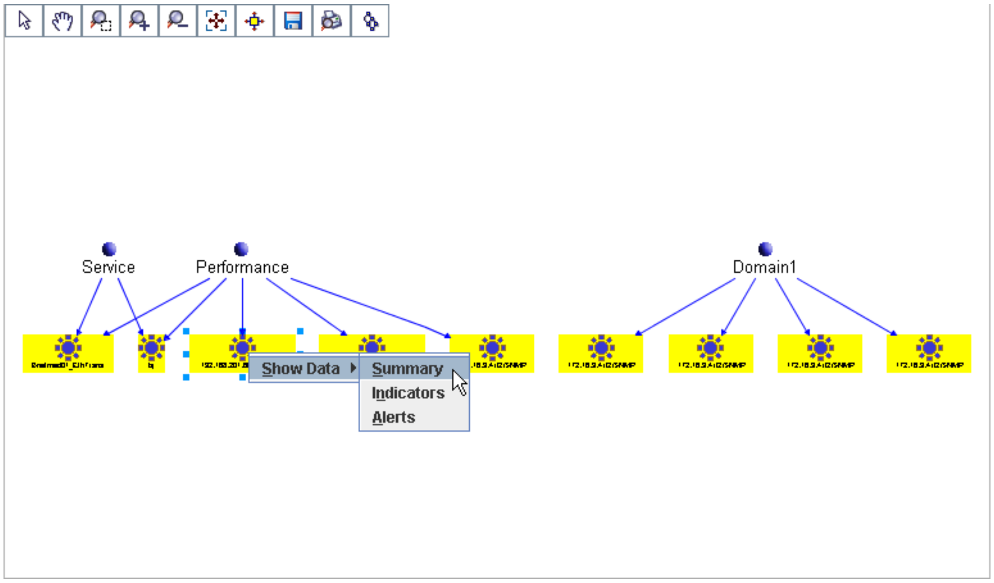
Display Summary Information for an Element

From Conformance View you can display summary information for a specific element. Use this feature to view the detailed information as it relates to a specific element with conformance issues.

To display summary information for an element

1 **Right-click** a specific element from the View window and select **Show Data > Summary** from the drop-down menu.

A screen similar to the following displays this action:



The following summary is displayed for the specific element under the View window:

Resource / Interface Summary

Name:	Bnetmed01_EthTrans
Connectivity Type:	Single Point
Resource / Interface Type:	Ethernet Transport
Domains:	• Performance • Service

The summary information lists the following element information:

- Name
- Connectivity Type
- Resource/Interface Type
- Domains

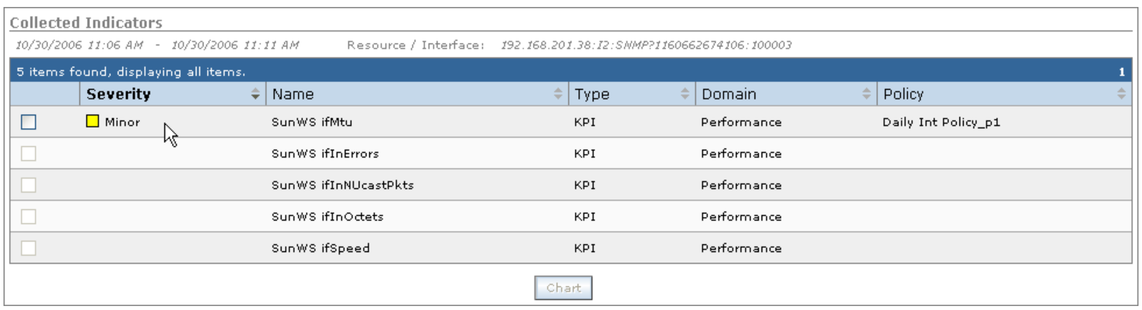
Display a List of Indicators for an Element

From Conformance View you can display a list of collected indicators for a specific element. Use this feature to view the violation indicators that relate to a specific domain with conformance issues.

To display a list of indicators for an element

1 Follow the same steps in “To display summary information for an element” except select **Show Data** > **Indicators** from the drop-down menu.

The following list of indicators is displayed for the specific element under the View window:



2 Select the check box next to the indicator that you want to view the details for. Click **Details**.

The details for the specific indicator are displayed.

The following rules apply when determining the list of indicators for an connection:

- If user provides both an **End Date** and a **Start Date**, then the system finds all expressions for the connection that exists in the specific date/time window for all policies.
- Otherwise, if user provides only an **End Date** then the system finds all expressions for the connection that exists in the time window where the start date is zero (0) and the end date is the user specified end date, for all policies.
- For any expression and policy, the latest violation (if any), in that interval is also determined.

From here you can:

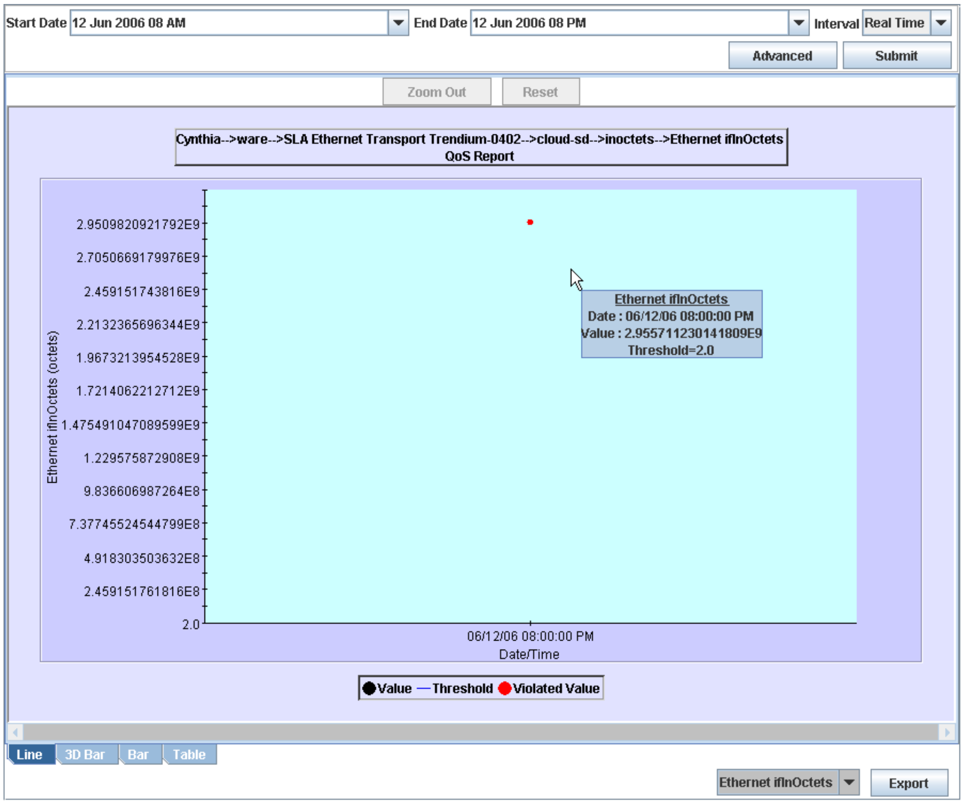
- Display a performance evaluation chart

Display Performance Evaluation Chart

To display a performance evaluation chart for a violation

1 Select the check box next to the violation that you want the performance evaluation chart to display for. Click **Chart**.

A screen similar to the following is displayed:



The interval for the chart is determined as follows.

- The **End Date** specified by the user is the end time for the chart.
- If user specifies a Start Date, it is the start time for the chart. If not specified then the start date defaults to the previous day.

The Performance Evaluation Chart feature works in the same manner throughout the PerforMAX application. See “Working with Performance Evaluation Charts” in “Resources Monitoring” for detailed information.

Display an Alerts List for an Element

From Conformance View you can display a list of alerts for a specific element. Use this feature to view the alerts that relate to a specific domain with conformance issues.

To display an alerts list for an element

1 Follow the same steps in “To display a list of indicators for an element” except select **Show Data** > **Alerts** from the drop-down menu.

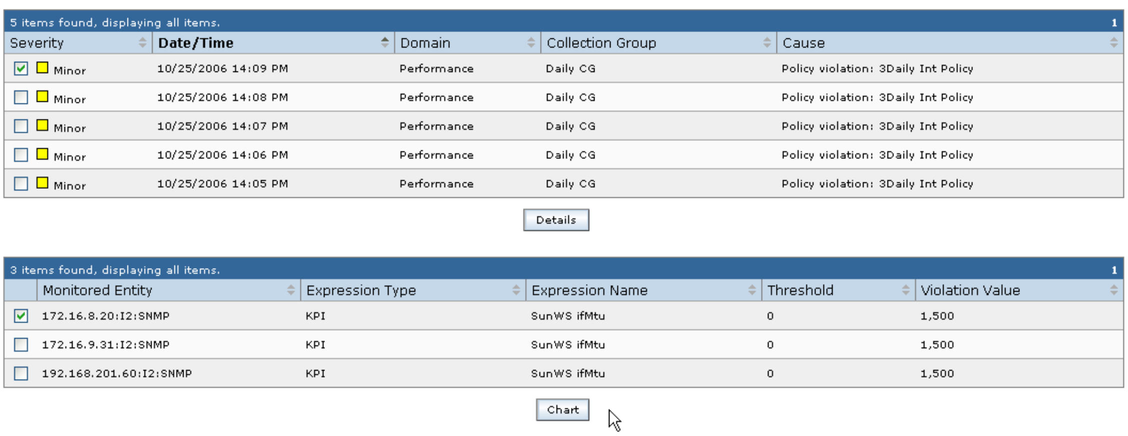
Display Alert Details for an Alert

From Conformance View you can display alert details for a specific alert. Use this feature to view the details of the alert.

To display alert details for an alert

1 From the Alerts list, select the check box next to the specific Alert that you want to view the details for. Click **Details**.

A screen similiar to the following is displayed:



The details for the specific alert are displayed.

Display a Chart for an Alert Detail

From Conformance View you can display a chart of specific alert details. Use this feature to view the details of a specific alert in chart format.

To display a chart for an alert detail

1 From the Alerts Details list, select the check box next to the specific alert detail that you want to view the chart for. Click **Chart**.

A chart is displayed for the specific alert detail.

Administration

9

The following documentation provides information and procedures for administrative tasks in PerforMAX.

Some of the topics in this documentation include the following:

- Admin Menu
- System Passwords
- Event Log Configuration
- System Event Logs
- Security Event Logs
- User Management
- Application Configuration
- Application Configuration Procedures
- FlushProxy Parameter

This documentation describes only some of the basic administrative tasks. For additional information about PerforMAX administration, see the *PerforMAX System Administrator Guide*.

Admin Menu

Administrative tasks are located under the Admin menu from the main screen Launch Pad.

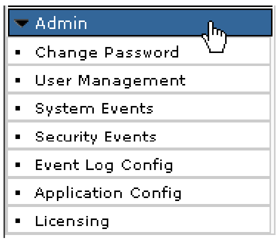


Figure 9-1: Admin Menu

From here, you can perform various tasks including:

- Change passwords
- Configure user permissions and access
- Configure the application
- Configure event logs
- View system and security logs

Viewing User Licensing

To view PerforMAX licensing

1 From the **Admin** menu, select **Licensing**.

The following table shows the license information in the Licensing screen.

Table 9-1: Licensing Information

FIELD	DESCRIPTION
Resource Type	Indicates the name of the feature type that PerforMAX monitors. A resource type may contain letters, numbers, and underscore characters.
Description	A description of a feature that PerforMAX monitors.
Expiration Date	The expiration date of a PerforMAX feature. The expiration date is expressed in a format as follows: <day-month-year> January 1, 2001 is 1-jan-2001.
Number of Licenses	The total number of licenses for the PerforMAX feature.
Used Licenses	The total number of used licenses for a selected PerforMAX feature.
Utilization	The percentage of feature license utilization in relation to the total number of licenses available for a feature.
Version #	Indicates the latest version number of the feature supported by your license file. The version number is in a floating point format, with a maximum of ten numeric characters.

System Passwords

The following information describes working with passwords.

- Password Rules
- Password Expiration
- Changing a User Password

Password Rules

You can specify the maximum length of passwords, as well as the minimum number of uppercase characters, special characters, and numbers that a password must contain.

For additional information about setting password rules, see the *PerforMAX System Administrator Guide*.

Password Expiration

Passwords must be changed by each User within pre-configured time period set by the System Administrator. Upon successful login, Users are prompted with an information dialog box reminding them that the password will expire in “*x*” number of days, followed by the prompt, *“Do you want to change your password now?”*.

If a User does not change their password within the expiration time, the password will be locked and the User password will have to be reset by an administrator.

The only exception to this is for members of the System Admin Group, who can still login after the expiration time.

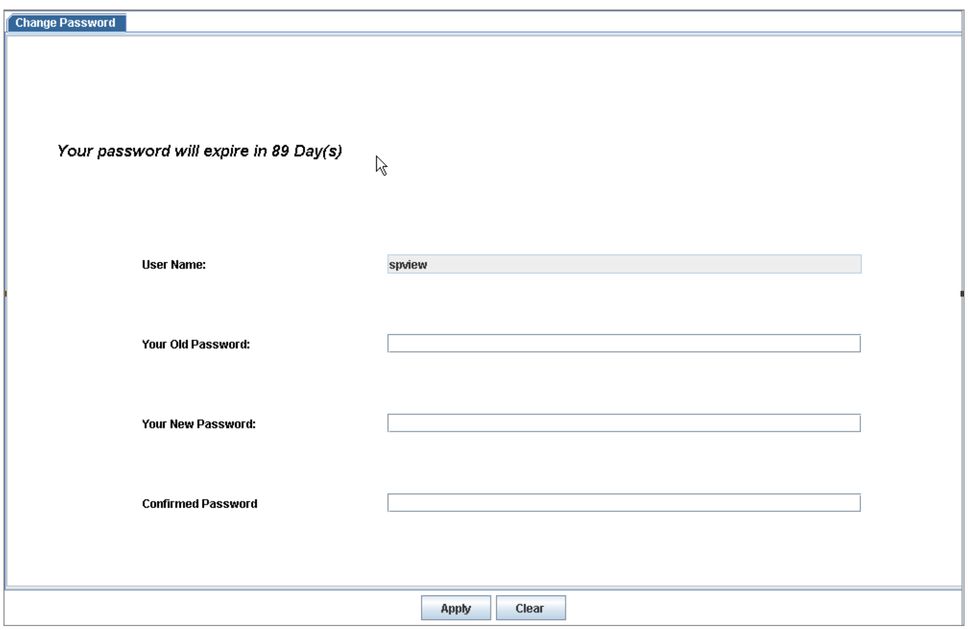
For additional information about setting password aging values, see the *PerforMAX System Administrator Guide*.

Changing a User Password

To change a user password

1 From the **Admin** menu, select **Change Password**.

The following screen is displayed:



2 Enter your old password, new password, and confirmation of the new password.

3 Click **Apply**.

The password is changed.

Event Log Configuration

The Event Log Configuration feature allows you to configure the system to log system/ security types of events using specific criteria.

You can display a list of system/security types of events from the default screen as well as add, and edit, and delete system/security event logs that were previously created.

System event logging consists of any system related incidents (for example, collection or communication failures, configuration problems, licensing issues, and so on).

Security event logging consist of audit type logging (for example, when someone adds/deletes/modifies templates, user or group actions, any changes to policies, report generation, and so on).

A System Event Log tracks the following types of events:

- **User events -** User actions, such as system log-ins, or events resulting from the adding SAPs and connections.
- **Processing events -** Internal processing events require administrator attention. A Processing event usually results from a component failure. PerforMAX categorizes processing events by the severity levels shown in the following table:

.

Table 9-1.

Severity	Level	Description
Critical	1	A system or application fault
Major	2	A major system or application fault
Minor	3	A minor system or application fault.
Warning	4	A system warning of potential error, exception, or abnormal state.
Cleared	5	A former system or application alarm, whose status was cleared.

- **Correlation events -** Events generated by one or more device adaptors connected to PerforMAX.
- **Data Collection events -** User actions that trigger data collection events, such as adding new resources, re-gathering, viewing or merging existing resources.
- **GUI events -** Any user generated events that involve the graphical user interface (GUI).

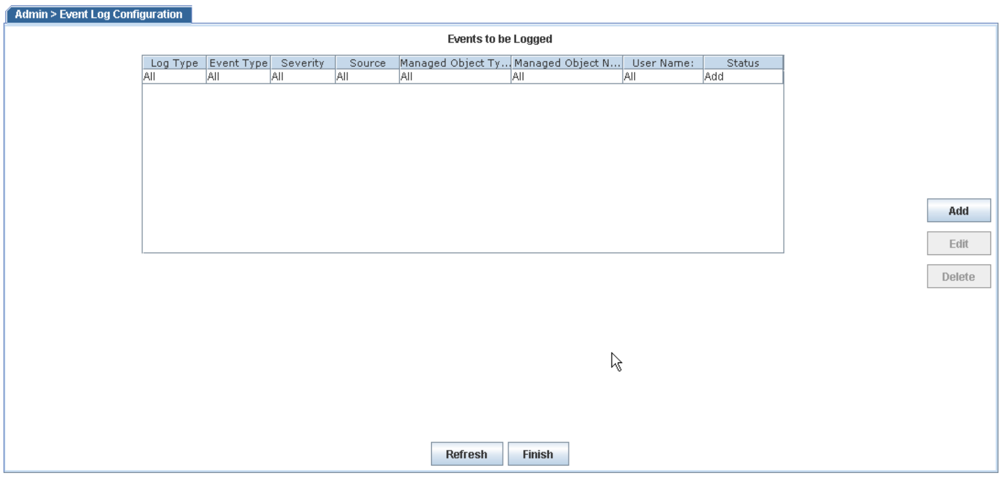
Security Event Logs are similar to System Event Logs except that Security Logs display only security-related events including, Integrity Violations, Operational Violations, Server Security Violations, Time Domain Violations, and so on.

Event Log Configuration Procedures

To display an event log configuration list

1 From the **Admin** menu, select **Event Log Config**.

A screen similar to the following is displayed:



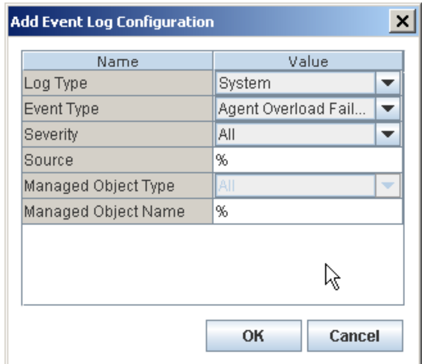
This screen displays a list of previously created system/security event logs. The events to be logged include the following information:

- Log Type
- Event Type
- Severity
- Source
- Managed Object Type
- Managed Object Name
- User Name
- Status

To add a system/security event log

1 From the default screen, click **Add**.

The following dialog box displays:



2. From the Value column, select the **Log Type, Event Type,** and **Severity** from the drop-down list.
3. Enter the **Source** of events to log (**%** defaults to all).
4. Select the **Managed Object Type** (Security event only) from the drop-down list or accept **All** as the default.
5. Enter a **Managed Object Name** for logged events (**%** defaults to all).
6. Enter a **User Name** (Security event only) for logged events (**%** defaults to all).
7. Click **OK**.

   The event is added to the list of events to be logged.

To edit a system/security event log

1. Select an item from the list of events to be logged.
2. Click **Edit**.

   The Edit Event Log Configuration dialog box displays.
3. Edit the criteria as necessary and click **OK**.

   The edited information displays in the list.

To delete an event log

1. Select an item from the list of events to be logged.
2. Click **Delete**.

   The event that was selected is deleted and shows "deleted" in the Status column.

System Event Logs

System event logging consists of all system related incidents (collection failures, communication failures, configuration problems, licensing issues, and so on).

System Event Log Procedures

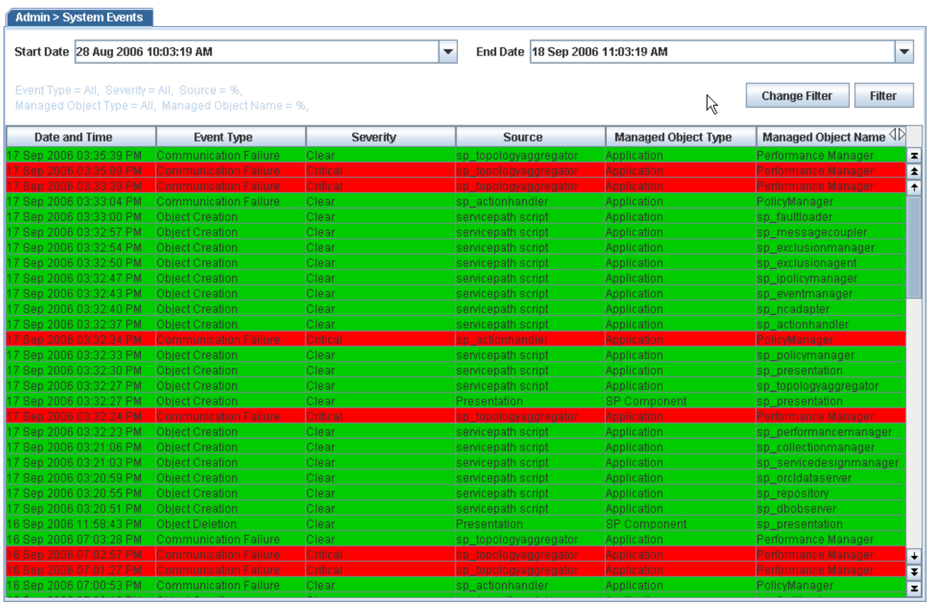
To display system events

1 From the **Admin** menu, select **System Events**.

2 Select a **Start/End** date from the drop-down list for the event dates that you want to display.

3 Click **Filter**.

A screen similar to the following is displayed:



Date and Time	Event Type	Severity	Source	Managed Object Type	Managed Object Name
17 Sep 2006 03:35:39 PM	Communication Failure	Clear	sp_topologyaggregator	Application	Performance Manager
17 Sep 2006 03:35:09 PM	Communication Failure	Critical	sp_topologyaggregator	Application	Performance Manager
17 Sep 2006 03:33:39 PM	Communication Failure	Critical	sp_topologyaggregator	Application	Performance Manager
17 Sep 2006 03:33:04 PM	Communication Failure	Clear	sp_actionhandler	Application	PolicyManager
17 Sep 2006 03:33:00 PM	Object Creation	Clear	servicepath script	Application	sp_faultloader
17 Sep 2006 03:32:57 PM	Object Creation	Clear	servicepath script	Application	sp_messagecoupler
17 Sep 2006 03:32:54 PM	Object Creation	Clear	servicepath script	Application	sp_exclusionmanager
17 Sep 2006 03:32:50 PM	Object Creation	Clear	servicepath script	Application	sp_exclusionagent
17 Sep 2006 03:32:47 PM	Object Creation	Clear	servicepath script	Application	sp_ipolicymanager
17 Sep 2006 03:32:43 PM	Object Creation	Clear	servicepath script	Application	sp_eventmanager
17 Sep 2006 03:32:40 PM	Object Creation	Clear	servicepath script	Application	sp_ncadapter
17 Sep 2006 03:32:37 PM	Object Creation	Clear	servicepath script	Application	sp_actionhandler
17 Sep 2006 03:32:34 PM	Communication Failure	Critical	sp_actionhandler	Application	PolicyManager
17 Sep 2006 03:32:33 PM	Object Creation	Clear	servicepath script	Application	sp_policymanager
17 Sep 2006 03:32:30 PM	Object Creation	Clear	servicepath script	Application	sp_presentation
17 Sep 2006 03:32:27 PM	Object Creation	Clear	servicepath script	Application	sp_topologyaggregator
17 Sep 2006 03:32:27 PM	Object Creation	Clear	Presentation	SP Component	sp_presentation
17 Sep 2006 03:32:24 PM	Communication Failure	Critical	sp_topologyaggregator	Application	Performance Manager
17 Sep 2006 03:32:23 PM	Object Creation	Clear	servicepath script	Application	sp_performancemanager
17 Sep 2006 03:21:06 PM	Object Creation	Clear	servicepath script	Application	sp_collectionmanager
17 Sep 2006 03:21:03 PM	Object Creation	Clear	servicepath script	Application	sp_servicedesignmanager
17 Sep 2006 03:20:59 PM	Object Creation	Clear	servicepath script	Application	sp_orcldataserver
17 Sep 2006 03:20:55 PM	Object Creation	Clear	servicepath script	Application	sp_repository
17 Sep 2006 03:20:51 PM	Object Creation	Clear	servicepath script	Application	sp_dbobserver
16 Sep 2006 11:58:43 PM	Object Deletion	Clear	Presentation	SP Component	sp_presentation
16 Sep 2006 07:03:28 PM	Communication Failure	Clear	sp_topologyaggregator	Application	Performance Manager
16 Sep 2006 07:02:57 PM	Communication Failure	Critical	sp_topologyaggregator	Application	Performance Manager
16 Sep 2006 07:01:27 PM	Communication Failure	Critical	sp_topologyaggregator	Application	Performance Manager
16 Sep 2006 07:00:53 PM	Communication Failure	Clear	sp_actionhandler	Application	PolicyManager

From here, click an item to read specific details of the system event.

Optionally you can:

4 Select/clear the fields to be displayed on the screen. Click the **Hide Column** button.

The Hide Column dialog box is displayed.

5 In the **Hide** column, select or clear the check box next to the field that you want to display in the results list. Click **OK**.

The list displays only the fields that you selected.

To find a specific element(s) in the list:

6 Click **Change Filter** to define more specific criteria that you want to display.

The Event Logs dialog box displays.

7 Select or type in the specific criteria that you want to display and click **OK**.

The screen displays only the specific criteria you selected/entered.

Security Event Logs

Security event logging consists of Audit logging (Add/Delete/Modify templates, users, groups, policies, report generation, and so on).

Security Event Logs are similar to System Event Logs except that the Security Logs display security-related events including, Integrity Violation, Operational Violation, Security Server Violation, Time Domain Violation, and so on.

Security Event Log Procedures

All of the Security Event Log procedures work in the same manner as in the System Event Log feature. See “System Event Log Procedures” for detailed information.

User Management

This section describes the administration options and functions of User Management in the Dashboard. You can create the following:

- User Groups
- Users

User Management lets you configure User Groups and User access to the PerforMAX application. You can add, edit, view and delete various types of User Groups and User permissions and information using these procedures.

The following topics include:

- User Groups
- Users
- Types of User Groups and Users
- User Management Procedures
- User Group Procedures
- User Procedures

User Groups

User Groups are a collection of Users configured in “groups” with similar system permissions and system access. User Groups are defined by:

- Group name
- Group type
- List of permissions
- Accessibility to one or more domains

Users

Users are part of a “User Group.” Users are assigned to one User Group only. Various individual Users make up a “User Group.”

Types of User Groups and Users

The following types of User Groups and/or Users exist in PerforMAX:

- System Administrator
- System User
- Domain Administrator
- Domain User

System Administrator

A System Administrator is the highest level of access permissions and privileges to the Dashboard. A System Administrator might perform the following configuration and management tasks:

- Configure, edit, remove, and view users.
- Configure the system. The System Administrator and System User groups are the only user entities with this permission level.
- Edit or remove system access or resource permissions for service providers or domain entities.

System User

A System User is the second highest level of access permissions and privileges in the Dashboard. A System User might perform the following configuration and management tasks:

- Configure, edit, remove, and view users assigned to their domain.
- Configure the system. The System Administrator and System User group are the only user entities with this permission level.
- Edit or remove system access or resource permissions for service providers or domain entities.

Domain Administrator

A Domain Administrator is also the third highest level of access permissions and privileges in PerforMAX. A Domain Administrator might perform the following configuration and management tasks:

- Configure, edit, remove, and view the following types of users within their assigned domain:
  - Other Domain Admin accounts
  - Domain User

- Manipulate domain entities (for example, create groups of equal or less privilege/permission)
- Create and assign users to a permission group within the Domain Administrator infrastructure

Domain User

A Domain User might perform the following configuration and management tasks:

- Configure user groups of equal or less privilege and permission, within their assigned domain
- Create and assign users to a permission group within the same domain

User Management Procedures

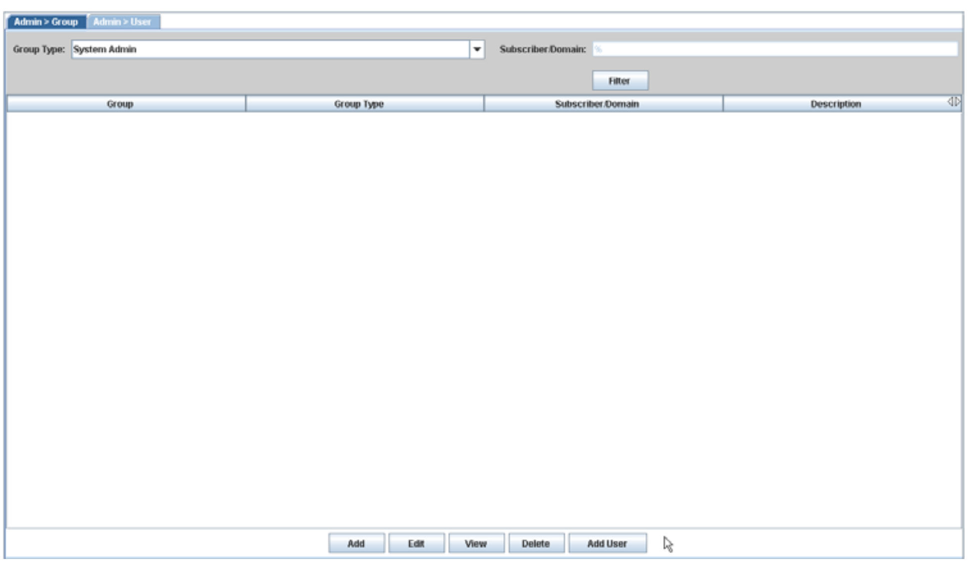
Accessing the User Management Screen

The User Management screen provides all the options for adding, editing, viewing and deleting User Groups and Users and for controlling system access.

To access the user management screen

1 From the **Admin** menu, select **User Management**.

A screen similar to the following screen displayed:



From here, click the **Admin** > **Group** (for User Groups) or **Admin** > **User** (for Users) tab. After selecting the appropriate tab you can perform the following tasks for User Groups or Users:

- Add

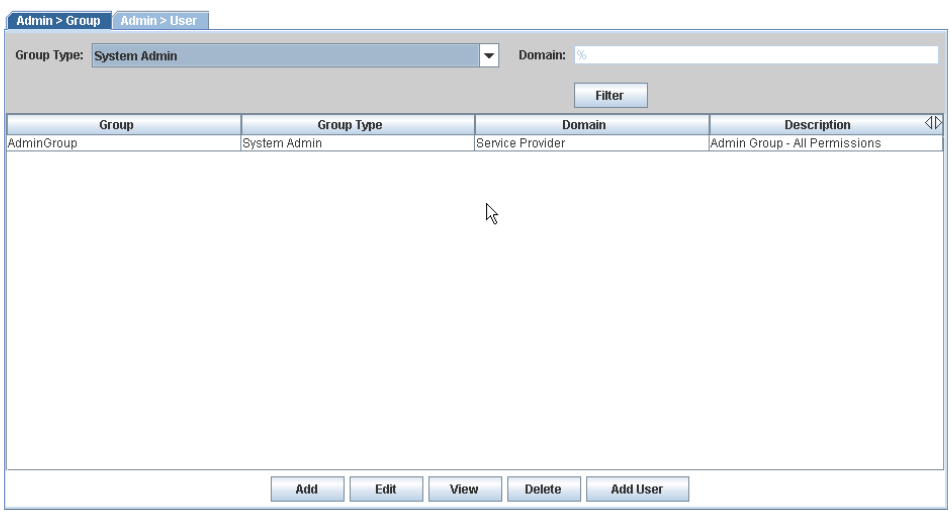
- Edit
- View
- Delete
- Add User

User Group Procedures

To display a user group list

1. From the User Management default screen. Click the **Admin** > **Group** tab.
2. From the **Group Type** drop-down list, select a group type.
3. If necessary enter the **Domain** name, or accept the default (**%**) to display User Groups for all domains.
4. Click **Filter**.

A list of User Groups is displayed:

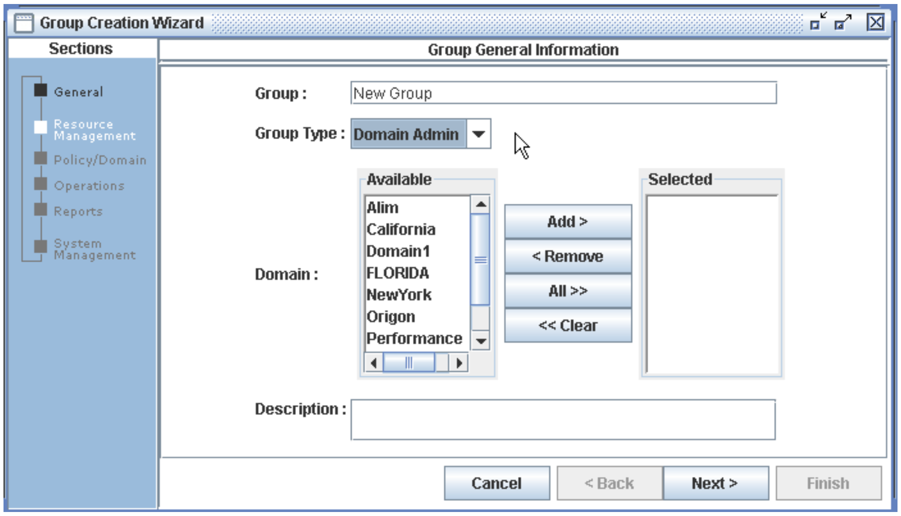


To add a user group

1. From the User Management default screen. Click the **Admin** > **Group** tab.
2. Click **Add**.

   Note: Avoid using the login name or group name of “Dummy.” This is a reserved name to be used only by Trendium, Inc. Using this name will cause technical errors.

A screen similar to the following screen displayed:



3 Type a **Group** name.

4 From the **Group Type** drop-down list, select a group type.

5 From the **Domain Available** field, select one or mutiple domain(s) and perform one of the following:

Note: If you select System Admin or System User Groups as the **Group Type**, the Domain list (Selected pane) defaults to Service Provider and cannot be changed.

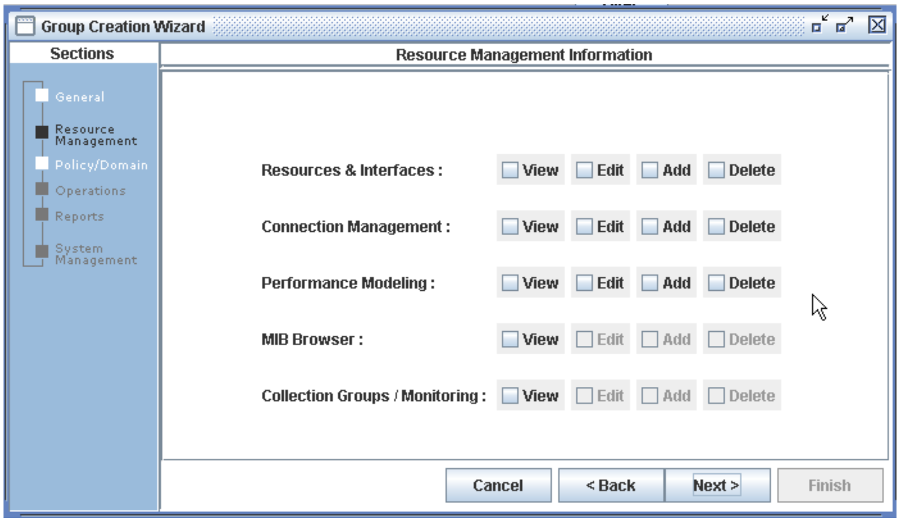
a. click **Add** to add the domain(s) selected
b. click **All** to add all
c. click **Remove** to remove
d. click **Clear** to clear all

After you have added your selections they will appear in the **Selected** pane. These will be the domains whose data can be accessed by the Users that belong to this Group.

6 Type a **Description** of the group being created.

7 Click **Next**.

A screen similar to the following is displayed:



8 Go to "To create user group permissions".

To create user group permissions

These procedures describe how to create User Group permissions in PerforMAX.

1 From the *Resource Management Information* screen, assign the following User Group levels of access (permissions):

- View
- Edit
- Add
- Delete

to the following System Components:

- Resources & Interfaces
- Connection Management
- Performance Modeling
- MIB Browser
- Collection Groups/Monitoring

2 When finished, click **Next**.

The Policy/Domain Information screen is displayed.

3 From the *Policy/Domain Information* screen, assign the valid User Group levels of access (permissions) to the following system components:

- Policy Management

4 When finished, click **Next**.

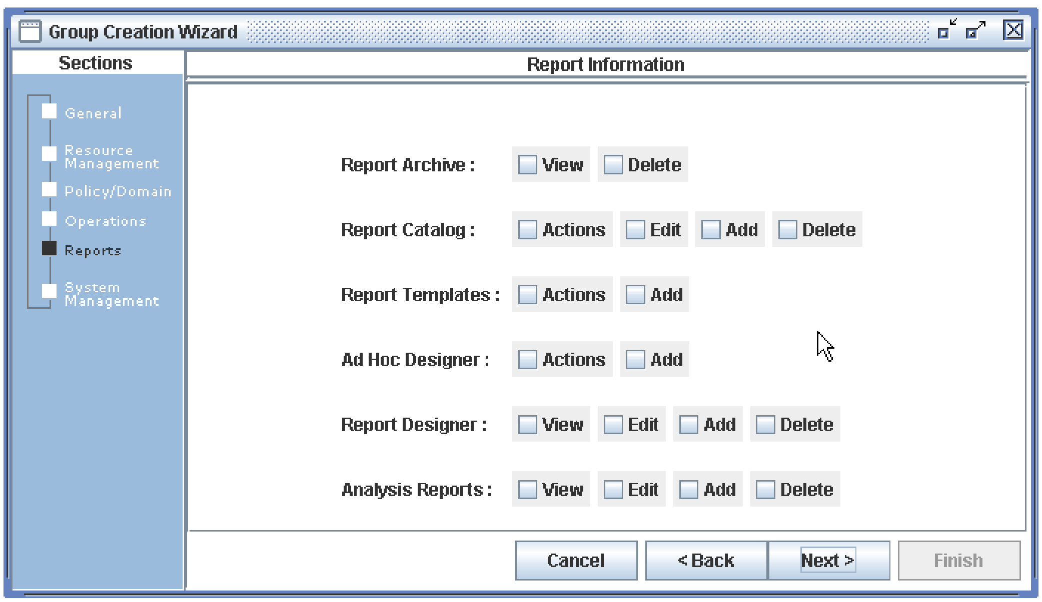
The Operations Information screen is displayed:

5 From the *Operations Information* screen, assign the valid User Group levels of access (permissions) to the following system components:

- Connection Group/Monitoring
- Domain Viewer
- Alert Viewer

6 When finished, click **Next**.

The following screen is displayed:



7 From the *Report Information* screen, assign the valid User Group levels of access (permissions) to the following system components:

- Report Archive
- Report Catalog
- Report Templates
- Ad Hoc Designer
- Report Designer
- Analysis Reports

8 When finished, click **Next**.

The System Administrator Information screen is displayed.

9 From the *System Administrator Information* screen, assign the valid User Group levels of access (permissions) to the following system components:

- User Management

- Licensing & Event Log

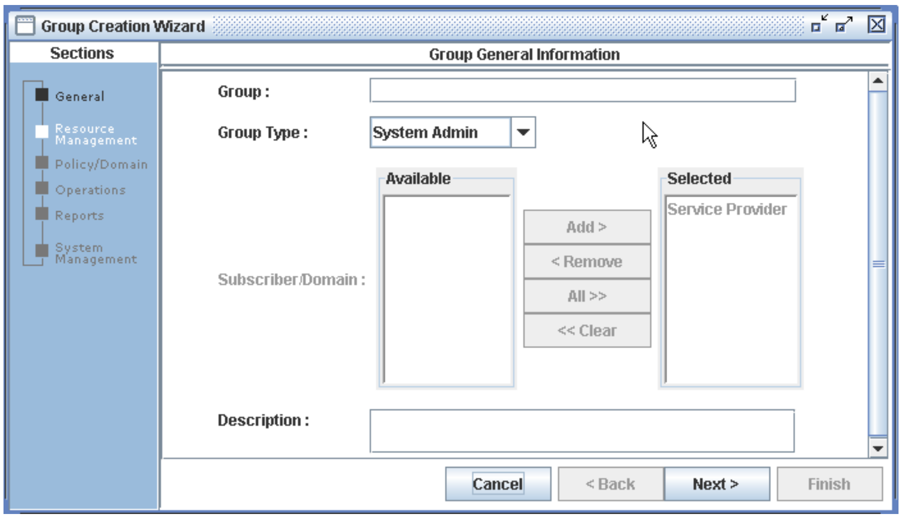
10 When finished, click **Finish**.

The new User Group is added to the system.

To edit a user group

1. From the User Group list, select the User Group that you want to edit.
2. Click **Edit**.

   A screen similar to the following is displayed:



Note: You can edit all sections of the User Groups while using this feature. From here you can edit the general information and all previously assigned permissions.

3. Edit all necessary information. Click **Next** and **Back** to navigate through the screens.
4. Click **Finish** when you are done.

   The default User Group list is displayed.

To view a user group

1. From the User Group list, select the User Group that you want to view.
2. Click **View**.

   The User Group that you selected displays in read-only format.

3. Click **Next** and **Back** to navigate through the screens.
4. Click **Cancel** when you are done.

To delete a user group

1. From the User Group list, select the User Group that you want to delete.
2. Click **Delete**.

   A delete confirmation message displays.

3. Click **Yes** to delete the User Group or **No** to cancel the deletion process.

Note: Before deleting a User Group you must first delete all Users that are assigned to the User Group that you want to delete.

User Procedures

To display a users list

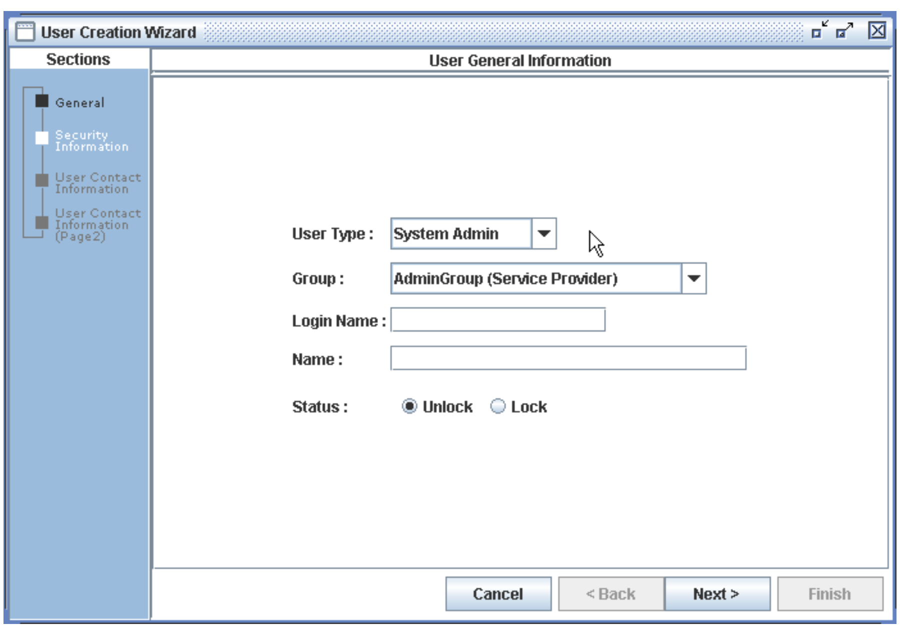
You can display a Users list by selecting specific filtering criteria. This feature works in the same manner as in displaying a User Group list. See "To display a user group list" for detailed information.

To add a user

1. From the User Management screen, click the **Admin** > **User** tab.
2. Click **Add**.

   A screen similar to the following is displayed:

   .



3. From the **User Type** drop-down list, select an Administrator Group type.
4. From the **Group** drop-down list, select the User Group that the User will be assigned to.

5 In the **Login Name** field, type the User's login name.

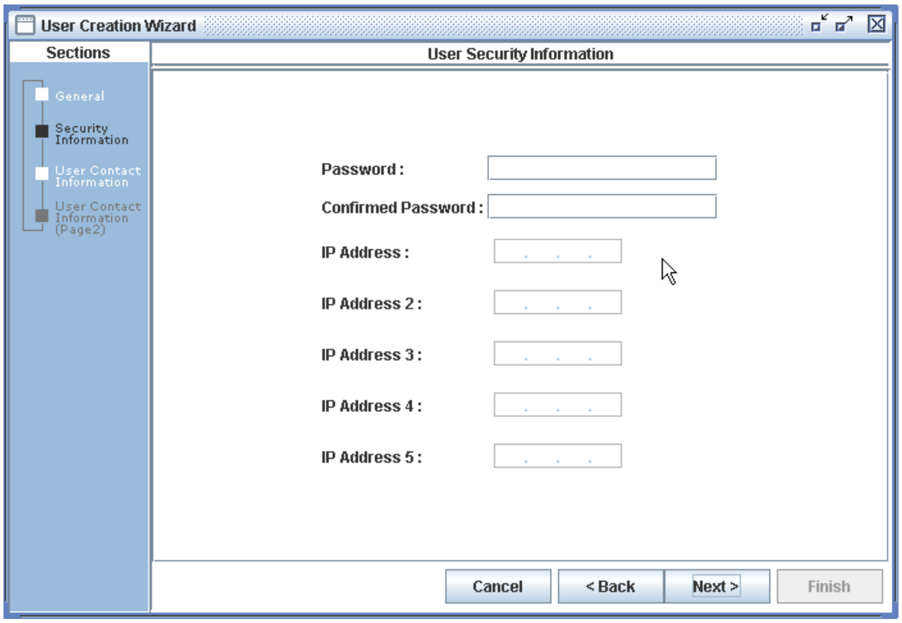
Note: Avoid using the login name or group name of "Dummy." This is a reserved name to be used only by Trendium, Inc. Using this name will cause technical errors.

6 In the **Name** field, type the User's name.

7 Select the **Unlock** or **Lock** option button for the User account **Status**.

8 Click **Next**.

A screen similar to the following is displayed:



9 In the **Password** field, type the initial User password.

10 In the **Confirmed Password** field, type the re-type the initial User **Password** for confirmation.

Note: Password Expiration Date cannot be modified in the User Management GUI. The Password Expiration Date is established by the Security Policy in the server configuration (*ServicePath.config*) file.

11 Click **Next**.

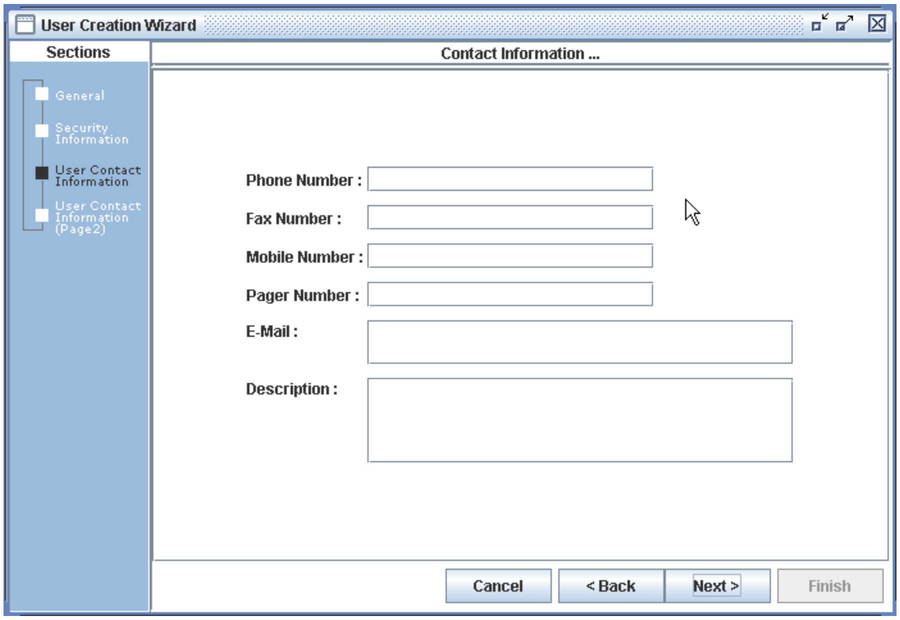
For administrator users only

If you are adding an Administrator User type then you may need to populate the IP Address fields.

12 In the **IP Address(es)** fields, enter up to five addresses for Administrator User access to administrative functions from specific computers.

Note: This step is performed only for **Administrator Users** and when access is to be limited to specific computers. Otherwise, leave the IP Address fields empty.

A screen similar to the following is displayed:



13 In the **Phone Number** field, type the User's telephone number.

14 In the **Fax Number** field, type the User's fax number.

15 In the **Mobile Number** field, type the User's mobile telephone number.

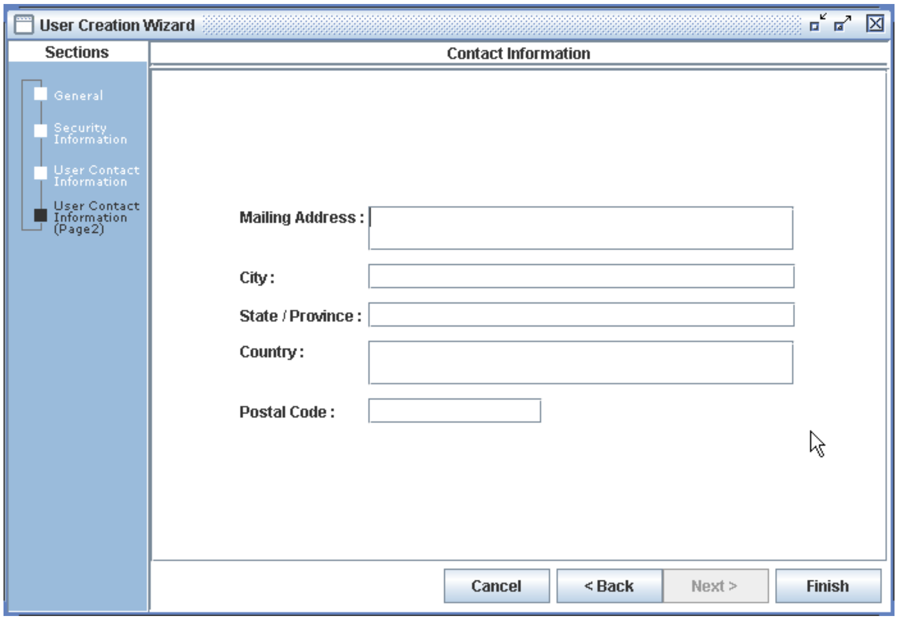
16 In the **Pager Number** field, type the User's pager number.

17 In the **E-mail** field, type the User's e-mail address(es).

18 In the **Description** field, type any additional contact information (optional).

19 Click **Next**.

A screen similar to the following is displayed:



20 In the **Mailing Address** field, type the User’s mailing address.

21 In the **City** field, type the User’s city.

22 In the **State / Province** field, type the User’s state or province.

23 In the **Country** field, type the User’s country.

24 .In the **Postal Code** field, type the User’s postal zip code.

25 Click **Finish**.

The User is added and displayed on the respective Users list.

To edit a user

You can edit a User by selecting a specific user from the User list. This feature works in the same manner as in editing a User Group. See “To edit a user group” for detailed information.

Note: You can edit all User information while using this feature. From here you can edit the general information and all other detailed information.

To view a user

You can view a User by selecting a specific user from the User list. This feature works in the same manner as in viewing a User Group. See “To view a user group” for detailed information.

To delete a user

You can delete a User by selecting a specific user from the User list. This feature works in the same manner as in deleting a User Group. See "To delete a user group" for detailed information.

Application Configuration

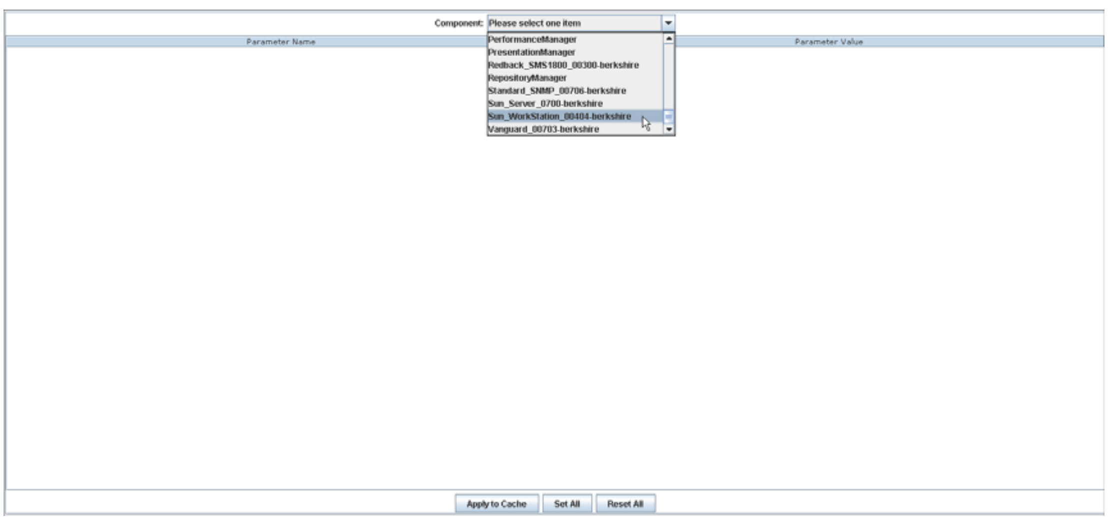
The Application Configuration screen can be used to modify many dynamic application parameters.

Application Configuration Procedures

To view application configuration parameters

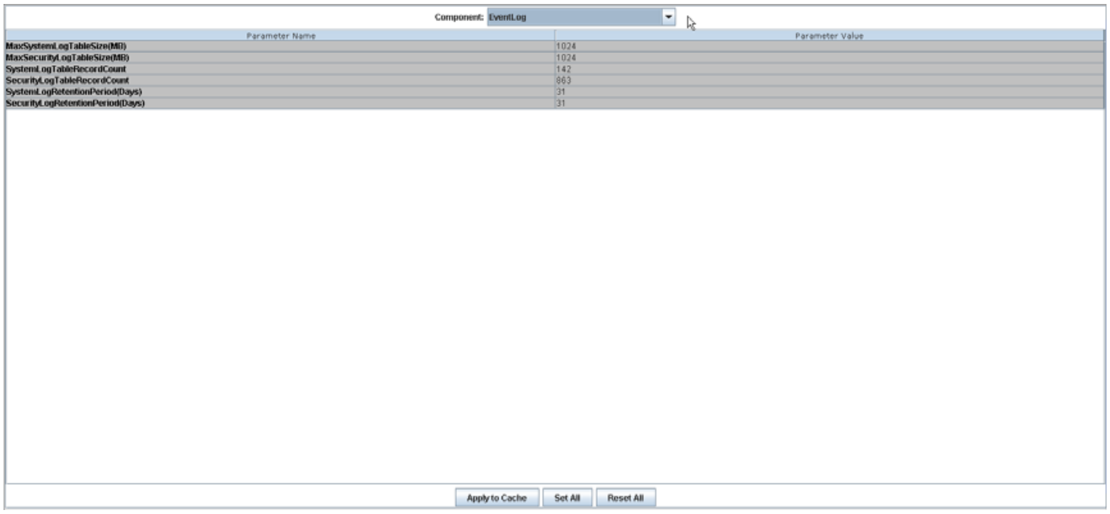
1 From the **Admin** menu, select **Application Config**.

The following screen displays:



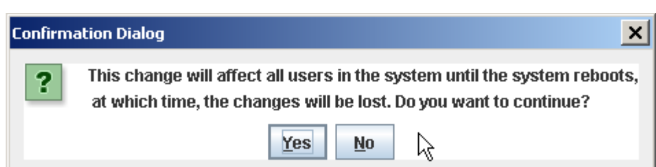
2 Select the desired **Component** from the drop-down list.

The following screen displays when the Event Log component is selected:



To edit application configuration parameters

1. From the **Admin** menu, select **Application Config**.
2. Select the desired **Component** from the drop-down list.
3. Click on a **Parameter Value** to change the value.
4. Type or select the new value.
5. Click on the **Parameter Name** portion of the screen to update the value.
6. When you are finished with all changes:

If you...	then...
Click **Apply to Cache**	*The following confirmation message appears:*  Click **Yes** to save changes or **No** to the return to the screen without saving the changes.
Click **Set All**	*A confirmation message displays.* Click **Yes** to save changes or **No** to return to the screen without saving the changes.

If you...	then...
Click **Reset All**	All changed parameter values before being saved will be reset to default values. **Note:** You must click **Reset All** before clicking **Apply to Cache** or **Set All** for the parameter values to be reset to their default values.

Note: For all Collection parameters, when changing the **Parameter Value** from the GUI, the change is only temporary. Permanent changes must be done in the Collection Agent configuration file.

For example, if you select the **Parameter Value** “Evidence” for the “If running counter has negative value” **Parameter Name** and click **Apply to Cache** or **Set All** then the change will be temporary only.

FlushProxy Parameter

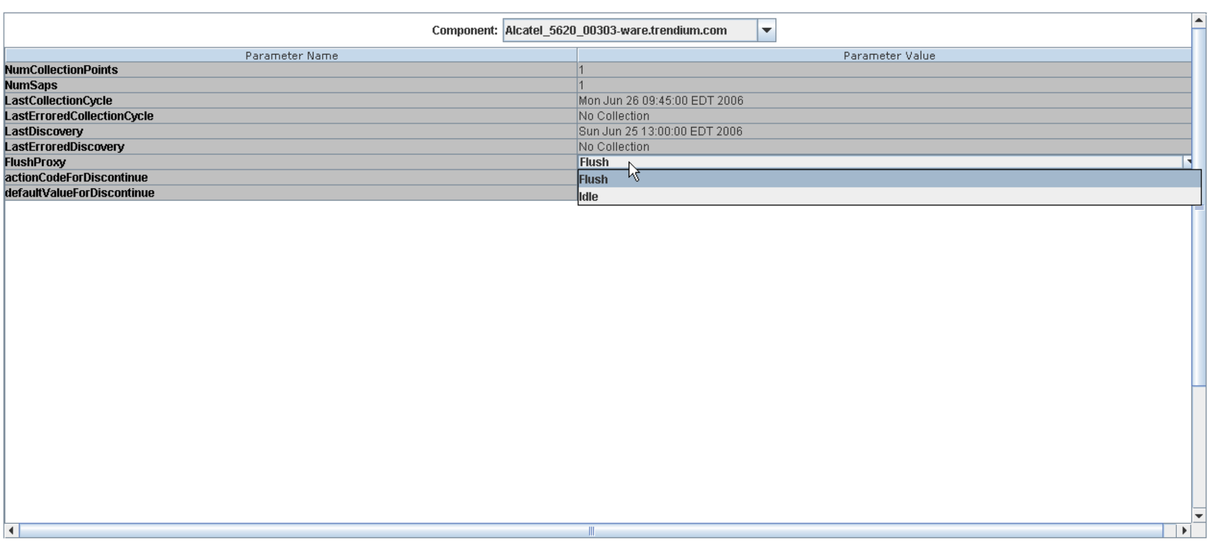
Setting the **FlushProxy** parameter lets you manually flush the cache and send data to the Performance Manager. This allows you to immediately send data rather than wait for the collection cycle.

To set the FlushProxy parameter

1 From the **Admin** menu, select **Application Config**.

2 Select the desired **Component** (device adapters type only) from the drop-down list.

A screen similar to the following displays:



3 Select the **FlushProxy** parameter value.

4. Select **Idle** or **Flush** from the drop-down list where:

   Idle = no action (default value)

   Flush = ready to flush

5. Follow step 6 in “To edit application configuration parameters” to complete the procedure.

Error Handling and Recovery

A

Error messages with clear category and apprehensible context contributes to a user-friendly interface of a system. PeforMAX incorporates error codes with the following attributes:

- Error Code

The error code distinctively identifies each error scenario.

- Error Message

Each error message maps with a unique error code.

- Error Context

The error context provides precise context information regarding the error scenario, such as IDs, name, and so on.

The topics in this documentation include the following:

- Presentation Error Codes and Enums
- Topology Manager Error Codes and Enums
- Repository Manager Error Codes and Error Enums
- Service Design Manager Error Codes and Error Enums
- Data Recovery

Presentation Error Codes and Enums

The error scenarios from internal presentation operations are listed in the following table:

Error Code	Error Enum	Default Error Message
10100001	EX_PRT_UNEXPECTED	Unexpected exception caught in Presentation Manager
10100002	EX_PRT_OBJ_NOT_FOUND	Object $1 not found
10100003	EX_PRT_CORBA_INTF_UNAVAIL	Corba Interface $1 not available

Topology Manager Error Codes and Enums

The error scenarios from topology management interfaces (PeforMAX: CollectionMgr:ITopology) are listed in the following table:

Error Code	Error Enum	Default Error Message
1020001	EX_TPM_UNEXPECTED	Unexpected exception caught in Topology Manager
1020002	EX_TPM_ALREADY_EXISTS	Topology Item $1 already exists
1020003	EX_TPM_FAIL_GET_CPREF	Failed to get CORBA reference to Topology Item $1
1020004	EX_TPM_FAIL_SET_TOPOLOGY	Failed to set Topology Item $1 to repository
1020005	EX_TPM_FAIL_GET_ID	Failed to get database ID(s) from set Topology item $1
1020006	EX_TPM_FAIL_CONFIG	Failed to configure Topology Item $1
Error Code	Error Enum	Default Error Message
1020007	EX_TPM_FAIL_REGATHER	Failed to regather Topology Item $1
1020008	EX_TPM_NOT_FOUND	Topology Item $1 not found in system
1020009	EX_TPM_FAIL_DELETION_ACTIVE	Topology Item $1 cannot be deleted due to active data collection subscription
1020010	EX_TPM_FAIL_DELETION_REPO	Topology Item cannot be deleted due to Repository error $1
1020011	EX_TPM_FAIL_ACTIVATE	Failed to activate Topology Item via Repository operation $1.
1020012	EX_TPM_FAIL_DEACTIVATE	Failed to deactivate Topology Item via Repository operation $1.
1020013	EX_TPM_FAIL_ADD_SD	Failed to add connection $1 to the service topology cache.
1020014	EX_TPM_FAIL_ADD_SLA	Failed to add SLA $1 to the service topology cache
1020015	EX_TPM_FAIL_REMOVE_SD	Failed to remove connection $1 from the service topology cache
1020016	EX_TPM_FAIL_REMOVE_SLA	Failed to remove SLA $1 from the service topology cache

Repository Manager Error Codes and Error Enums

The error scenarios from repository management interfaces are listed in the following table:

Error Code	Error Enum	Default Error Message
1030001	EX_REP_UNEXPECTED	Unexpected exception caught in Repository Manager
1030002	EX_REP_ORALCE_NOTAVAIL	Cannot connect to oracle data server
1030003	EX_REP_INVALID_PARAM_LENGTH	Parameter $1 length exceeding column definition
1030004	EX_REP_FAIL_INSERT_DSS	DSS $1 insertion failed
1030005	EX_REP_FAIL_INSERT_DSS_ORACLE	DSS insertion failed due to oracle error $1
1030006	EX_REP_FAIL_SET_DEVDSS	Failed to attach devices to DSS
1030007	EX_REP_FAIL_SET_DEVDSS_ORACLE	Failed to attach devices to DSS due to oracle error $1
1030008	EX_REP_FAIL_DELETE_DSS	DSS $1 deletion failed
1030009	EX_REP_FAIL_DELETE_DSS_ORACLE	DSS deletion failed due to oracle error $1
1030010	EX_REP_FAIL_INSERT_DEV	Device $1 insertion failed
1030011	EX_REP_FAIL_INSERT_DEV_ORACLE	Device insertion failed due to oracle error $1
1030012	EX_REP_FAIL_DELETE_DEV	Device $1 deletion failed
1030013	EX_REP_FAIL_DELETE_DEV_ORACLE	Device deletion failed due to oracle error $1
1030014	EX_REP_FAIL_ACTIVATE_DEV	Device $1 activation failed
1030015	EX_REP_FAIL_DEACTIVATE_DEV	Device $1 deactivation failed
1030016	EX_REP_FAIL_INSERT_INTF	Interface $1 insertion failed
1030017	EX_REP_FAIL_INSERT_INTF_ORACLE	Interface insertion failed due to oracle error $1
1030018	EX_REP_FAIL_DELETE_INTF	Interface $1 deletion failed
1030019	EX_REP_FAIL_DELETE_INTF_ORACLE	Interface deletion failed due to oracle error $1
1030020	EX_REP_FAIL_INSERT_SAP	SAP $1 insertion failed
1030021	EX_REP_FAIL_INSERT_SAP_ORACLE	SAP insertion failed due to oracle error $1
1030022	EX_REP_FAIL_DELETE_SAP	SAP $1 deletion failed
1030023	EX_REP_FAIL_DELETE_SAP_ORACLE	SAP deletion failed due to oracle error $1
1030024	EX_REP_FAIL_GET_ALL_DSS	Failed to retrieve entire DSS topology
1030025	EX_REP_FAIL_GET_ALL_DEV	Failed to retrieve entire Device topology

Error Code	Error Enum	Default Error Message
1030026	EX_REP_FAIL_GET_DEVSPEC	Failed to retrieve entire device specific infor
1030027	EX_REP_FAIL_GET_DEVPARAM	Failed to retrieve all device params
1030028	EX_REP_FAIL_GET_APPPARAM	Failed to retrieve all application params
1030029	EX_REP_FAIL_GET_DEVCONFIG	Failed to retrieve all device configuration strings
1030030	EX_REP_FAIL_GET_DSSCONFIG	Failed to retrieve all DSS configuration strings
1030031	EX_REP_FAIL_NOTIFY_TM	Failed to notify TM regarding updated Dev/ DSS Configuration
1030032	EX_REP_FAIL_CHECKOUT_SAP_LIC	Failed to check out additional SAP license

Service Design Manager Error Codes and Error Enums

The error scenarios from service design manager interfaces are listed in the following table:

Error Code	Error Enum	Default Error Message
1040001	EX_SDM_UNEXPECTED	Unexpected exception caught
1040002	EX_SDM_BAD_ID	$1 Bad ID, 0 is reserved by system
1040003	EX_SDM_OP_NOTPERMIT	Service topology operation not permitted
1040004	EX_SDM_EMPTY_NAME	$1 Name cannot be empty
1040005	EX_SDM_NULL_ID	$1 ID cannot be NULL
1040006	EX_SDM_FAIL_ADD_CP_MISMATCHEXP	Failed to add service template, expression list mismatch with summarization script
1040007	EX_SDM_FAIL_ADD_CP_MISMATCHPARAM	Failed to add service template, application parameter list mismatch with analysis script
1040008	EX_SDM_FAIL_ADD_CP_BADEXP	Failed to add service template, invalid expression $1, check for preload data
1040009	EX_SDM_FAIL_ADD_CP_BADPARAM	Failed to add service template, expression $1 has parameter $2 not defined
1040010	EX_SDM_FAIL_ADD_CP_BADSUMMSC RIPT	Failed to add service template, invalid sum- marization script
1040011	EX_SDM_FAIL_ADD_CP_BADANASCRI PT	Failed to add service template, invalid analy- sis script
1040012	EX_SDM_ORACLE_ERROR	Service Topology operation $1 failed, oracle error $2
1040013	EX_SDM_FAIL_NOTIFY_SLAMGR	Unable to notify SLAManager

Error Code	Error Enum	Default Error Message
1040014	EX_SDM_BAD_EXP	Invalid Expression $1, check the expression string
1040015	EX_SDM_FAIL_ADD_PT_ANALYSISTYPEMISMATCH	Failed to add policy template, analysis type $1 mismatch the given expression $2
1040016	EX_SDM_FAIL_MODIFY_SD_CLONE	Cloned connection $1 cannot be modified
1040017	EX_SDM_FAIL_INSERT_SD_TUNNEL	Failed to insert new tunnel connection, 2 SAP are required.
1040018	EX_SDM_FAIL_INSERT_SD_SP	Failed to insert new Single Pont connection, more than 1 SAPs are specified.
1040019	EX_SDM_FAIL_INSERT_SD_NOFOCAL	Failed to insert new connection, no focal pts
Error Code	Error Enum	Default Error Message
1040020	EX_SDM_FAIL_INSERT_SD_FUNNEL_EXTRAFOCAL	Failed to insert new connection, funnel detected more than one focal point
1040021	EX_SDM_FAIL_INSERT_SD_DIFFSERV	Failed to insert new connection, SAP $1 has different id than the connection's
1040022	EX_SDM_FAIL_INSERT_SD_DUP_SAP	Failed to insert new connection, duplicate SAP id $1 detected.
1040023	EX_SDM_FAIL_INSERT_SD_FUNNEL	Failed to insert new Funnel connection, at least 2 SAPs are required.
1040024	EX_SDM_FAIL_SETSTATUS_BADSTATUS	Failed to set status, bad status code detected
1040025	EX_SDM_FAIL_SETSTATUS_ALREADYDELETED	Failed to set status because already deleted
1040026	EX_SDM_OBSERVER_NIL	Cannot contact to the DB observer
1040027	EX_SDM_INVALIDPARENT_SELF	Service $1 cannot be parent of itself
1040028	EX_SDM_INVALIDPARENT_HASPARENT	Service $1 cannot be parent service of service $2 since it has a parent service $3
1040029	EX_SDM_INVALIDCHILD_HASPARENT	Service $1 cannot be child service of service $2 since it has a parent service $3
1040030	EX_SDM_INVALIDCHILD_HASCHILD	Service $1 is a parent of child service $2 and cannot act as a child to service $3
1040031	EX_SDM_EXPID_MISMATCH_COND_POLICY_TEMPLATE	Expression ID in condition template mismatches that in policy template
1040032	EX_SDM_NEGATIVE_COLLECTIONCYCLE	Negative value detected in the collection cycle
1040033	EX_SDM_INVALIDTIMESTAMP	$ timestamp is invalid

Data Recovery

The system provides data level recovery mechanisms that focus on the integrity of the user data, such as user data recovery in the data collection adapters, the summarization data recovery in performance manager.

However, not all user data is protected in the PeforMAX system.

There are two approaches to protect and recover data flow in case the receiving end is not available: the file caching and the retry.

The collection proxy uses the file caching mechanism. It synchronizes the data to a local file when the receiving party is out of scope. The data flow recovery happens when the receive party is detected alive by the subsequent data publishing.

The retry approach, keeps the data in the local memory when the receiving party is detected offline, and to initiate a retry timer to trigger republishing of the data assuming that the receiving party will be back online soon.

A combination of both retry and file caching approaches have been implemented so that both performance and data integrity concerns are addressed.

Uneven Collection Cycles

B

Uneven collection occurs when a collection is triggered for a Connection with a collection cycle that is a multiple of the collection cycle supported by at least one endpoint.

Example 1

For example, if the type of devices on either end of a tunnel connection, support different collection cycles (for example, A=10 minutes and Z=30 minutes), then the data must be aggregated for endpoint (SAP) that is collected more often (for example, 10 minutes in our example) so that analysis can occur. The aggregation method is specified by the Vertical Script in the Connection Template.

Using the above referenced tunnel connection, the collection cycle will be set to 30 minutes. Collection will occur for end **A** of the circuit every 10 minutes or three times within the collection cycle, collection for end **Z** will occur one time in the same collection cycle.

Example 2

Two single point connection's which support different collection cycles (for example, 10 mins., 30 mins.) then the data must be aggregated for the first connection's endpoint (SAP). The collection cycle will be 30 mins. (or another higher multiple, such as 1 hr. and so on).

Vertical Script

PeforMAX uses a Vertical Script inside the Connection Template to perform the aggregation of the collected values for a SAP when the Uneven Collection is triggered.

The Vertical Script is used when the user intends to add a connection to:

- a Connection Group

At least one Connection Template must be created to perform the above actions.

Prerequisites

- **KPIs -** In order to be able to create a Connection Template, KPIs must be selected for the Performance Class and Connectivity type.
- **Policy Templates -** If a Policy is to be included in the Connection Template, a Policy Template must be created for the same Performance Class as the Connection Template.

Adding Connection Template Records

To add a connection template

1 From the **Catalog** menu, select **Connection Templates**.

The Connection Templates default screen is displayed.

2 Click **Add**.

The Connection Template General Information screen is displayed.

Note: For complete details on adding a new Connection Template, see “Adding Connection Template Records”.

3 Complete each field in the General Information screen.

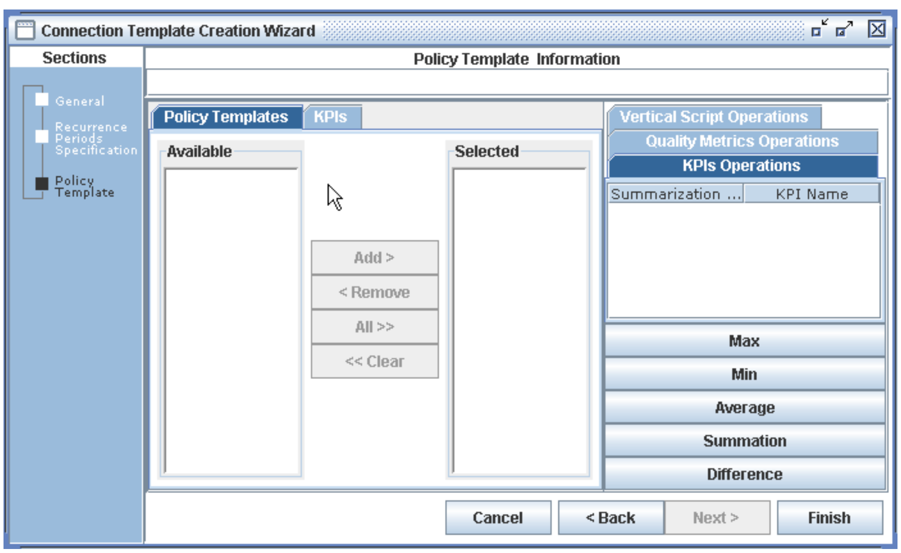
4 Click **Next**.

The Recurrence Periods Specification Information screen is displayed.

5 Complete each field on the Recurrence Periods Specification Information screen.

6 Click **Next**.

A screen similar to the following is displayed:



7 Select the **Vertical Script Operations** tab.

- Possible options for the Vertical Script include:
- Max - of all values being aggregated
- Min - of all values being aggregated
- Average - of all values being aggregated

- Summation - of all values being aggregated
- Difference - of all values being aggregated

8 For each KPI, select the **KPI** in the KPI Operations window and select a **Vertical Script** option.

9 Select **Next**.

The Policy Template Conditions Thresholds Information screen is displayed.

10 Complete the applicable information in the Policy Template Conditions Thresholds screen.

11 Click **Finish**.

The new Template is added to the list.

Contacting Trendium Corporation

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Technical Support

When contacting technical support, please have the following information available:

- Support contract ID
- Company name
- Your name and phone number

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